

ES4204 : Structural Geology and Petrology Fieldwork



By 22MS DES

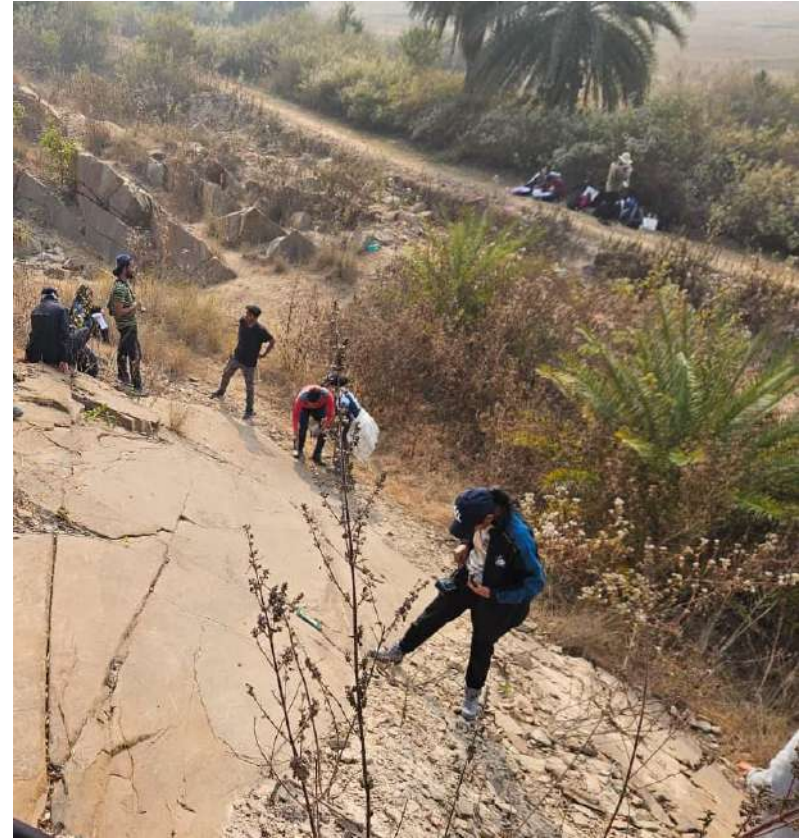


What do we mean by ‘fieldwork’?

It is the essential practice of observing, mapping, and recording geological data directly from the natural environment.

It analyses rock outcrops, structures, and landscapes to understand geological history, assess the hazards, and locate natural resources.

It acts as a crucial bridge between theoretical knowledge and practical applications.



Why is 'fieldwork' important?

- Observation and Data Acquisition
- Geological Mapping
- Resource Exploration
- Hazard Mitigation
- Environmental Assessment
- Hypothesis Testing



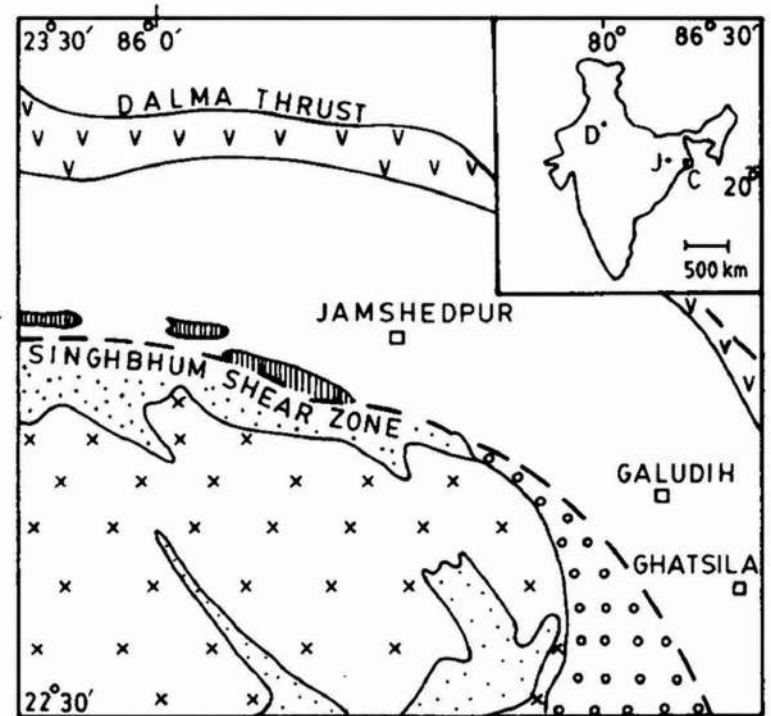
Why did we do the ‘fieldwork’? Our aim?

- **To establish structural geometry through field mapping:** To accurately reconstruct the Chorinda region’s structural map by identifying key marker beds and measuring the orientations of various foliations, lineations, and joint sets.
- **To analyze deformational mechanisms and uniformity:** To determine the mechanisms of fold formation and identify evidence of superposed folding to see if structural styles remain uniform across the region from North to South.

A very brief overview of the study area:

Geological Setting: Our study area is centred on the Galudih-Ghatshila sector, which exposes the Chaibasa Formation.

Structural Significance: To study multiple phases of deformation within the Singhbhum Shear Zone which allows for the observation of superposed folding and the development of distinct foliations.



Source: Ghosh and Sengupta 1987

Geological setting of study area

- Location of Singhbhum Shear zone - Odisha, Jharkhand and parts of eastern India covering approximately ($\sim 50,000 \text{ km}^2$) .
- Precambrian Shear zone consisting of 3 major tectonic units stacked -
 - ➔ **Singhbhum Granite Craton**, the rigid Archaean basement ($\sim 3.5\text{--}2.5 \text{ Ga}$)
 - ➔ **North Singhbhum Mobile Belt (NSMB)** — Proterozoic metasedimentary and volcanic rocks
 - ➔ **Chotanagpur Gneissic Complex** in the north
- **Singhbhum Shear Zone (SSZ)** — the tectonic boundary zone between the two blocks

Sources-
Roy & Matin (2020)
Mazumder (2005)
Mamtani et al. (2004)

Geological setting of study area

- Galudih area lies **just north of the SSZ**, between the Dalma volcanics and the shear zone
- Comprises **alternating quartzites and schists** of the Proterozoic Singhbhum Group
- Rocks here show **southeasterly plunging folds** with NW-SE trending axial planes — directly related to the compressional event that formed the SSZ

Sources-

Roy & Matin (2020)

Mazumder (2005)

Mamtani et al. (2004)

Deformation framework (Expected from Literature)

- **Non-coaxial, progressive simple shear** as the dominant deformation style within the SSZ (*Ghosh & Sengupta 1987; Roy & Martin 2020*)
- A strong **down-dip stretching lineation** parallel to transport direction, consistently present across all lithologies
- Predominantly **reclined, isoclinal asymmetric folds** with hinges plunging steeply down-dip (*Ghosh & Sengupta 1987*)
- Folds expected to be **non-cylindrical** with curved, discontinuous hinge lines
- **NE-SW trending cross-joints perpendicular to fold hinges**, bearing **plumose structures** (Mode-I extension fractures), are predicted as a direct product of flexural-slip folding (*Mamtani et al. 2004*)

Structural and Petrological history

- Three generations of deformation (D1, D2, D3) are recognised, with all three sets of folds (F1, F2, F3) and associated fabrics (S1, S2, S3) excellently developed — particularly visible in the **Tetuldanga area** (~3 km east of Ghatshila) (Roy& Matin 2020)
- Three generations of reclined folds developed **successively within the same progressive deformation**
- Later folds refold earlier ones — **hook-shaped outcrop patterns** result from interference of second and third generation reclined folds

Petrological

- **Grain boundary migration (GBM) recrystallization** dominant over subgrain rotation in quartz and feldspar — indicates deformation temperature **~600°C**
- Near-complete recrystallization of feldspar and full recrystallization of quartz grains into **granoblastic texture**.

Our questions.....in detail.

Broad objective:

1. Prepare a structural map of the Galudih region
2. What is the general structural geometry of the Galudih area?
3. How did you constrain it?
4. Are there any marker beds? Why?
5. Classify the folds based on plunge, closure, interlimb angle, dip isogon. Are the folds overturned?
6. Can you comment on the mechanism that may have led to the formation of these folds. Please justify your answer.
7. Is the structure uniformly developed in the region? Why?
8. Is the structural style remaining uniform from South to North in the region? Why?
9. What was the original orientation of the paleocurrent direction?
10. Map the Joint sets, and try to correlate them with the fold.

River section:

1. How many types of foliations did you recognize? How did you recognize them? What are their orientations?
2. How many types of lineations did you recognize? How did you recognize them? What are their orientations? What are the rakes of these lineations on the surfaces that they are exposed?
3. Can you comment on the structural facing of these rocks?
4. What are the different orientations of hinge lines that you inferred? What are the criteria that you used?
5. What types of minor folds did you observe? Do their asymmetry change looking from the same direction?
6. How do the orientations of the minor folds correlate with the hinge line orientations that you measured earlier?
7. Did you recognize any evidence of superposed folding? Can you infer the type of interference pattern?
8. Which lithology develop the cleavage better? Do you see evidence of cleavage refraction?
9. Did you find any igneous rocks? Can you comment if these are concordant or discordant rocks. Please justify.

Broad questions

1. What is the structural architecture of the region, and how is it established from field evidence?
2. What are the characteristics and kinematics of deformation?
3. What is the deformation history and how does the lithology and structural overprinting control it?

Study area

The whole study area Group will show some pictorial and quantitative analysis from some literature review on our zone of interest and leave on some questions to be answered for the later sections.

Study Area



Chorinda

22°40'20.58"N

86°25'54.61"E



Tetuldanga

22°33'40.11"N

86°30'29.49"E



Subarnarekha
river section

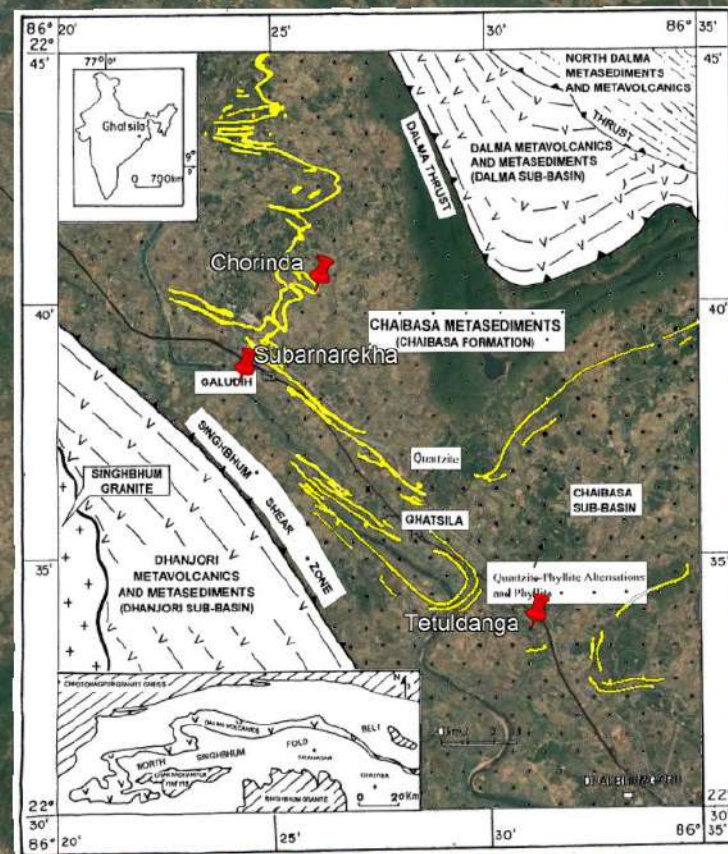
22°38'32.23"N

86°24'17.49"E



Study Area

Regional Geological setting



Modified from:

Bhattacharya, H. &

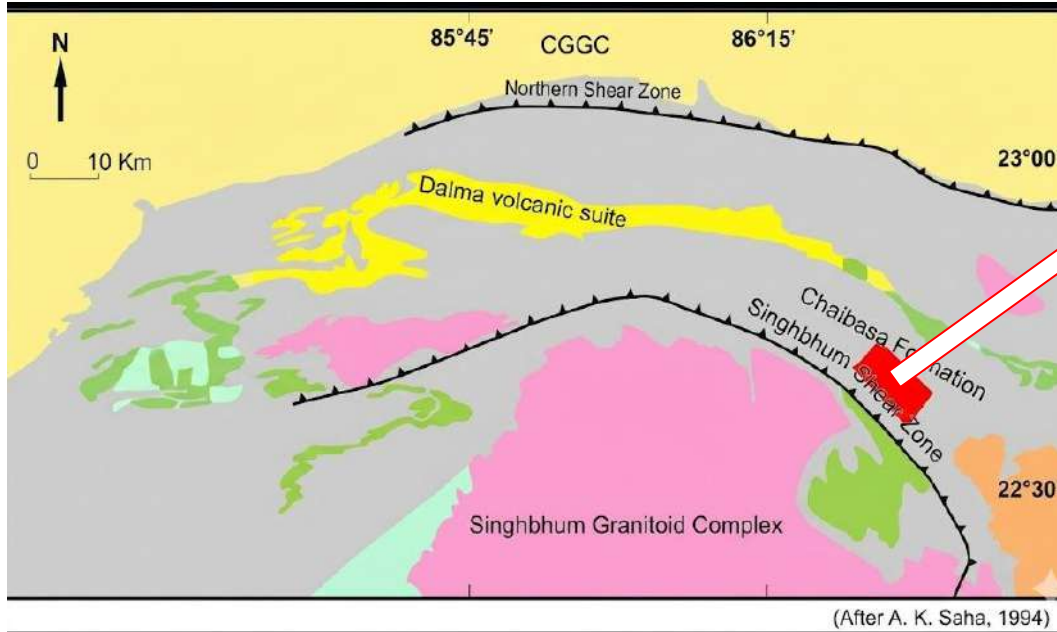
Bandyopadhyay, Sandip. (1998)

Overlapped:

Google Earth

Study Area

Regional Geological setting

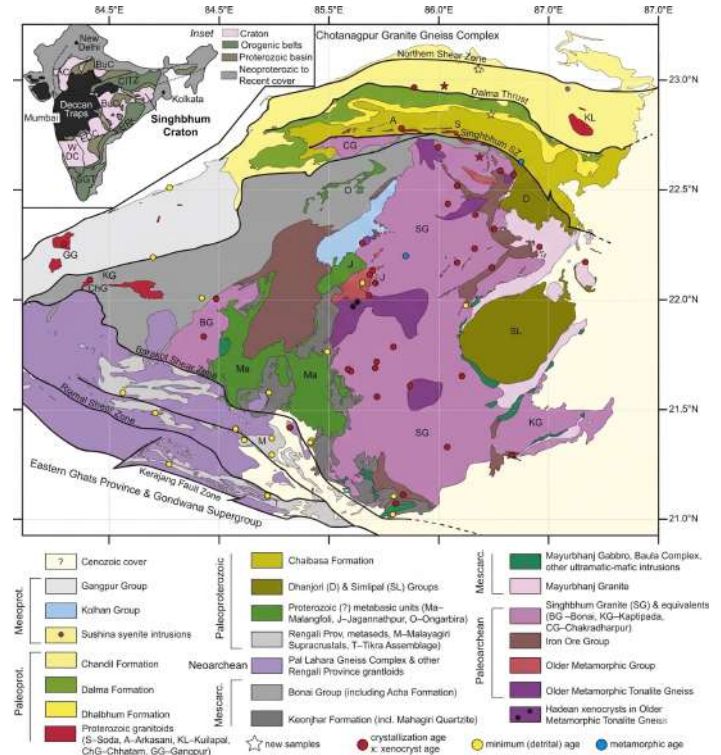
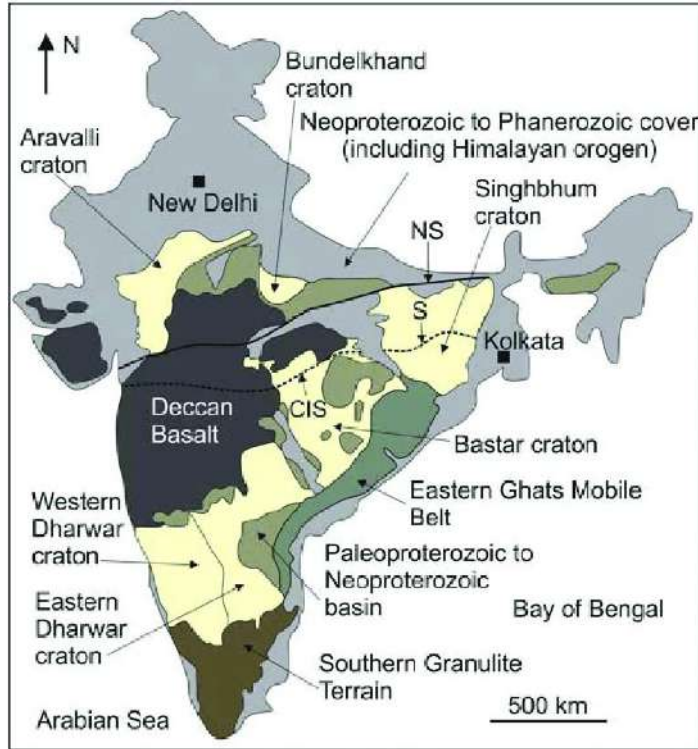


Our
Study Area

Singhbhum Craton one of the
oldest Archaean craton

Study Area

Broader Geological setting

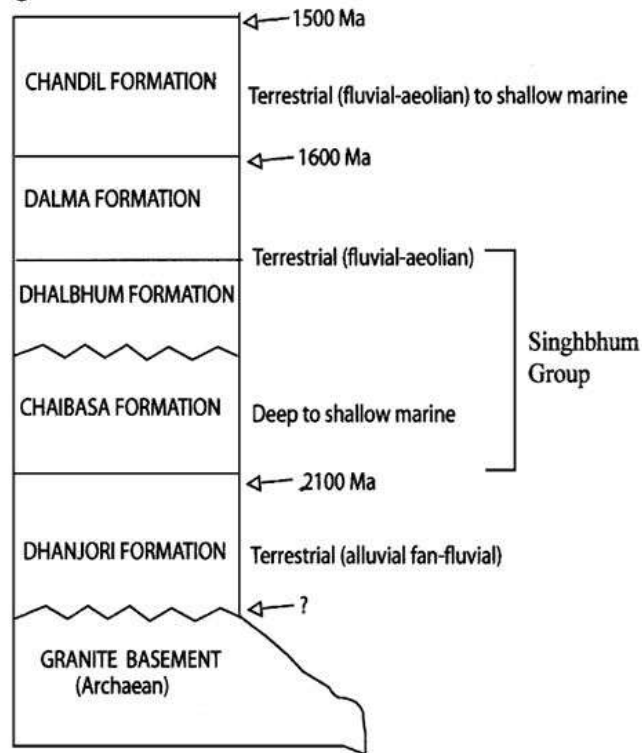


Sukanta Dey et al.(2019)

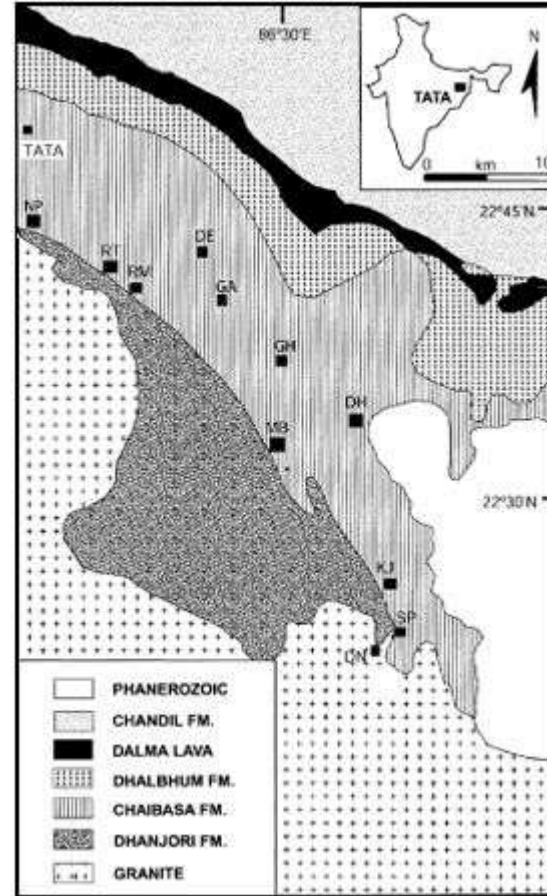
Olierook et al.(2019)

Study Area

Stratigraphic Framework



Source-Mazumder 2005



Source- Mazumder 2005

Chandil Formation



Marks **younger sedimentary** phase dominated by **sandstone and shale**.

Dalma Formation



Composed of **volcanic and volcanoclastic rocks**(basaltic, tuffs) representing a major phase of intense volcanism and tectonic activity in the Singhbhum region.

Singhbhum Group



Singhbhum Group :- (a) **Dhalbhum Formation**
b) **Chaibasa Formation** : Mainly **Sedimentary** rock.

Dhanjori Formation



Represents **early volcano-sedimentary basin** [volcanic rocks (basalts, tuffs) interbedded with sedimentary rocks], indicating initial crustal instability and basin formation.

Granite Basement



The Singhbhum Granite is the oldest unit in the region and forms the **Archaean basement core**, providing a stable continental foundation over which all younger volcano-sedimentary sequences were deposited.

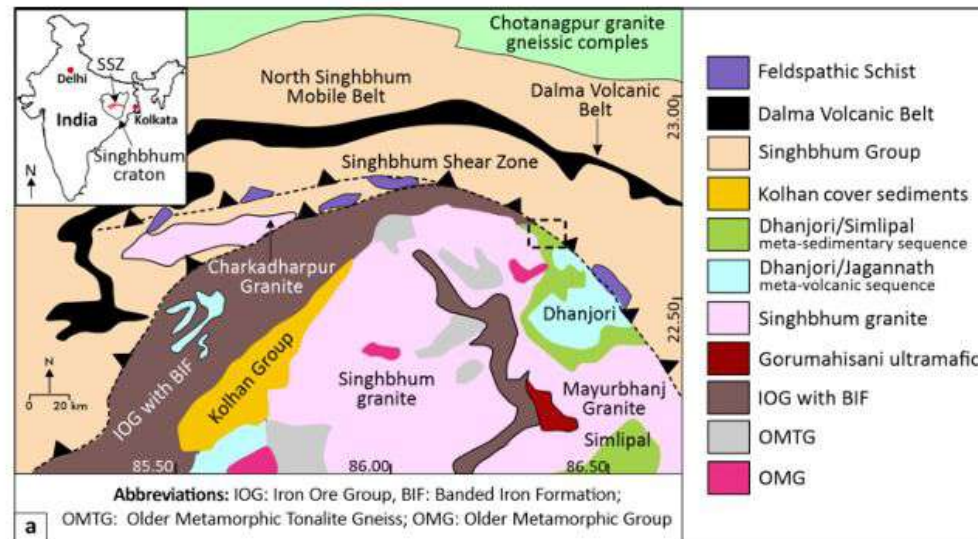
Importance of Chaibasa Formation(~2000 to 2100 Ma)

- ❖ Chaibasa is an ideal area for the students of geology background .These areas are ideally suited for geological field works and research. We can better understand the rock deformation and its manifestation due to lithological contrast.
- ❖ Chaibasa Formation is dominantly sedimentary and represents the early basin phase, whereas Dhalbhum Formation shows volcanic influence and reflects a later tectonically active stage of basin evolution.
- ❖ Chaibasa is known for economic resources like Uranium and Copper deposits.
- ❖ The formation consists of low to medium grade mica-schists, garnet muscovite and quartzites .

Study Area

Stratigraphic Framework

Stratigraphic Units	Age in Ma
----- UNCONFORMITY -----	
Newer Dolerite dykes and sills	1600–950
Mayurbhanj Granite	2100
Ultramafics (Gabbro-Anorthosite)	2100–2200
Kolhan Group	2600–2500
----- UNCONFORMITY -----	
Dhanjori and Simlipal Volcanics, granitoid plutons	2700
Quartzite-Conglomerate (Dhanjori Group)	2800
Pelitic and arenaceous metasediments (Singhbhum Group/Chaibasa Formation)	2900
----- UNCONFORMITY -----	
Singhbhum Granite (SBG - III)	3000
Iron Ore Group	3100
Singhbhum Granite (SBG - I and II)	3300
Older Metamorphic Tonalite Gneiss (OMTG)	3700
Older Metamorphic Group (OMG)	4000



source: Understanding the kinematics of the Singhbhum Shear Zone , Swagata Singha et al,2026

Lithology

1. Chorinda Region - Quartzite is a hard, quartz-rich rock formed from sandstone, where quartz grains have recrystallized showing strong deformation and high strength.
2. Subarnarekha River Section - garnet–muscovite–quartz schist showing strong schistosity; evidence of metamorphism .
3. Subarnarekha River - mafic–intermediate dyke cutting across mica schist; represents intrusive event.
4. Tetuldanga - Quartz–mica rich schist with well-developed foliation and compositional banding; records multiple deformation phases.



**Chorinda
(Quartzite)**



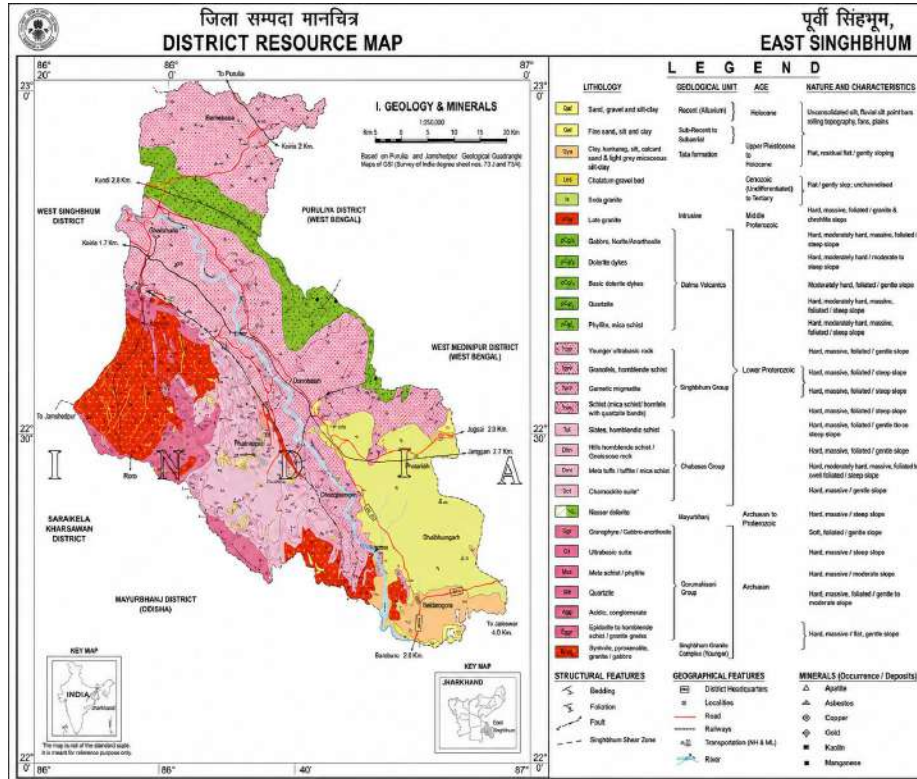
**Riverside Section
(Garnet bearing Mica
schist)**



**Tetuldanga
(Mica-schist)**

Study Area

Lithology

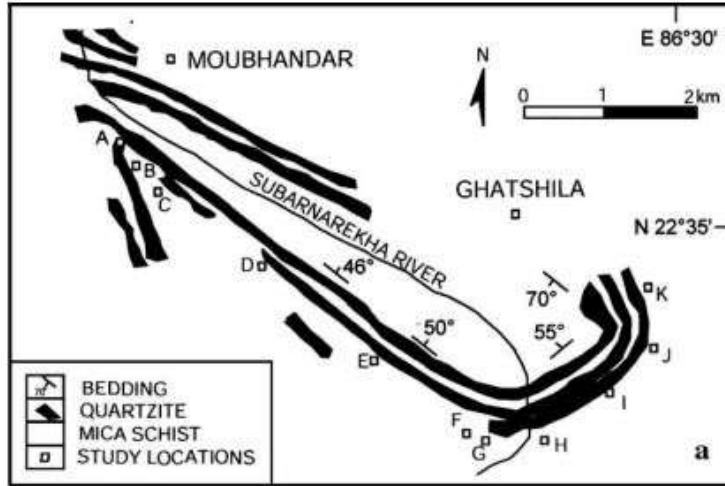


- Our Study area Primarily have Mica and Quartz rich rocks.
- Chorinda has Quartzite beds
- We have seen Mica-rich outcrops in Tetuldanga and Subarnarekha.
- Chaibasa we have seen meta sedimentary sequences.

source: Geological Survey of India

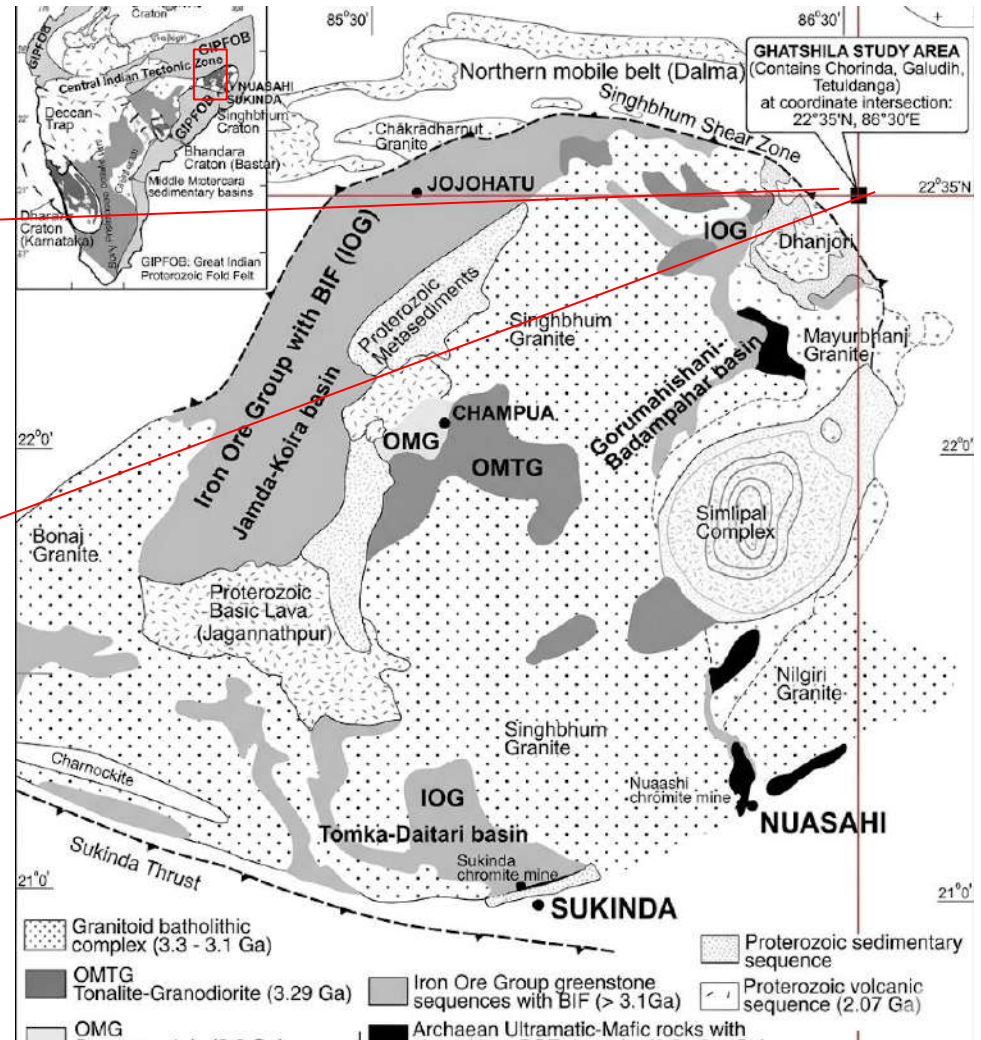
Study Area

Structural and Tectonic Importance



Geological map showing litho-units (Chaibasa Formation) in and around Ghatshila (modified after Naha, 1965).

Source : Majumdar, 2005



Study Area

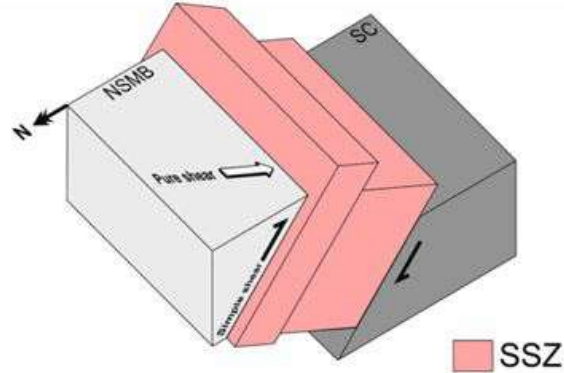
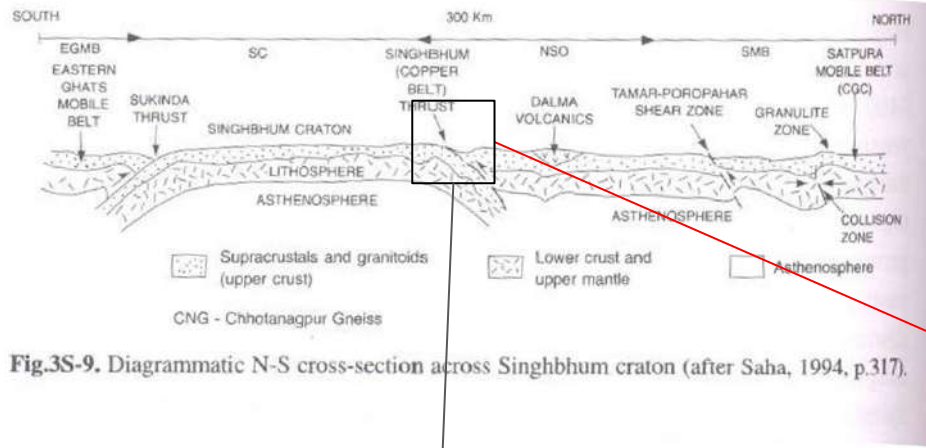
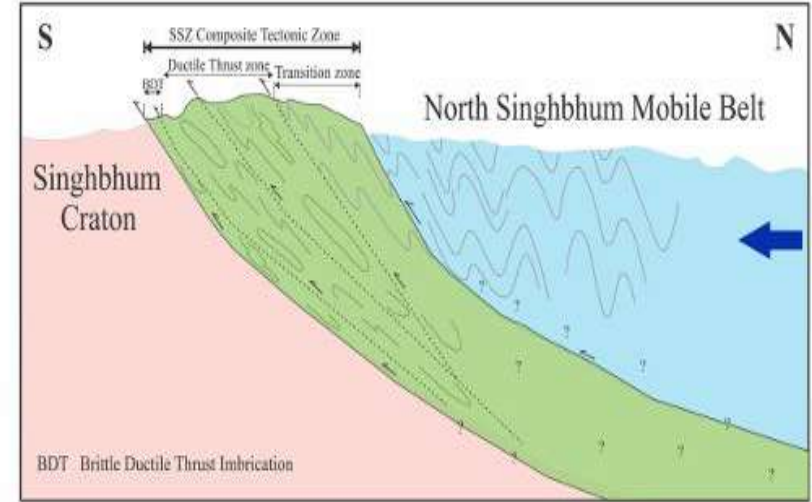


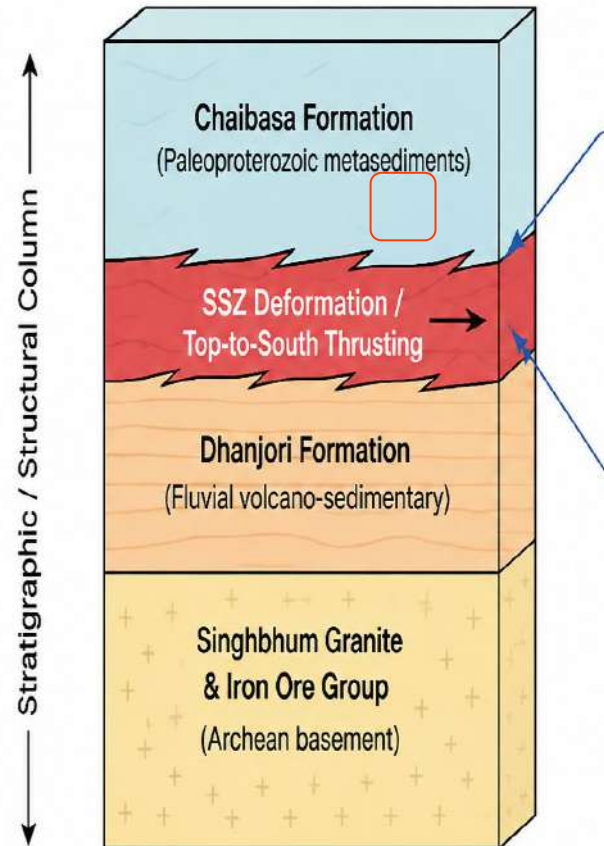
Figure 10. Schematic diagram of generalized 3D model of SSZ and adjoining areas marking the simple shear and flattening deformation in the SSZ.



schematic sketch of regional cross-section across Singhbhum shear zone (SSZ) mark the structural contact zone between Singhbhum Craton and North Singhbhum Mobile belt with the SSZ composite tectonic zone in between..

Source : Roy and Matin (2020)

Study Area



The Great Juxtaposition: Hanging Wall vs. Footwall

	Hanging Wall (North) North Singhbhum Mobile Belt (NSMB)	The Boundary Singhbhum Shear Zone (SSZ)	Footwall (South) Singhbhum Craton
Age	Paleoproterozoic	Strong L-S tectonites, mylonitized quartzite, soda granites, dynamic recrystallization	Meso- to Paleoproterozoic (3.3–3.5 Ga)
Stratigraphy	Chaibasa Formation		Dhanjori Group (intracontinental rift) & Singhbhum Granite
Lithology	Metasediments, mica schists, phyllites		Volcano-sedimentary successions, granitic batholith
Structural Role	The overriding thrust sheet		The subducting/underthrust basement

How then chaibasa ,Dhanjori , singhbhum craton, SSZ formed ?

Study Area (evolution)

Plume process (making of chaibasa)

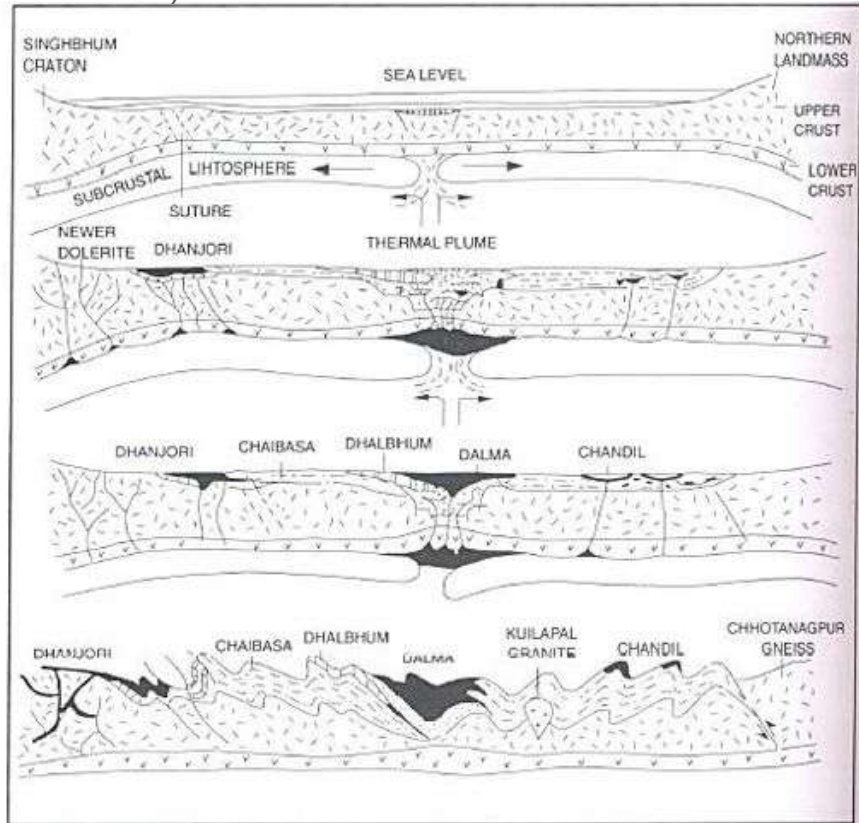


Fig.3S-11. Model of rifting and ensialic orogeny of Gupta and Basu (2000).

Subduction process (making of ssz)

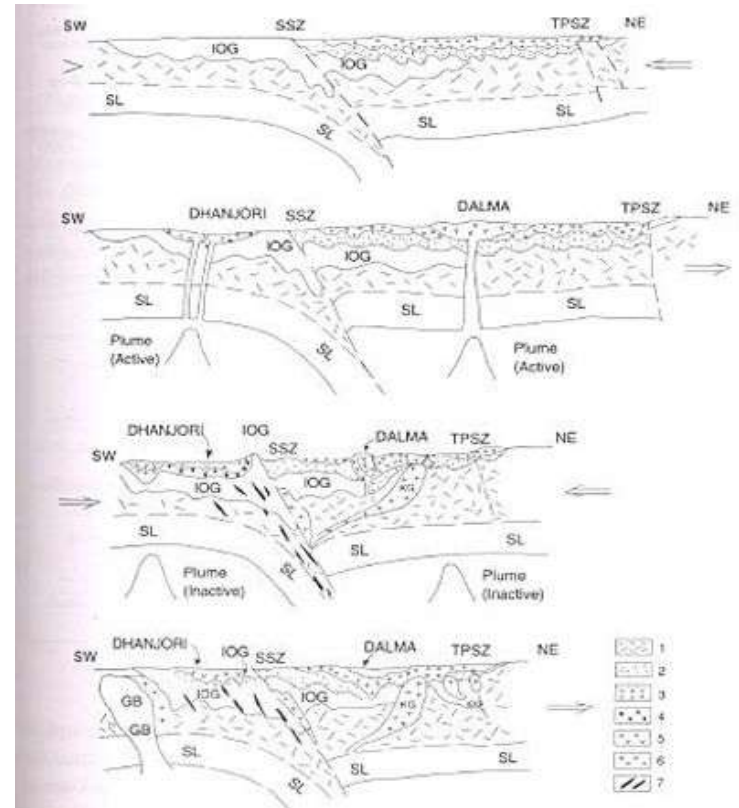
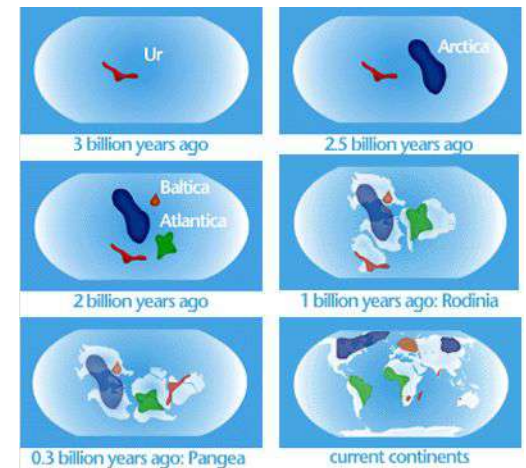
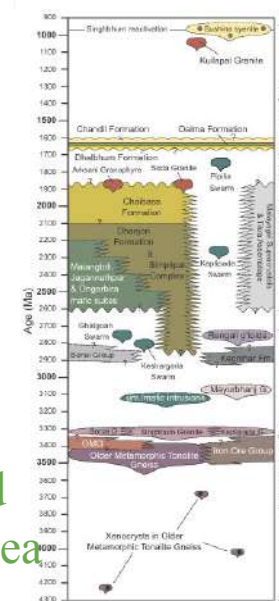
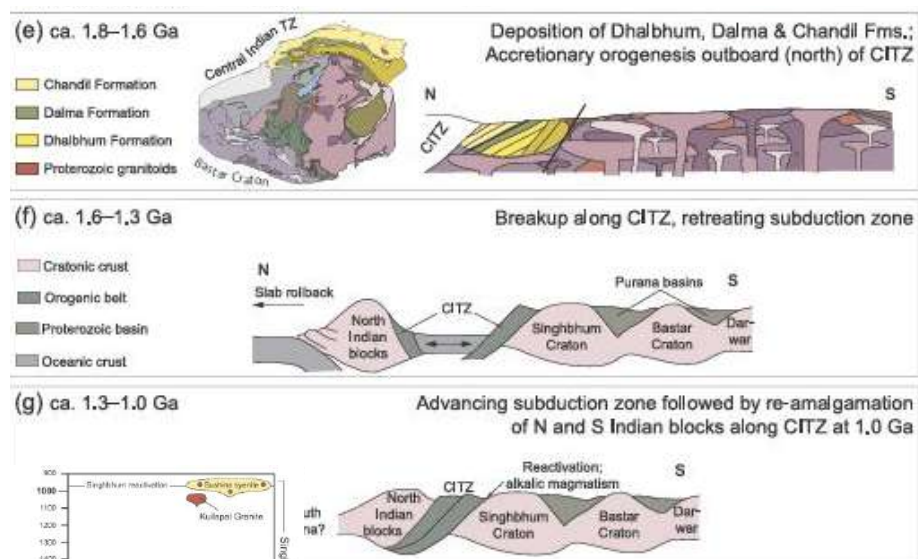
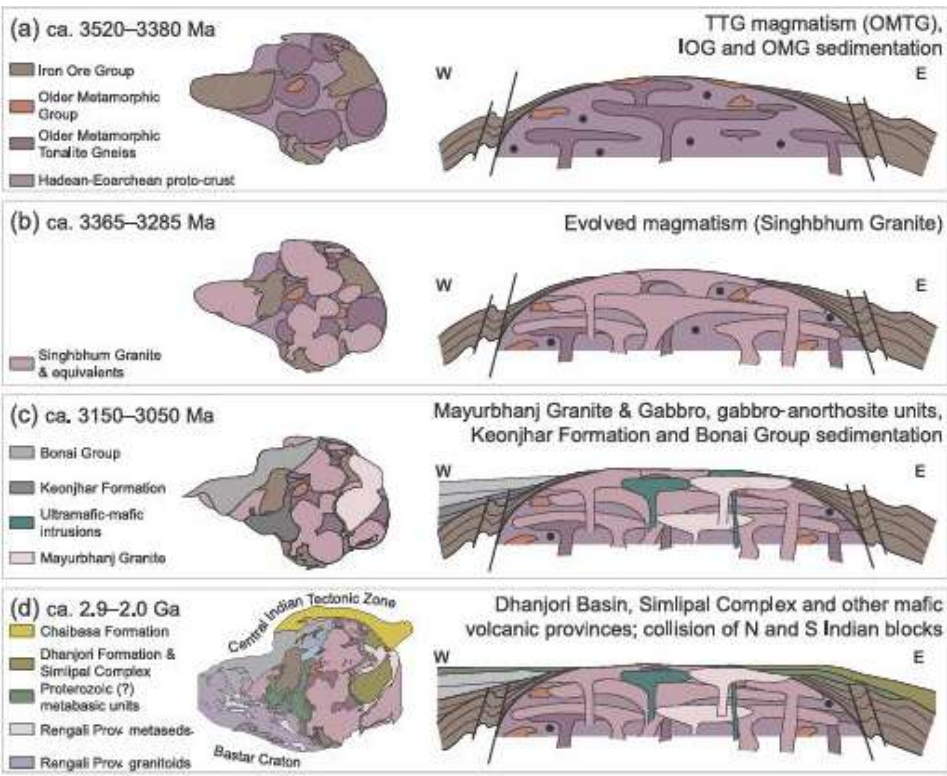


Fig.3S-10. Intraplate subduction model of Sarkar and Saha (1983). 1 - Continental lithosphere, 2 - Chaibasa Formation, 3 - Dhalbhum Formation, 4 - Dhanjori conglomerate, 5 - Dhanjori, Dalma lavas, 6 - Singbhum granite, 7 - Ultramafites. SL - Subcrustal lithosphere, IOG - Iron Ore group, GB - Gabbro, KG - Kailapal granite, SSZ - Singbhum shear zone, TPSZ - Tamar-Poropahar shear zone.



So we are seeing amalgamation breaking of columbia , rodinia involved with respect to our study area

So our study area deformed many times , do we have the data to defend from the field Olierook et al.(2019)

Location / Region	Key Minerals & Metals	Economic & Tectonic Significance
Ghatshila Area (Galudih, Moubhandar, Tetuldanga)	Copper (Cu) , Gold (Au), Silver, Nickel	The "Copper Belt." Why here? Hydrothermal fluids were trapped in the intense S2 and S3 fold hinges (like those at Tetuldanga) and sheared schist planes. Moubhandar serves as the primary smelting hub for the region.
Central Singhbhum Shear Zone (SSZ) (Jaduguda, Turamdih)	Uranium (U) , Apatite, Magnetite	India's premier Uranium Province. Why here? The SSZ acted as a deep-seated crustal conduit. The crushed, "mylonitized" rocks provided the perfect porosity for uranium-rich fluids to concentrate during the Proterozoic amalgamation.
Sukinda & Nuasahi (Southern/Eastern Craton)	Chromite (Cr) , Platinum Group Elements (PGE)	The Chromite Capital. Why here? These sites host ancient Archean ultramafic complexes. Massive chromite layers settled out of cooling magma chambers billions of years ago. Sukinda alone holds nearly 95% of India's chromite.
Iron Ore Group (IOG) Basins (Jamda-Koira, Noamundi)	Iron (Fe) , Manganese (Mn)	The Iron Ore Hub. Why here? These are massive "Banded Iron Formations" (BIFs). The rifting events discussed by Gupta and Basu (2000) created deep basins where iron-rich sediments could accumulate on a global scale.

Questions

Literature review gave us the tectonic history with lithological and stratigraphic successions

Now ,

- Does the field data shows lithological and stratigraphic signatures of Chorinda, Galudih , Tetuldanga as shown from the literatures
- Will the field data be coherent with tectonic evolution?
- Is Multigenerational deformation evident from the field data?

Methodology

Methodology

Field Analysis

Contents

1. Brunton Compass
2. Location and Bearing
3. Outcrop identification and lithology
4. Structure identification
5. Data collection
 - inner arc & outer arc
 - lateral distance measurement
 - Outcrop width measurement
6. Mapping
 - Fold Mapping
 - Fracture Mapping
7. Hinge Line determination using striping lineation

1. Capturing Data with the Brunton Compass



What do we Measure?

- The earth's crust is constantly being deformed by tectonic forces, causing flat rock layers to fold, tilt, and fracture.
- Allows us to measure the exact 3D orientation of these rock layers in the field so we can plot them accurately onto a 2D topographic map.

1. Measuring Horizontal Features (Bearing & Trend)

Tells the direction a linear feature and the direction of a distant landmark.

- **Leveling:** Hold the Brunton perfectly flat.
- Watch the **Round Level** (*the bullseye level*) and ensure the bubble is centered.

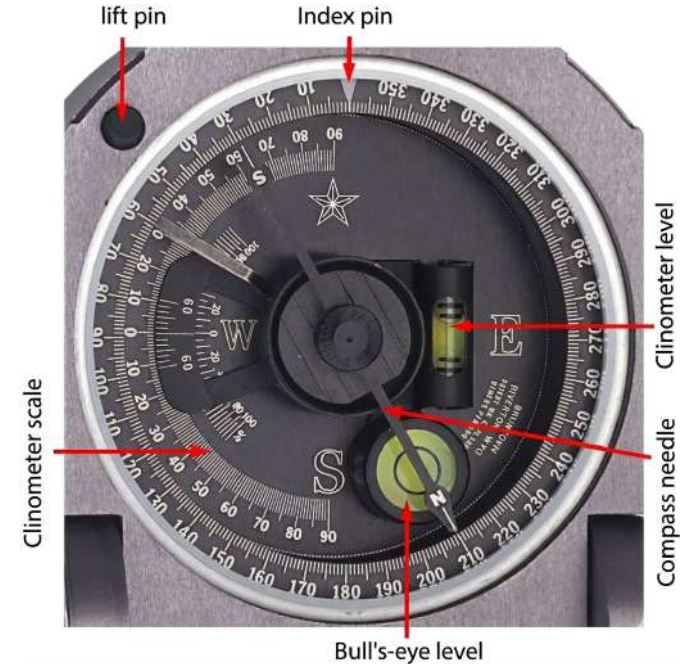
2. Measuring Vertical Features (Dip & Plunge)

For Dip

- **Orientation:** place the side edge of the Brunton
- **The Clinometer:** move the Vernier Arm slowly
- **Leveling:** Watch the Long Level.
- **Read the Angle:** Semicircular Scale

For Plunge (The angle of a line/pole)

- **Orientation:** align the **base** of the Brunton parallel to the linear feature
- **Execution:** use the **Vernier Arm** to center the bubble in the **Long Level**
- read the vertical angle from the semicircular scale.



2. Location and Bearing

Instrumentation & Tools

1. *Toposheet*
2. *Tracing paper*
3. *Brunton Compass*

What is a Topo Map?

A **topographic map** (or "topo map") is a detailed, accurate graphic representation of features on the Earth's surface. Use of **contour lines** to show elevation and terrain shape.



Contour map before plotting data

We took help of the help of brunton Compass for this.

2. selected two elevated natural or man made objects
3. measured the back-bearing of the two locations
4. fixed our position according to the true north
5. traced back the two back-bearings measured from the two selected objects on the topo map.
6. The intersection of the two back-bearing is our location on the topo map.

Points to be Noted:

1. make sure The North direction on the map aligns perfectly with the true north.
2. magnetic correction or the Declination correction to the measured values.
3. For our location the declination is very small(0.8-1).So this can be neglected.

We have measured our location with respect to the the Temple on the hill and the small hillock.

Master Location

In structural geology , a Master Location is a highly precise reference point used to determine the relative positions of other geological features or Outcrops.

We extended our knowledge of locating ourselves to locate the Master Location on the map.

The Northern and the Southern limb groups have located their Master locations and gradually moved towards the hinge zone measuring the Structural features w.r.t the master locations.

Now we Identify the Outcrops And carry Out Measurements With respect to Our Master Location.

3. Outcrop Identification & Lithology

- See if the rock is in-situ.
How do we know that ?
 - See if a rock is matching the general orientation of other rocks in the area.
 - Spatial Extent and Boundaries
- Look for the Lithology of the rock

-
- *buff-coloured*
 - *quartz-rich (quartzite)*
 - *competent*



4. Structure Identification

Bedding (S_0)

Variation in colour, texture, grain size



4. Structure Identification

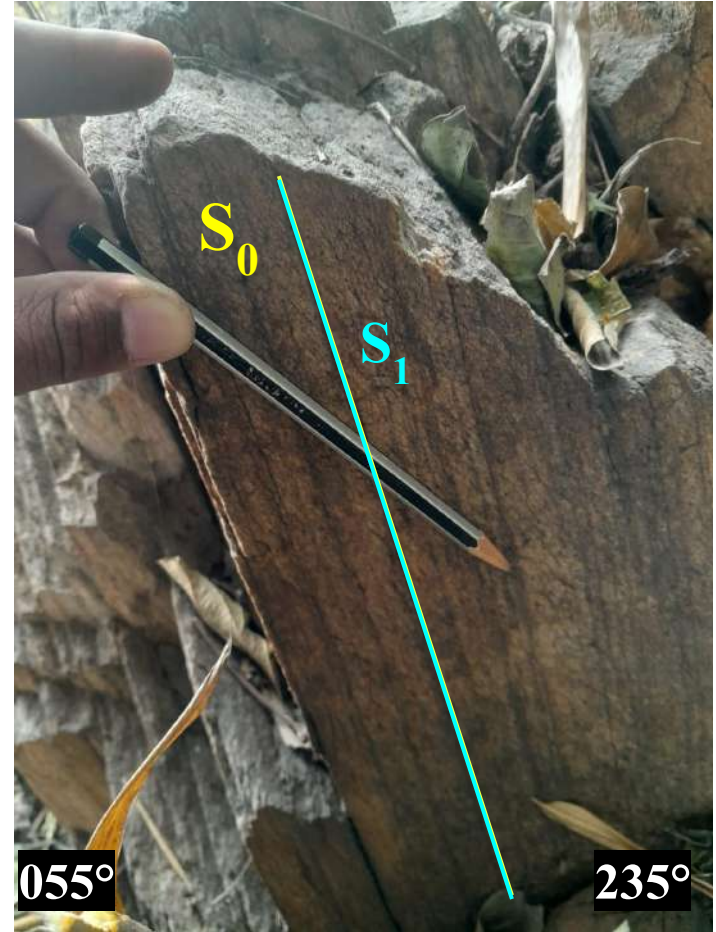
Bedding (S_0)

Primary Structure

Cleavage (S_1)

Secondary structure

Planes of weakness along which the rock can easily split.

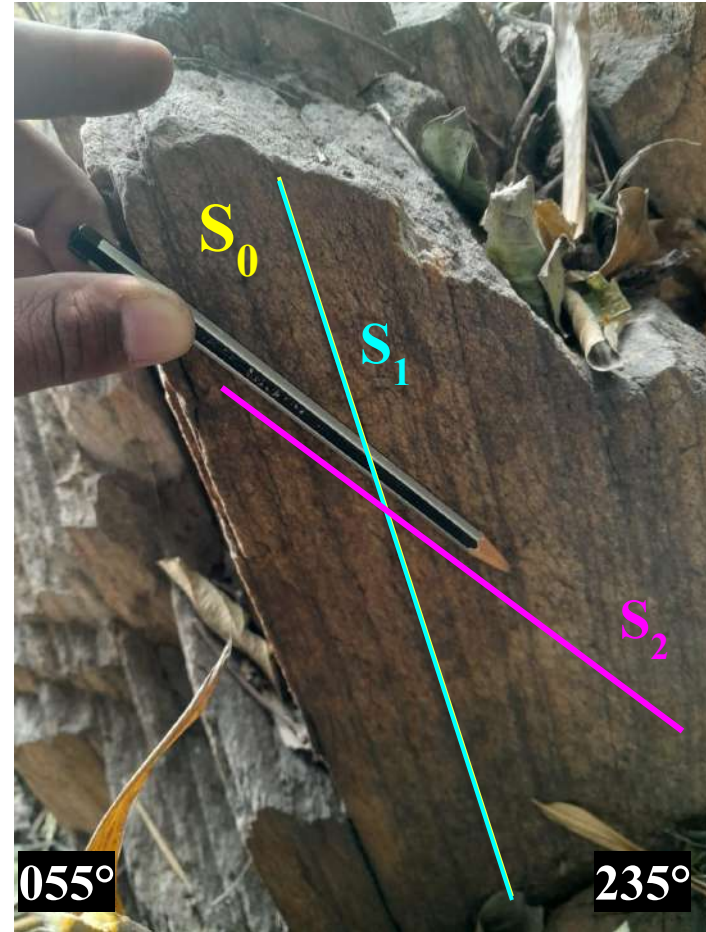


4. Structure Identification

Bedding (S_0)

Cleavage (S_1)

Cleavage (S_2)

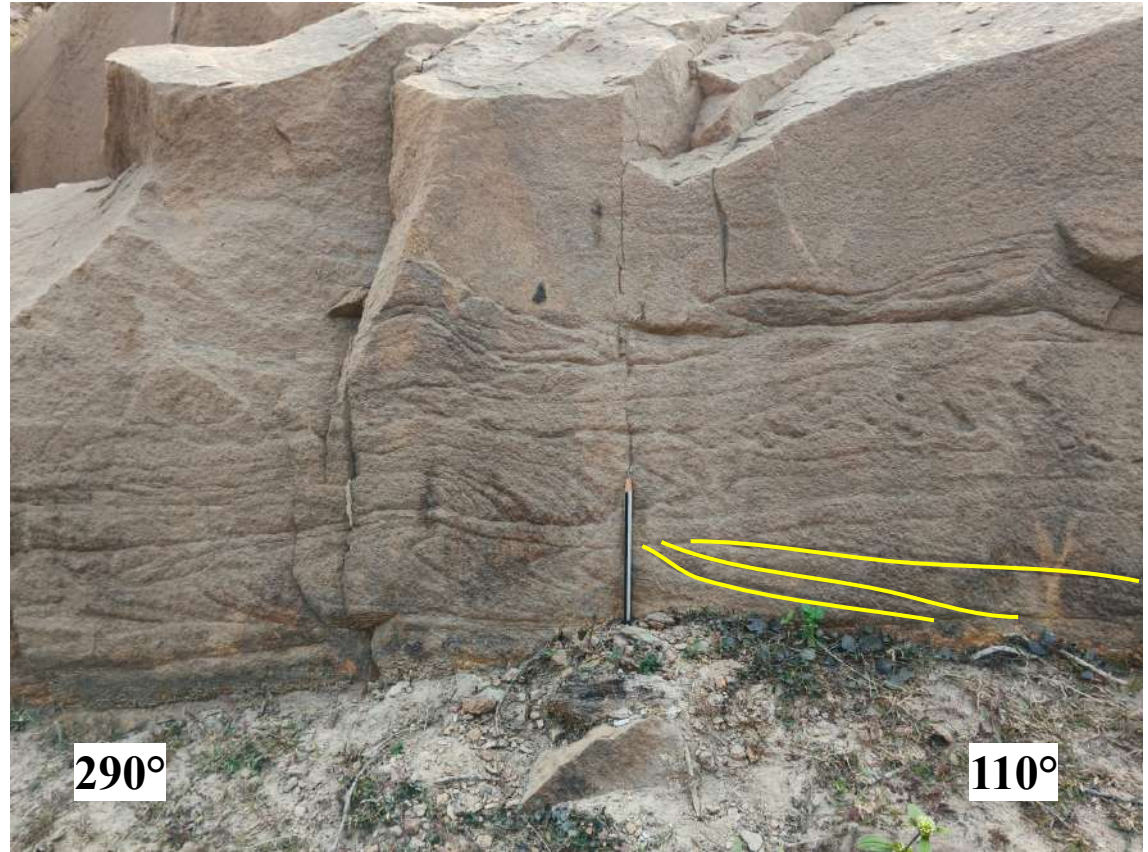


Primary Structures

Foreset

Gives us the Younging Direction

Can be later used to reconstruct paleoflow

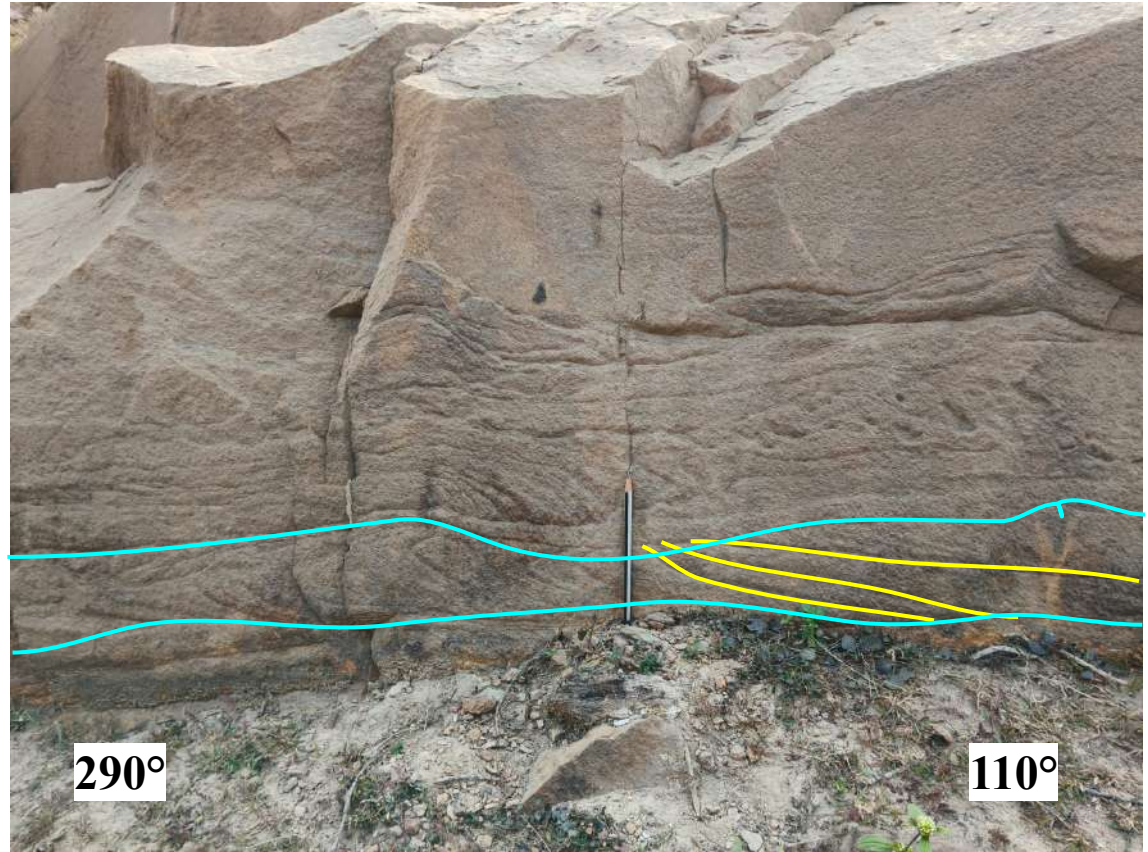


Primary Structures

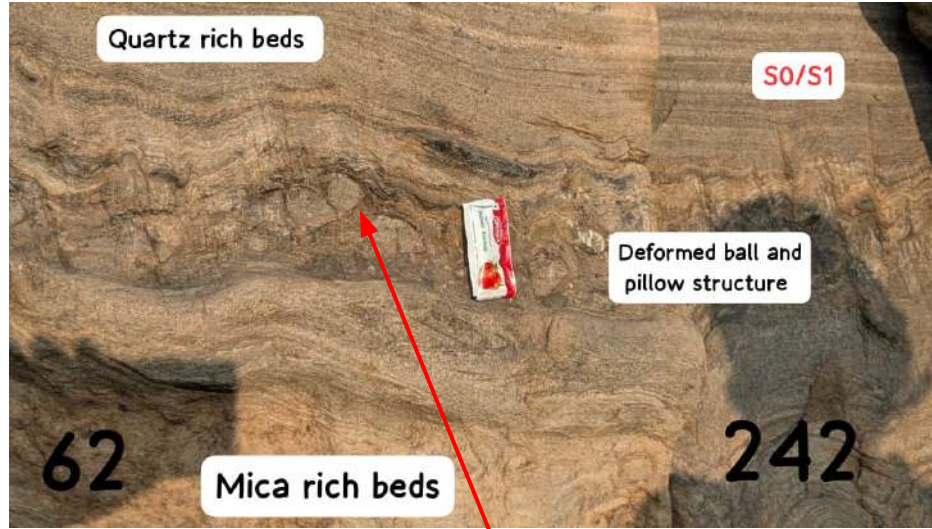
Foreset

Gives us the Younging Direction

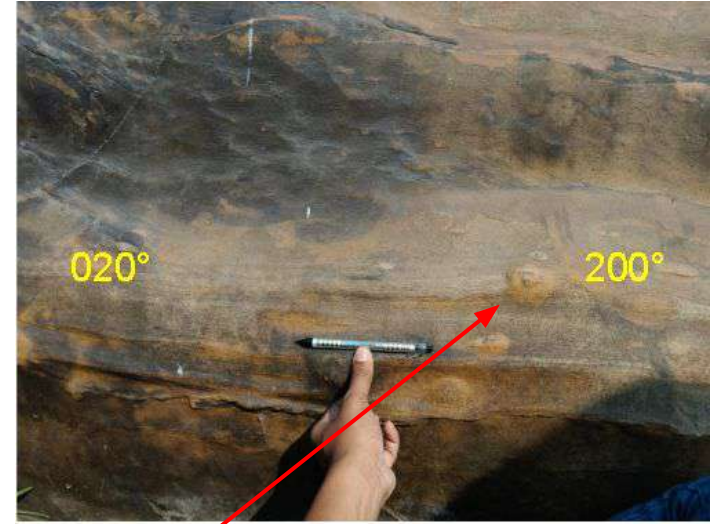
Can be later used to reconstruct paleoflow



Primary Structures



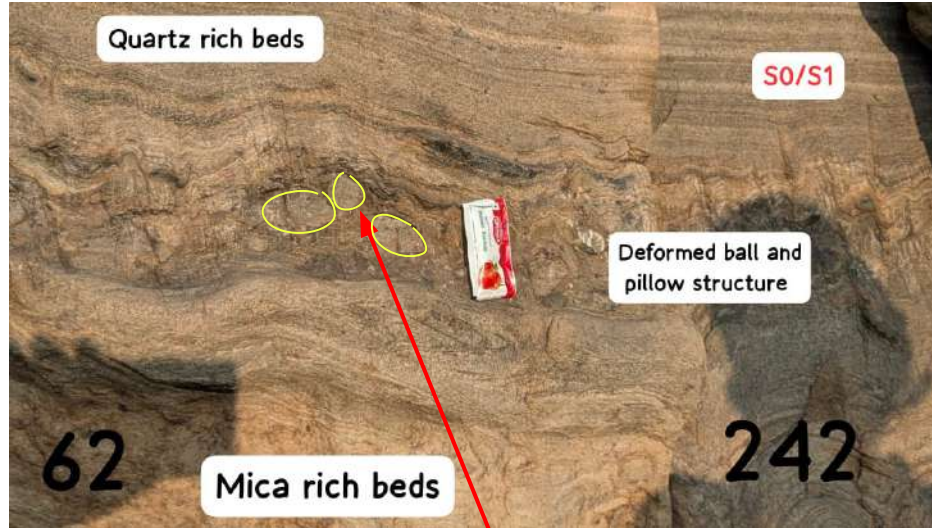
Ball and Pillow Structures



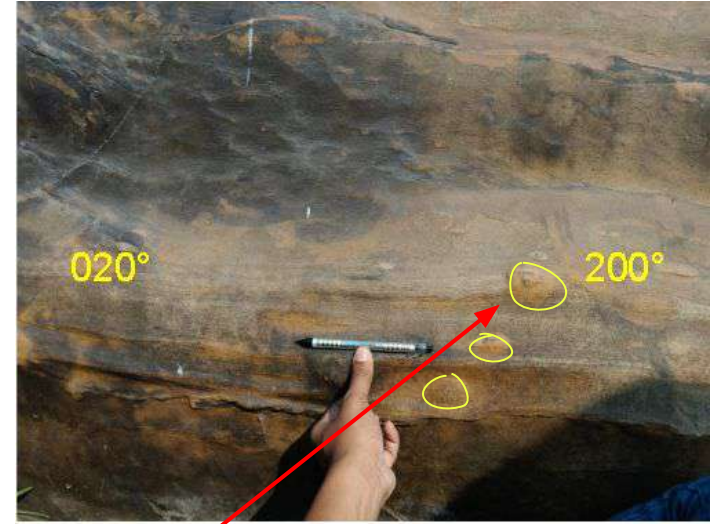
Bulbose Structure

detached masses of a denser material sinking into a softer matrix.

Primary Structures



Ball and Pillow Structures



Bulbose Structure

Fractures

Mode 1: Extension Fracture



Plumose Structure (Hackles)



Joints



Vein

Secondary Structures

Fractures

Mode 2: Shear Fracture

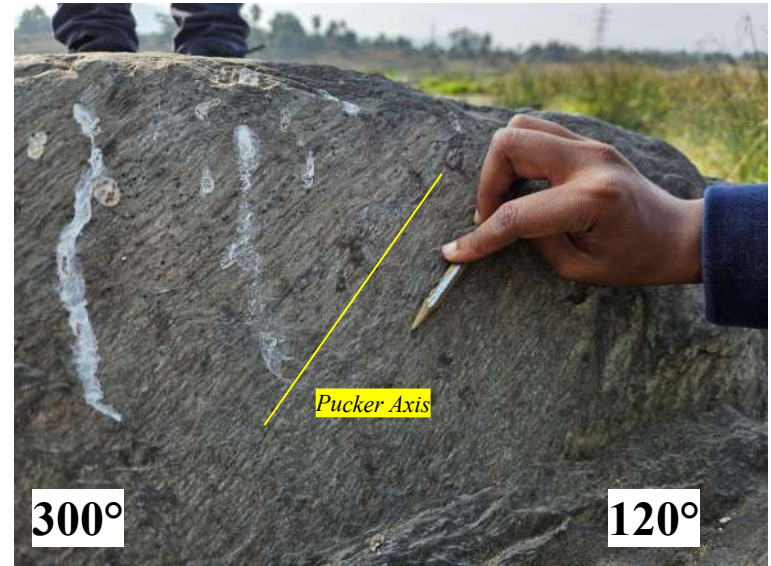
Identifiable with
Slickenlines.

Secondary Structures

Mineral Lineation

Pucker Axis :

Hinge line of a crenulation fold.
Indicates multistage deformation.



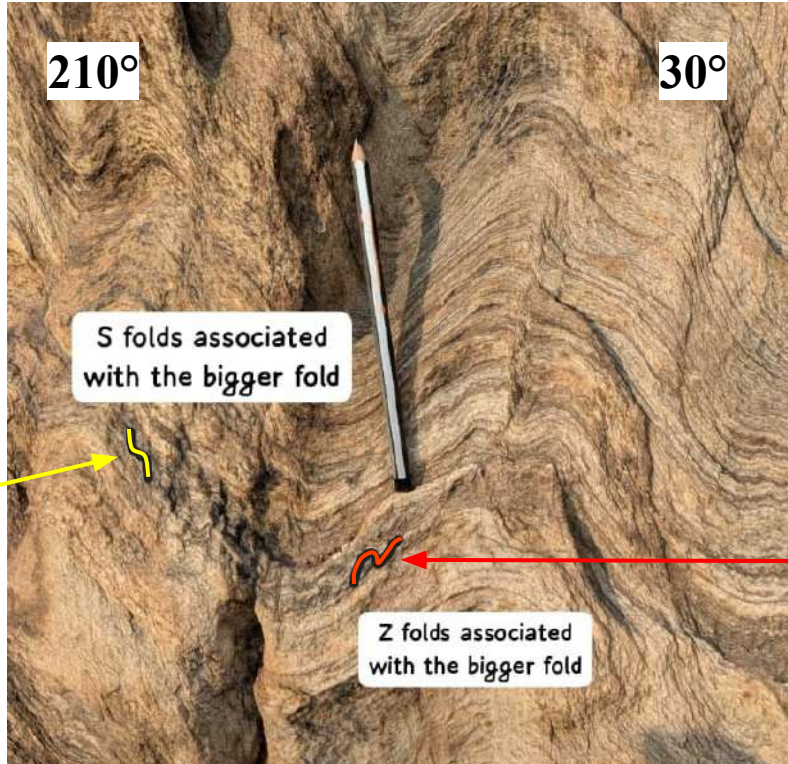
Folds

S and Z fold

Minor folds on limbs of macroscopic fold.

Shapes indicate the sense of tectonic shear and point towards the hinge.

S-fold



Z-fold

Isoclinal Folds

Fold having very low interlimb angle



Isoclinal Folds



Superposed Folds

Dome and Basin:

Type I Interference
Structure

Both Hinge lines and axial
planes are at high angle to
each other

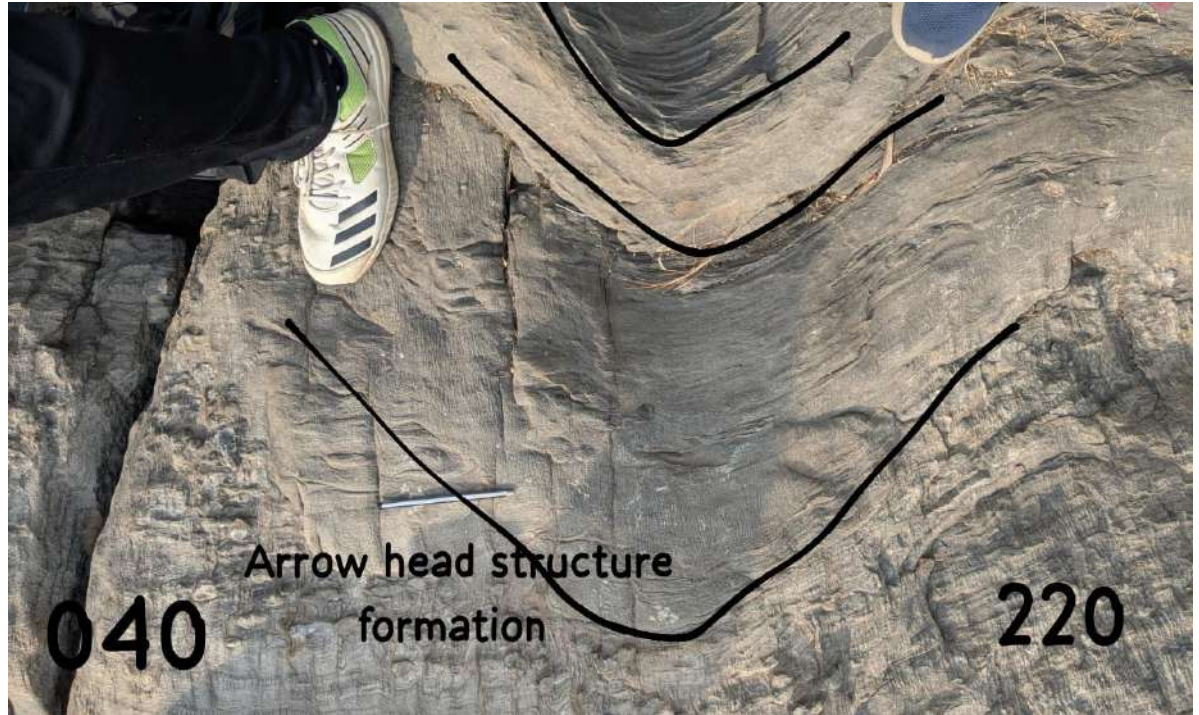


Superposed Folds

Arrow head :

Type II Interference
Structure

When hinge lines are at
high angle and axial
planes are at high angle

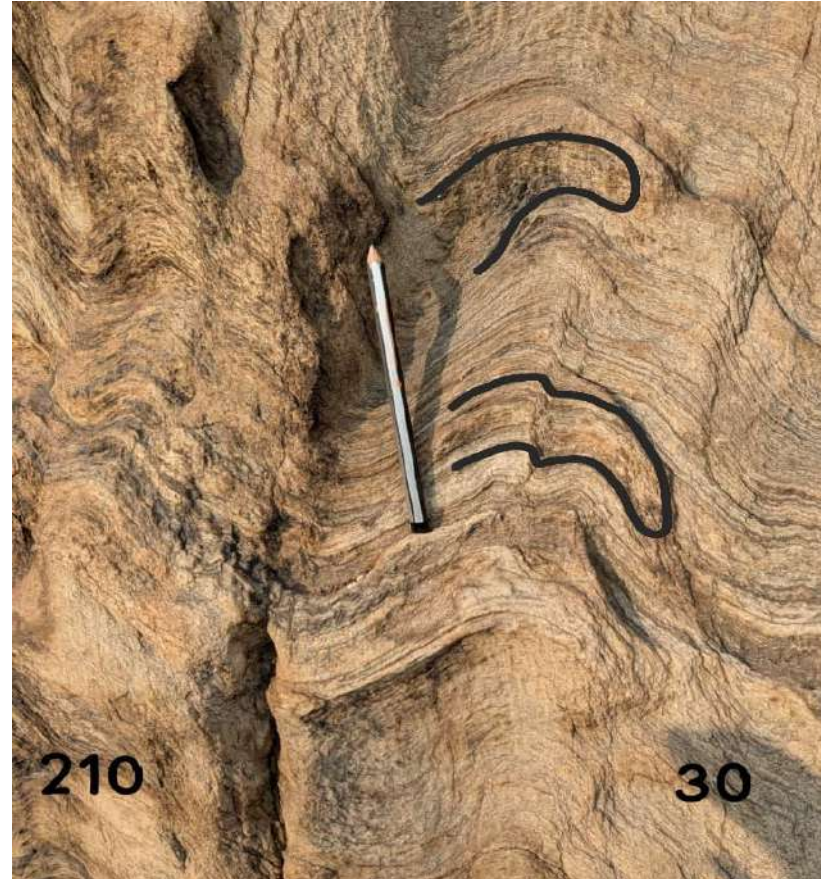


Superposed Folds

Hook structure :

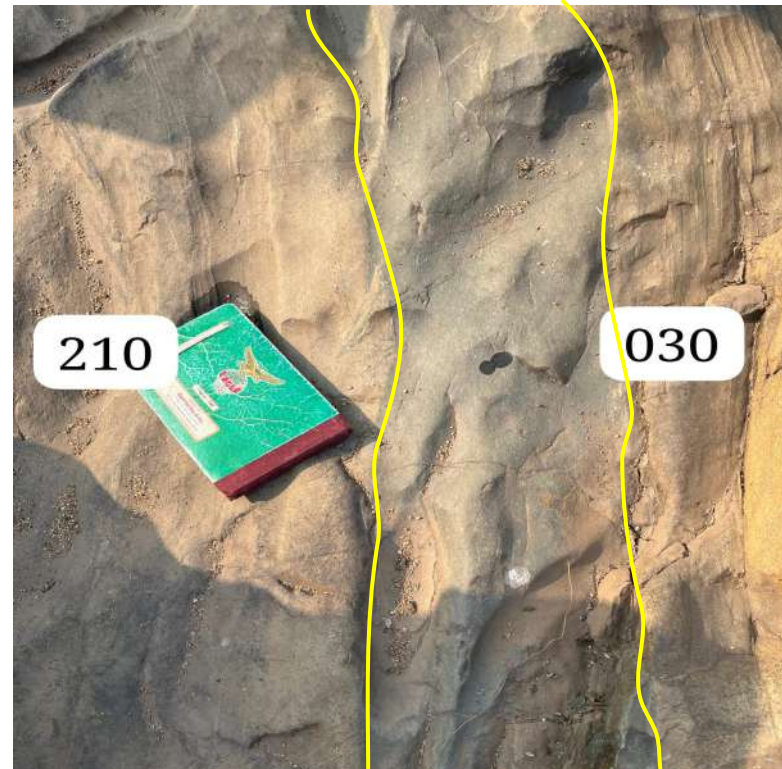
Type 3 Interference
Structure

Hinge lines are at low angle to
each other and Axial Planes are
at a higher angle to each other



Igneous Structure

Dyke



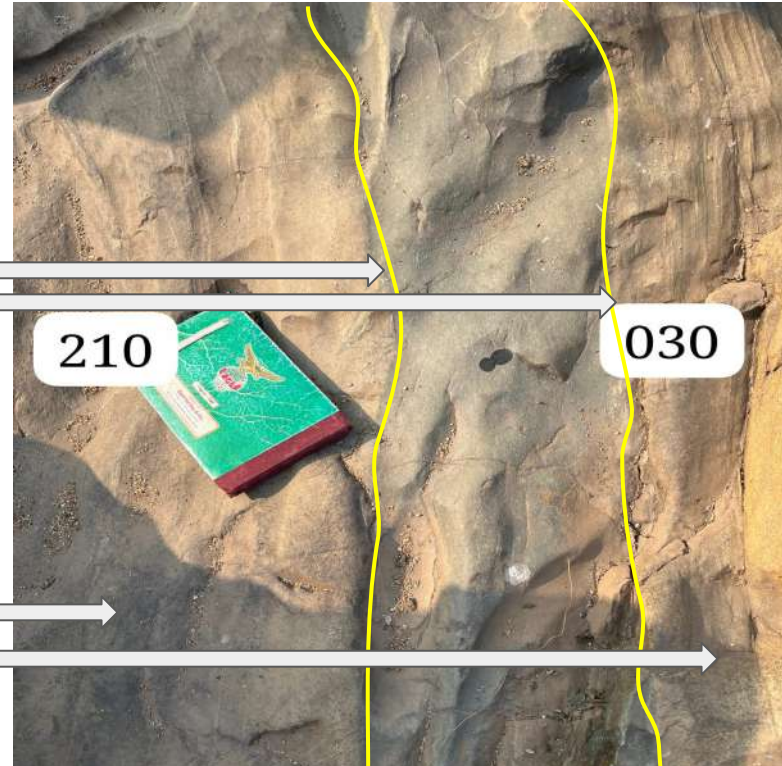
Dyke

Igneous Structure

Dyke

Distinguished by
chilled boundary

Country rock



Dyke

5. Data Collection

- Location
- Distance along limb
- Azimuth along limb
- S_0/S_1 , S_2 , S_3
- Outcrop width
- Width Azimuth
- Fracture set ($f_1 - f_7$)

Location	Dist along limb (cm)	azimuth along limb	S_0/S_1	S_2	S_3	OUTCROP WIDTH (cm)	WIDTH AZIMUTH
L1(0)	13.57	047°	84°/043° 80°/040°	84°/211°	36°/250°	17.15	153°
			84°/050°				

f_1	f_2	f_3	f_4	f_5	f_6	f_7
65/121	75/335 75/316	60/225	36°/143° 30°/140°	30°/029°	17°/240°	
	78/341					

Inner and Outer arc



Lateral distance measurement

- Walked along the strike of the bed, stopping at approximately 10-meter intervals. *Why ?*
- Used a tape to measure the distance.
- Used a compass to record the exact direction (azimuth) of the tape.



Measuring Outcrop Thickness

- At every ~ 10 -meter stop, we went along dip direction ($\perp$ to Strike)
- We did the width and azimuth measurement to the inner and outer edge (outcrop) of the fold.



Outcrop Width



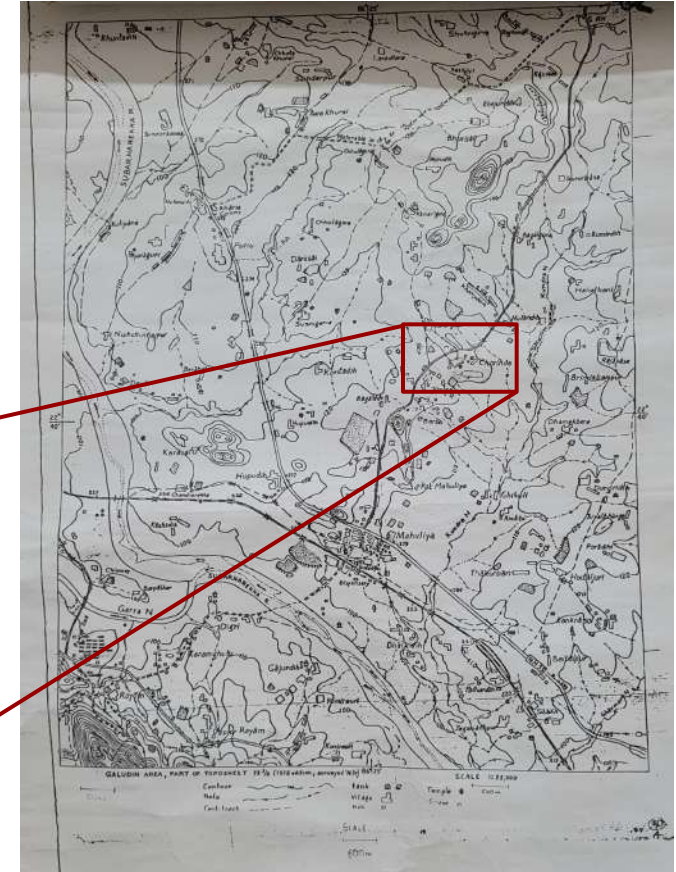
**The measuring tape needs to be straight and tight while measuring*

6. Mapping

- Find north using latitude and longitude marked on map and brunton in field.
- Locate master location using back bearing method.
- At each outcrop we measured outcrop width, bearing, S_0 (bedding attitudes), cleavage data, fracture data
- Scaled down distance according to map scale (**1:2500**) and plotted the outcrop locations.



Topo Map, Chorinda area (1:2,500 scale)

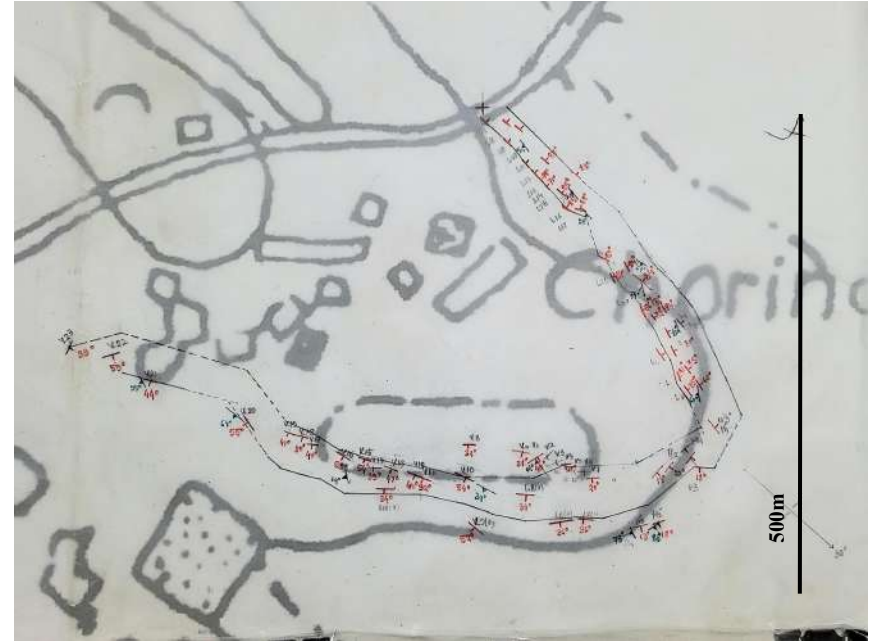
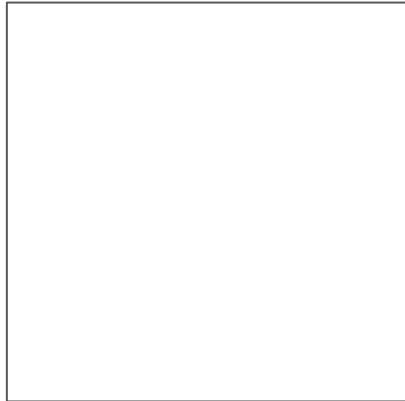


Topo Map, Galudih area (1:25,000 scale)

Fold Mapping

- Strike and dip of bedding and cleavage were marked for each location
- Fold was marked along strike direction, extrapolated according to strike angle.
- Fold width was marked along dip to give us inner and outer arc
- Areas without outcrops are connected using dotted lines

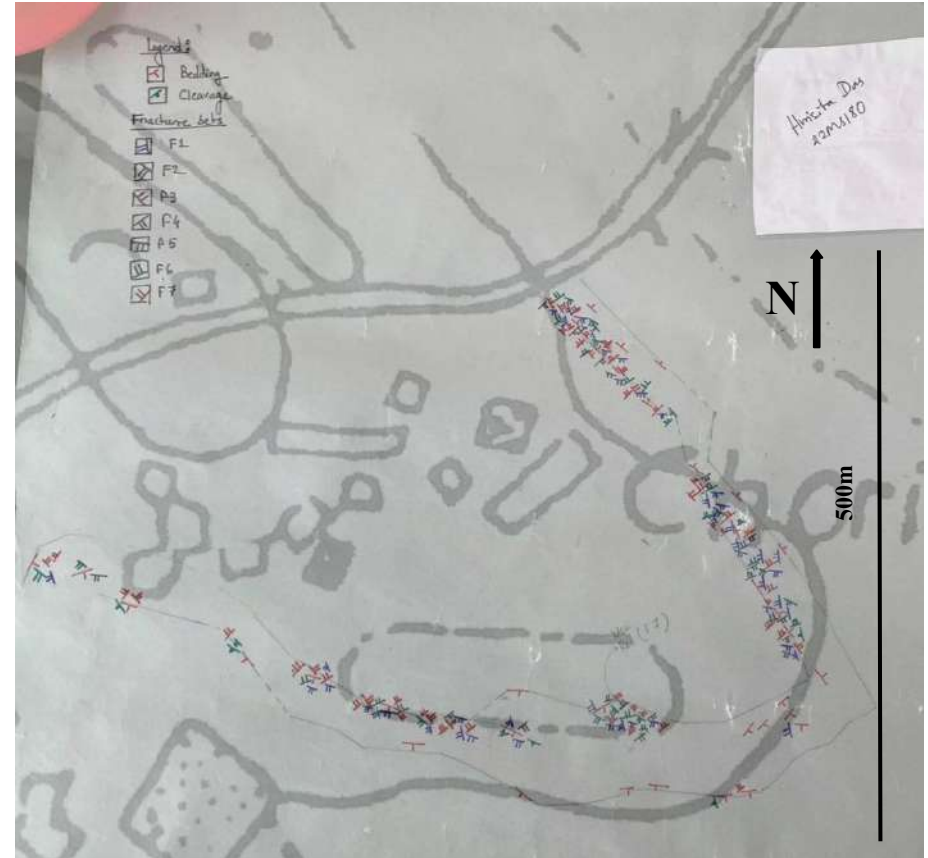
*Strike and dip
notation*



Fold mapped on topo sheet

Plotting the Fractures on Topo Map

- Fractures were sorted into 7 sets after Fracture characterization during lab analysis
- Fracture data was also plotted for each outcrop on the topo sheet



Fracture mapped on the topo sheet

Hinge Line Determination

- At the fold hinge, the lineation (L_1) is parallel to the fold axis.

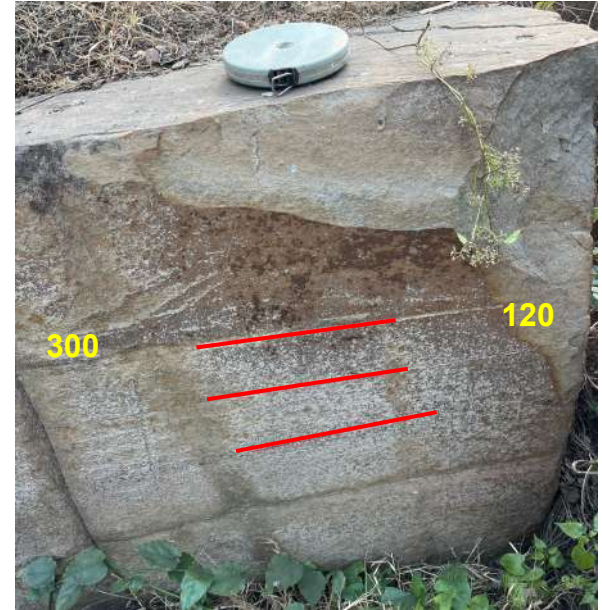
For example, in the River Section region, we located the fold axis of the isoclinal fold; stripping lineations parallel to it gave an idea about the hinge direction.

- Using the cleavage bedding angle (intersection between S_0 and S_1), the angle increases near the hinge zone. Dip changes in this region.
- Looking for M folds helps us to identify the hinge zone.



Striping lineations

- Formed by the intersection of the bedding plane (S_0) and axial planar cleavage (S_1).
- To identify the stripping lineations, we looked for parallel contrasting stripes (w.r.t. lithological layers) on the cleavage planes.
- Measured the plunge and trend using a Brunton compass directly on the foliation surface.



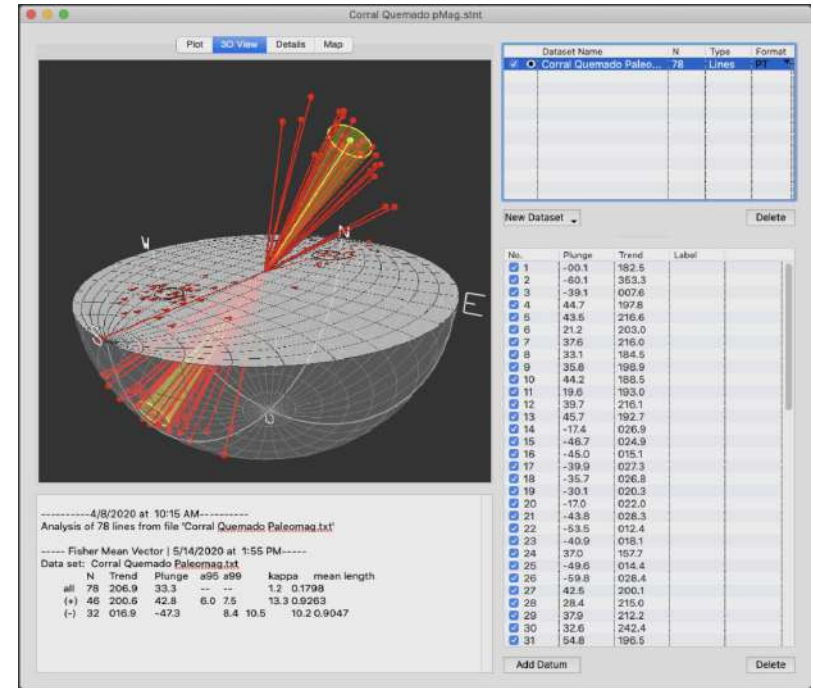
Striping lineations marked with red lines

Methodology

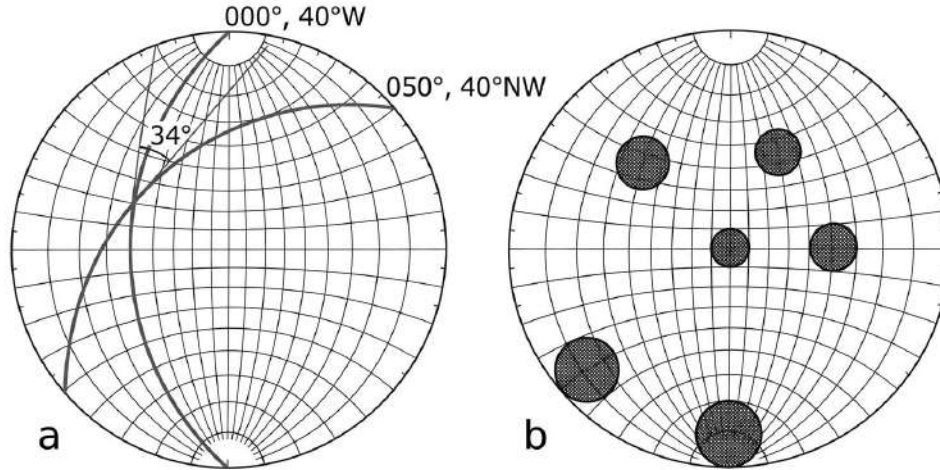
Lab Analysis

Stereographic Projections:

- 3D orientations $\rightarrow$ 2D representation
- Visualizes raw field measurements
- Reveals patterns and trends



Wulff Net: Equal Angle Projection



Dihedral angle between two planes = 34°

Figure 8-1. Properties of a stereographic or equal-angle projection. (a) The angle between two planes is the same as the angle between the tangents to the great-circle traces of the two planes; (b) identical circles on projection sphere project as circles of different sizes

- Angles are preserved
- Same circles $\rightarrow$ different sizes
- Area is distorted

Schmidt Net: Equal Area Projection

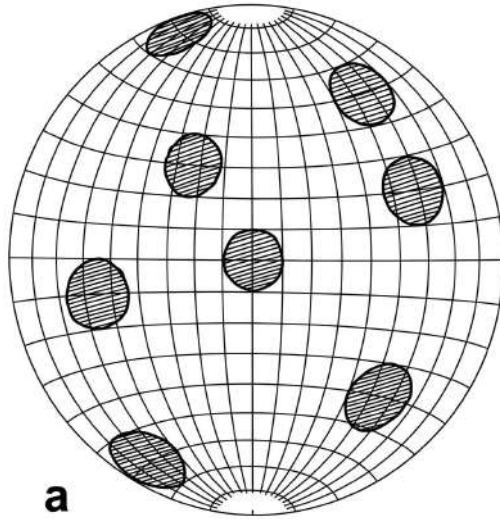


Figure 8-3. Properties of an equal-area projection. (a) Identical circles on projection sphere project as ellipses with various axial ratios but having the same area

- **Equal areas are preserved**
- **Shapes may change (circles → ellipses)**
- **Distribution is not distorted**

Fold Geometry Analysis

1.

Fold Hinge Line Orientation

Estimation of the orientation of the fold hinge using gathered bedding data through structural analysis.

2.

Interlimb Angle

Measurement and calculation of the angle between the two limbs of a fold to classify the fold type and tightness.

1. Estimation of Fold Hinge Orientation

Key techniques for estimating the hinge line using bedding data:

(a)

π - Plot

Using pairs of planes to find lines of intersection on a stereonet to define the fold axis.

(b)

Bedding-Cleavage Relationship

Analyzing geometric relationships between primary bedding and secondary cleavage planes.

(c)

Intersection of Limbs

Intersection of limbs is the line of intersection parallel to the fold hinge line.

(a) π - Plot Analysis

equal-area plot

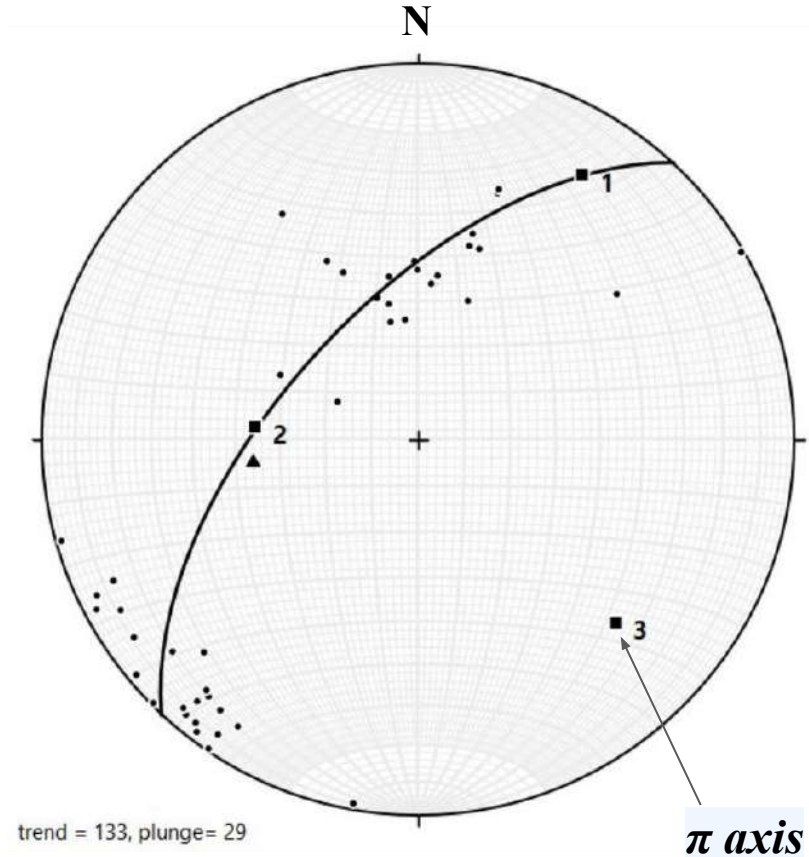
Methodology

1. Plot the Bedding (S_0) and their Poles.
2. Identify the π -Circle (The Girdle)
3. Find the π -axis

π axis = Fold Axis

Assumptions:

- S_1 is axial planar to the fold
- Cylindrical fold
- Single folding event



(b) Bedding - Cleavage Relationship

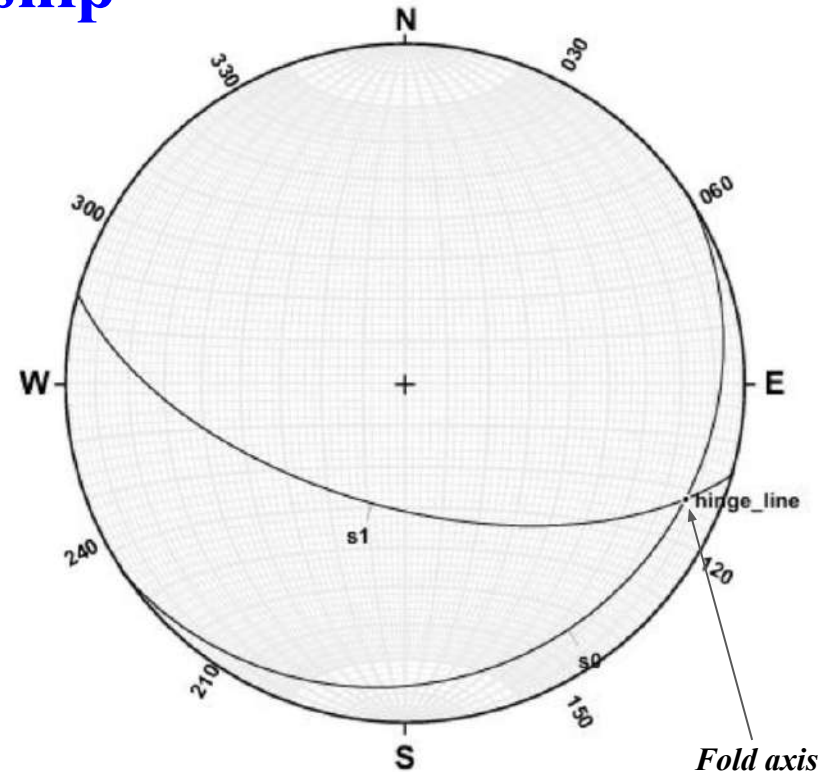
Methodology

1. Plot strike & dip of bedding (S_0) on the great circle
2. Plot strike & dip of cleavage (S_1) on the great circle
3. Intersection point defines a line

Intersection line = Fold axis

Assumptions:

- Axial Planar Cleavage
- Cylindrical Fold
- Single deformation phase

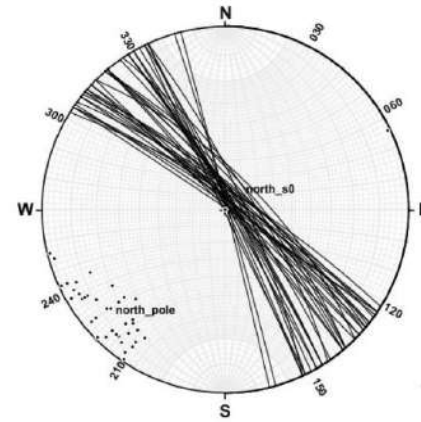


Bedding - Cleavage Stereographic Projection

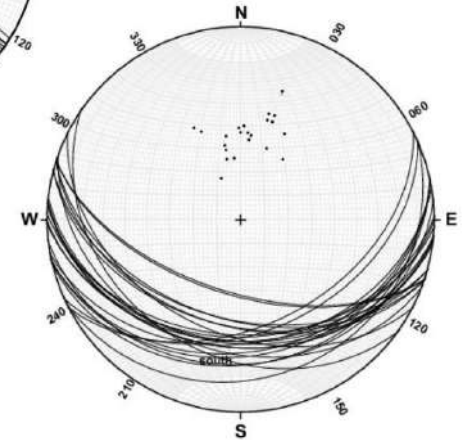
(c) Intersection of Limbs

Methodology

1. Plot the bedding data of both the limbs separately.
2. Plot the poles to the planes.



Northern Limb

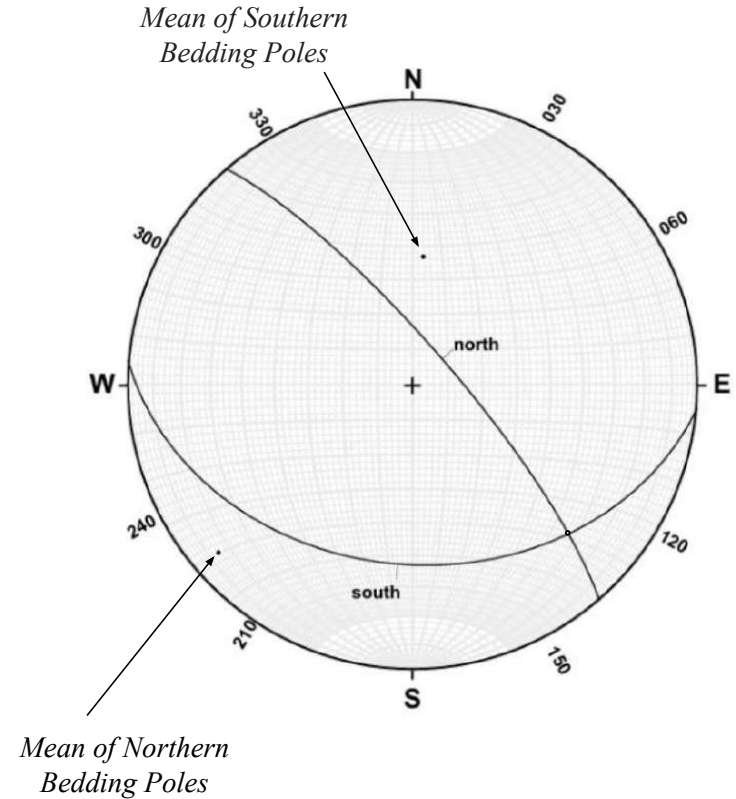


Southern Limb

(c) Intersection of Limbs

Methodology

1. Plot the bedding data of both the limbs separately.
2. Plot the poles to the planes.
3. Find the Mean Pole of the two limbs separately.
4. Then plot the plane to the mean pole.

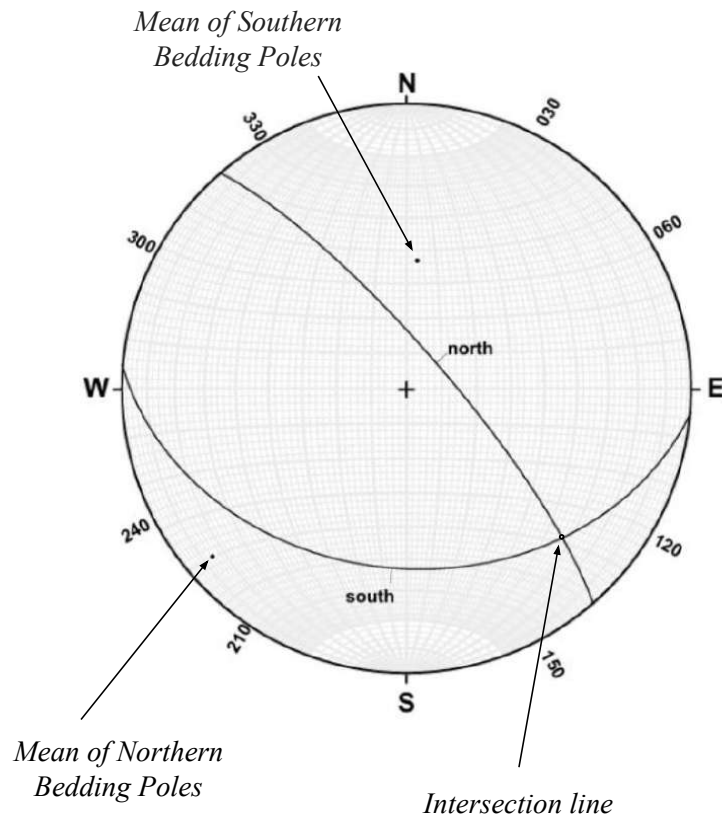


(c) Intersection of Limbs

Methodology

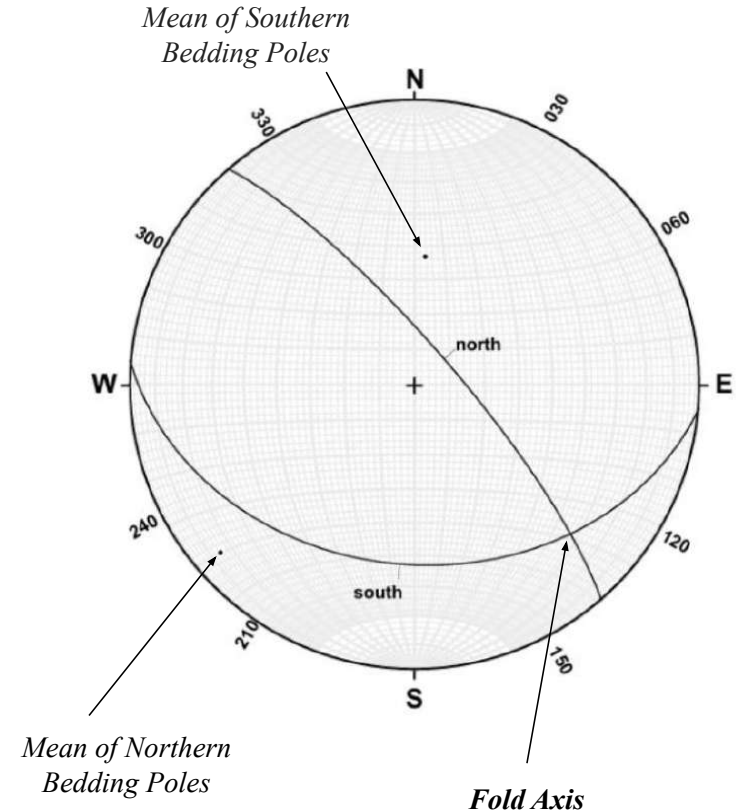
1. Plot the bedding data of both the limbs separately.
2. Plot the poles to the planes.
3. Find the Mean Pole of the two limbs separately.
4. Then plot the plane to the mean pole.

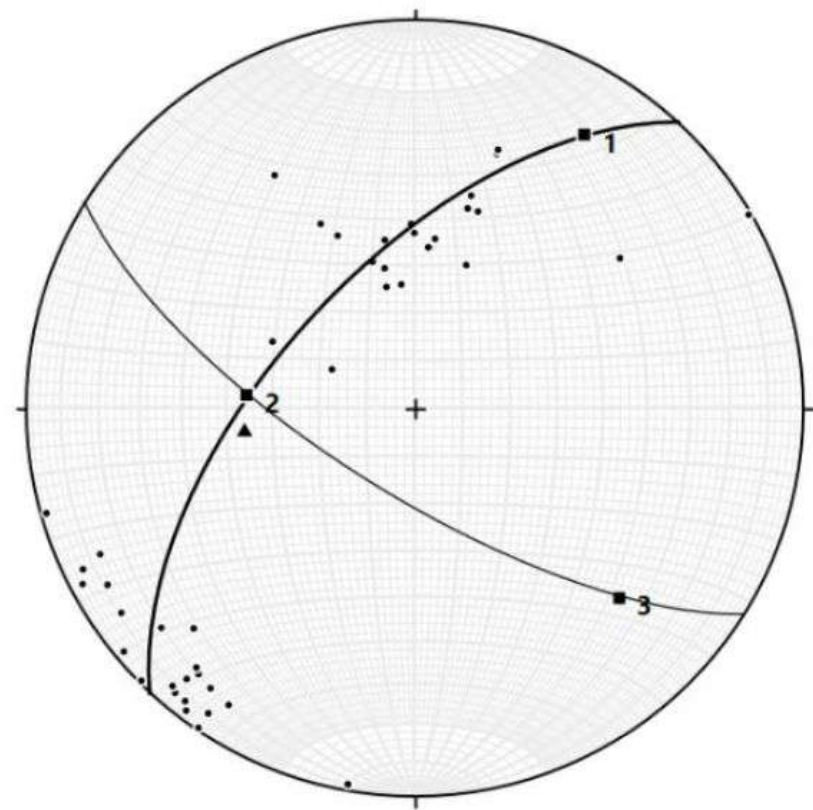
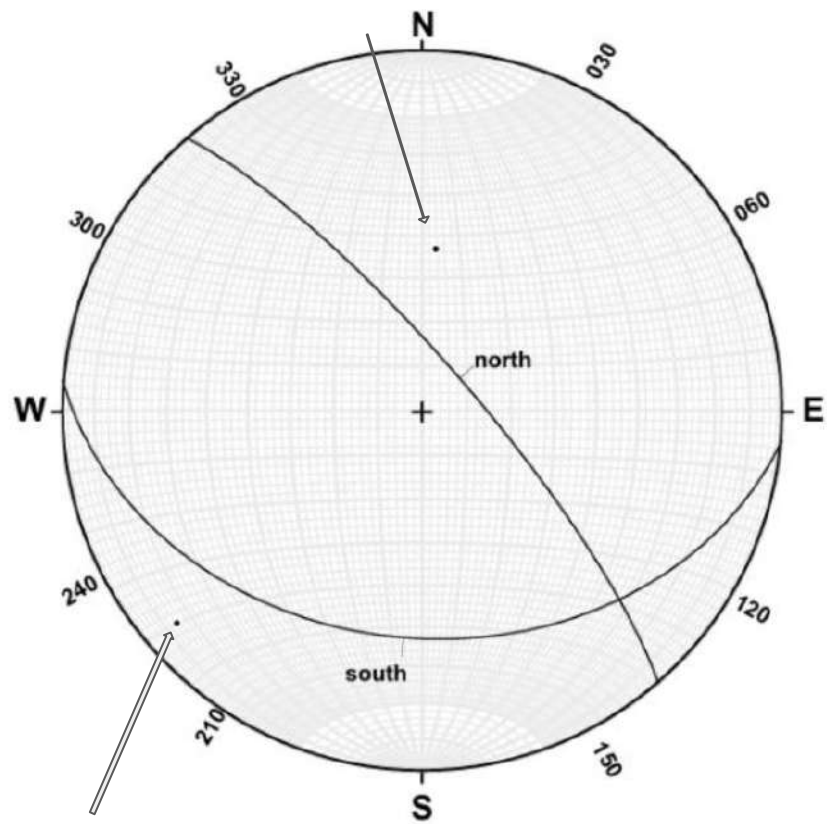
Intersection line = fold axis



2. Estimation of Interlimb angle

The angle between the two planes (Mean Northern Limb Bedding and Mean Southern Limb Bedding) give us the Interlimb Angle.



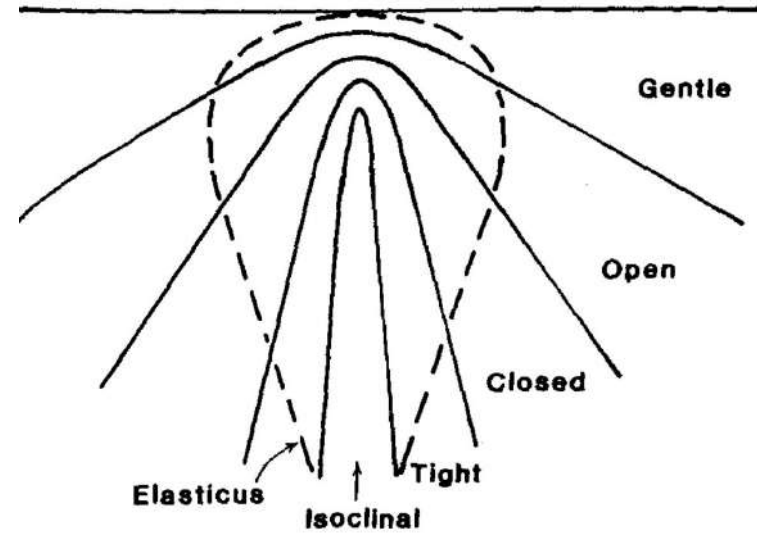


Hinge Surface
(passing through π -axis and bisecting interlimb angle)

Tightness of Fold

Tightness refers to the measure of the angle between the limbs (interlimb angle, θ) of the fold.

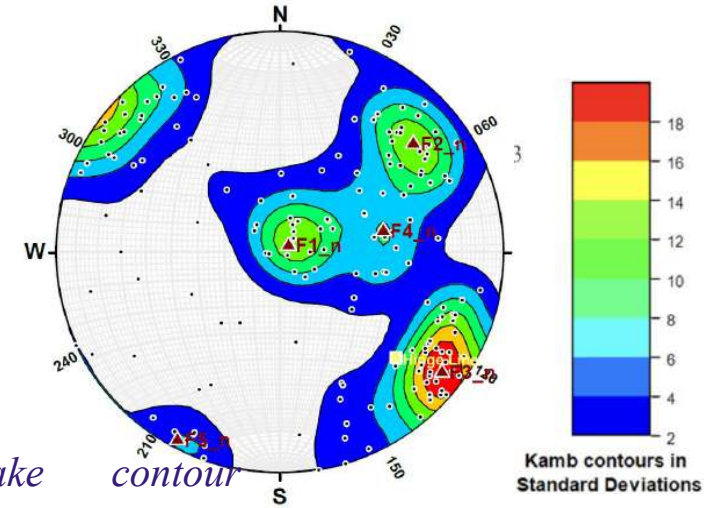
Interlimb angle (°)	Adjective
120 - 180	Gentle
70 - 120	Open
30 - 70	Close
10 - 30	Tight
0 - 10	Isoclinal
< 0	Elasticas (<i>Obtuse</i>)



(After Fleuty, 1964)

Plotting the Fracture on Stereonet

- Plot all the fractures on equal area net (Schmidt Net)
- Plot the poles to the fracture planes.
- Plot the fracture density contour using the stereonet software.
- Clustered plotted points indicate dominant fracture orientation present.



The software uses the Kamb method to make contour around regions with similar density of fractures.

Contour interval separates zones of different densities of fracture orientation.

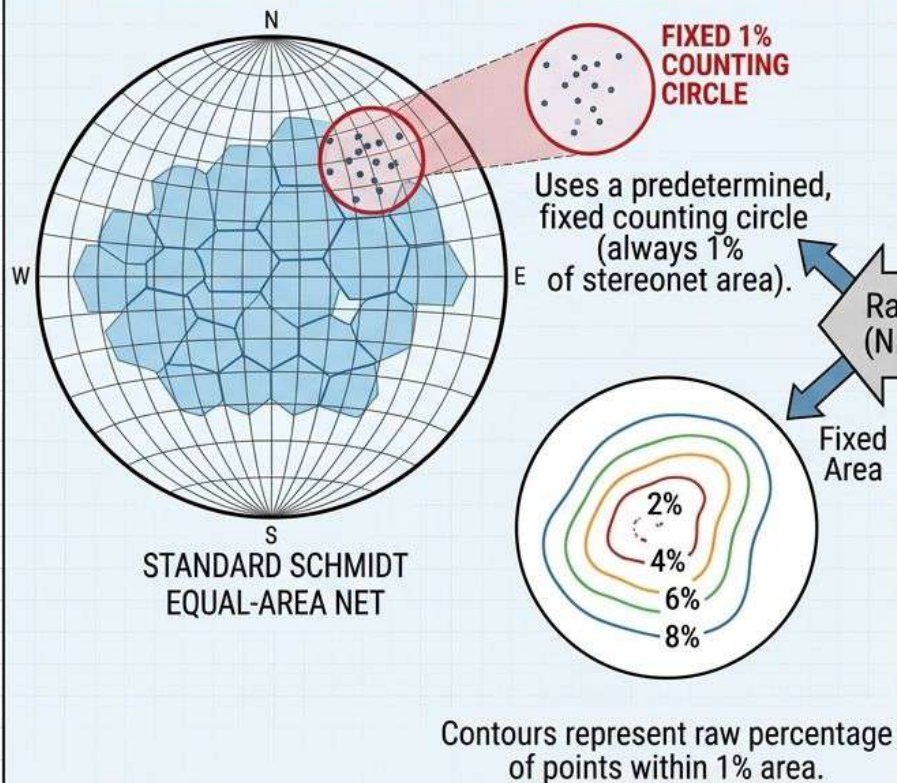
NOTE :

- Manual Kalsbeek net : shows the Raw percentage of fractures within the contour,
- Kamb method : statistically calculates and compares the actual fracture density in region w.r.t.. what data would look like if it were scattered randomly.

COMPARISON OF STEREONET CONTOURING METHODS: KALSBECK VS. KAMB

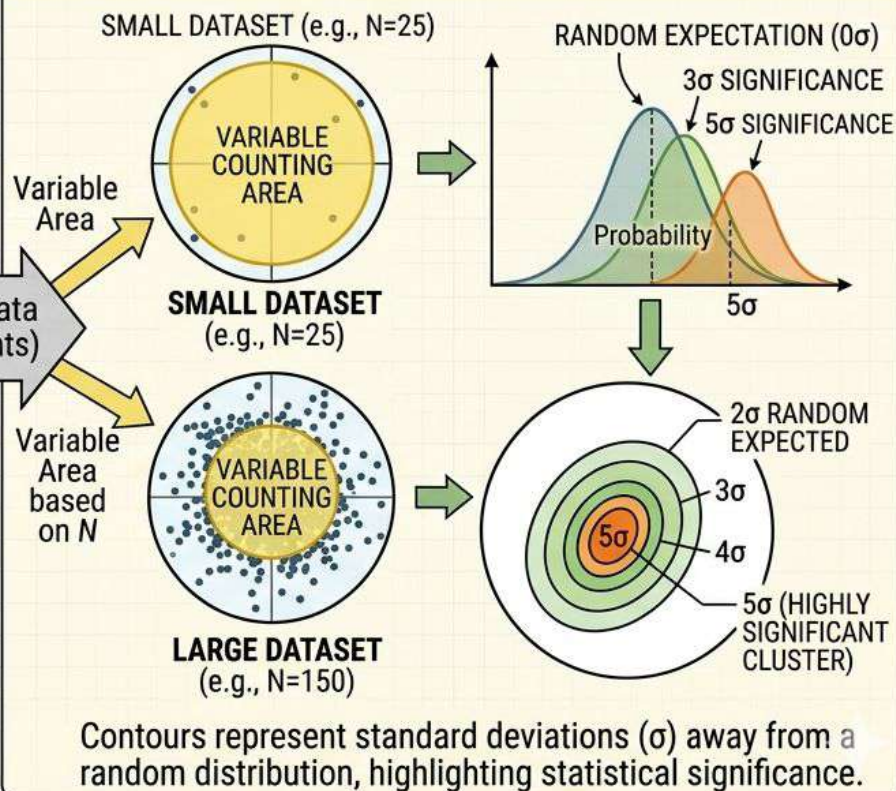
KALSBECK COUNTING NET METHOD

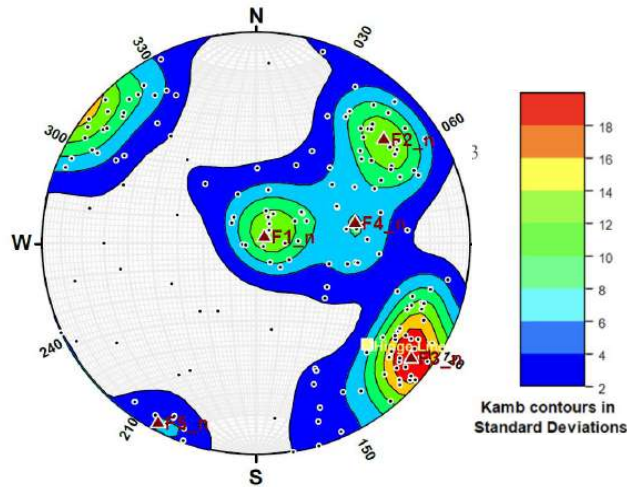
(Fixed Area/Raw Percentage).



KAMB METHOD

(Variable Area/Statistical Significance)





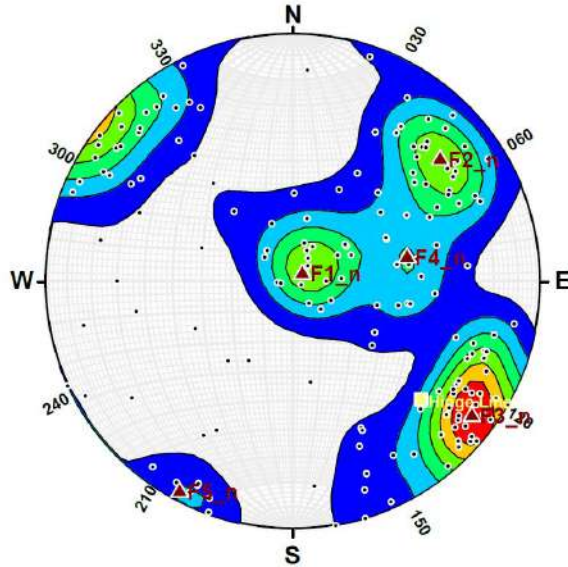
Kamb method helps to filter out random noise in data by dropping data points which are isolated from the clusters

From the density plot we can identify regions with maxima of density and several such fracture clusters were identified.

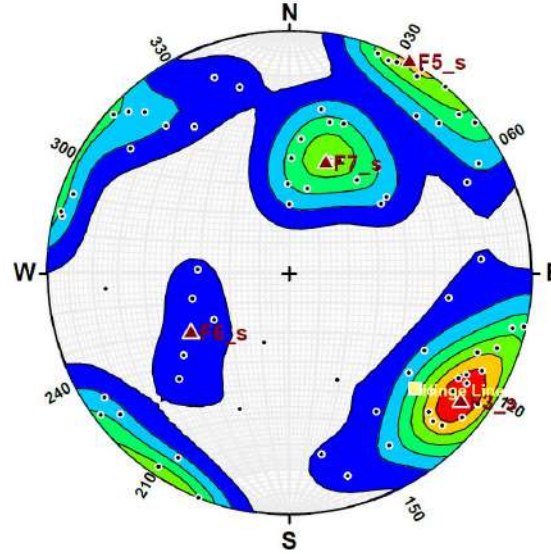
Mean of cluster gives mean attitudes.

Fracture Characterization

Methodology : Lab Analysis



Northern Limb Fracture
Density Contour



Southern Limb Fracture
Density Contour

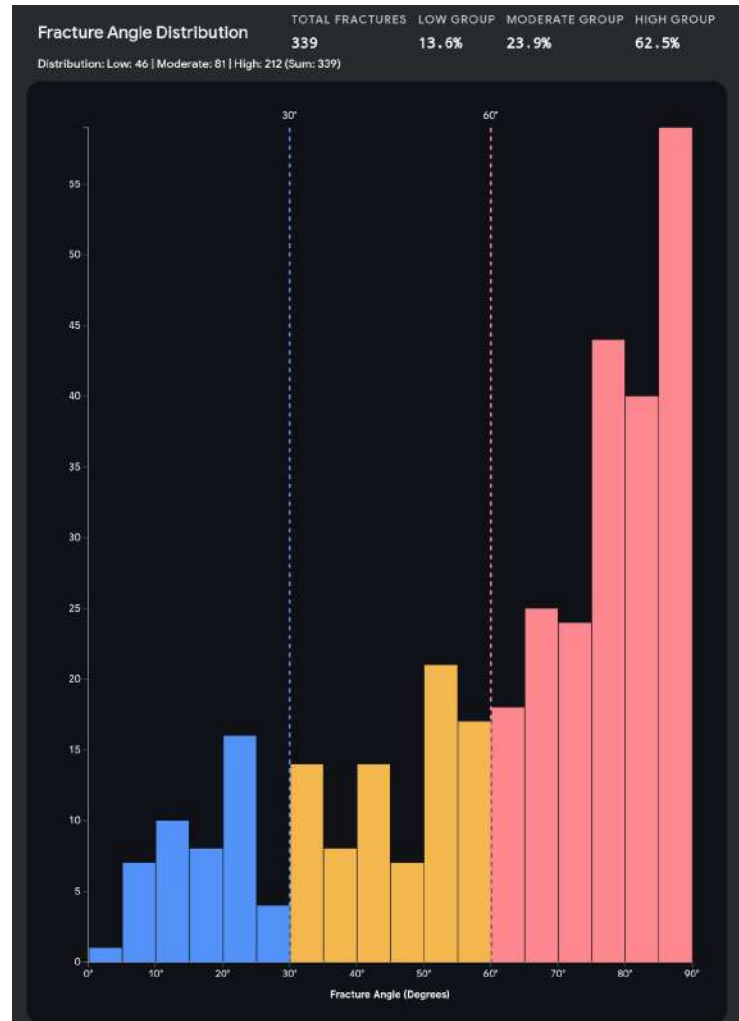
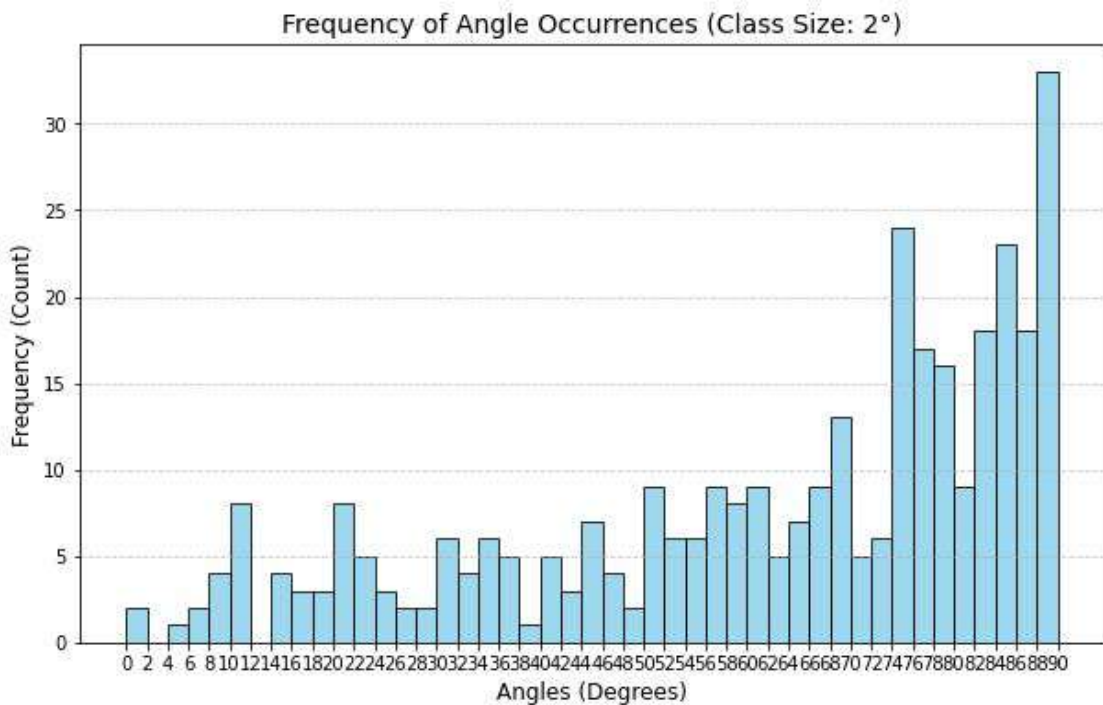
<i>Fracture Set</i>	<i>Present in</i>	<i>n</i>	<i>Mean Attitude</i>
F1	North	25	149/07
F2	North	29	141/66
F3	Both	125	217/80
F4	North	13	169/39
F5	Both	42	296/86
F6	South	14	328/39
F7	South	33	109/39

From the density plot we can identify regions with maxima of density and several such fracture clusters were identified. Mean of cluster gives mean attitudes.

Fracture classification

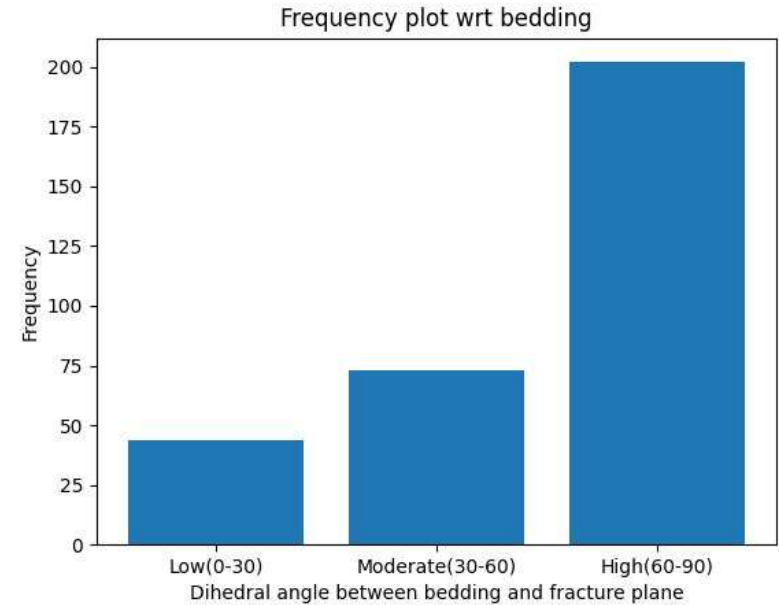
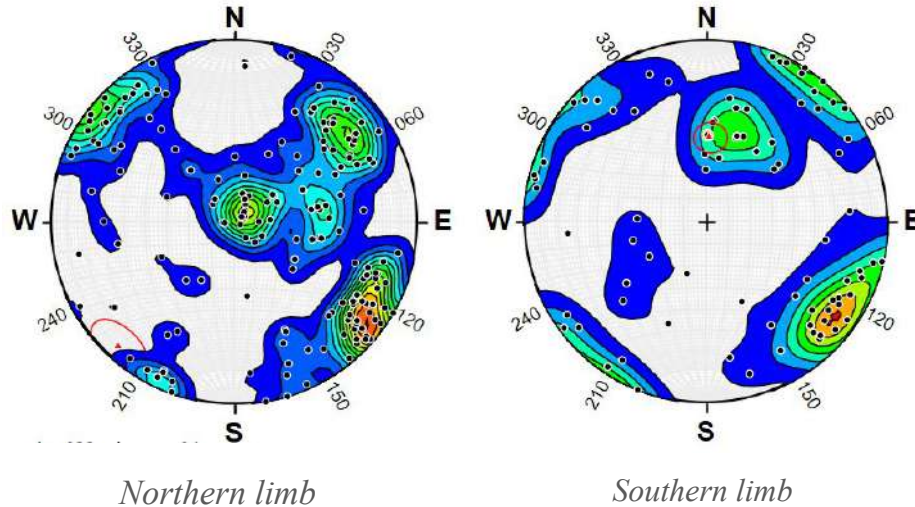
Now we classify fractures based on-

- The dihedral angle formed between the fracture plane and bedding plane.
 - low, moderate and high angle fractures.
- The dihedral angle between the fracture plane and hinge plane.
 - longitudinal, oblique and transverse fractures.



Fracture Analysis

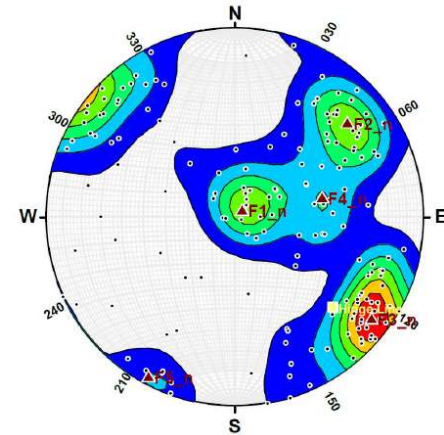
- Clusters of fracture are classified in different bins based on angularity such as *low*, *moderate* and *high angle wrt bedding* orientation.
- Then, Frequency plots were generated for the same.



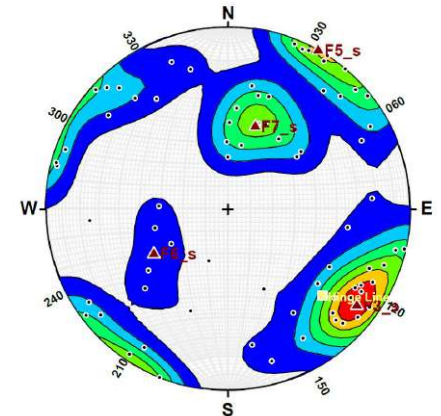
Fracture Analysis

The poles of the fracture set which make high angle with hinge line orientation are classified as longitudinal fractures and those which made low angle are classified as transverse fractures.

- Fracture that strike parallel to the trend of the regional structure(in this case fold) is called **Longitudinal Fracture**.
- Fracture that strike perpendicular to the trend of the regional structure is called **Transverse Fracture**.



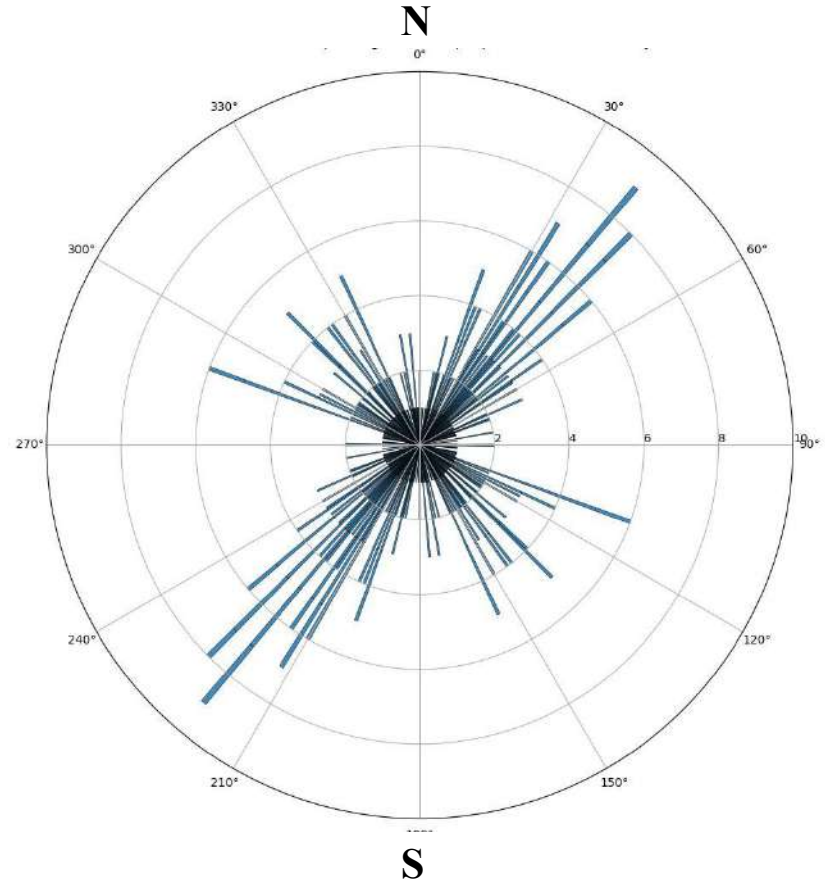
Northern Limb Fracture Density Contour



Southern Limb Fracture Density Contour

Rose Diagram

- To visualize the frequency of orientation of fractures
- A rose diagram makes these sets immediately visible as distinct "petals"
- The radial length of petal represents frequency of occurrence of fractures in that bin.
- Pinpoints dominant primary and secondary trends of features like fractures.



Structural Restoration

Why ? Significance ?

Steno's principle of Horizontality

Viewing structure in a horizontal reference frame.

Operations involve :

- Un-plunging
- Un-dipping.

Unplunging

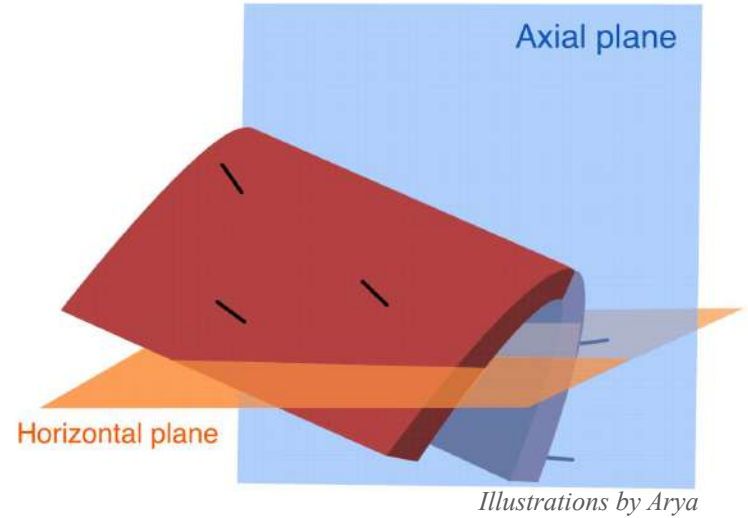
If the fold is cylindrical, these poles will fall along a single great circle called the pi-circle.

The pi-axis represents the plunge and trend of the fold axis.

If the pi-axis point lies exactly on the **primitive circle**, the fold axis is horizontal.

If not,

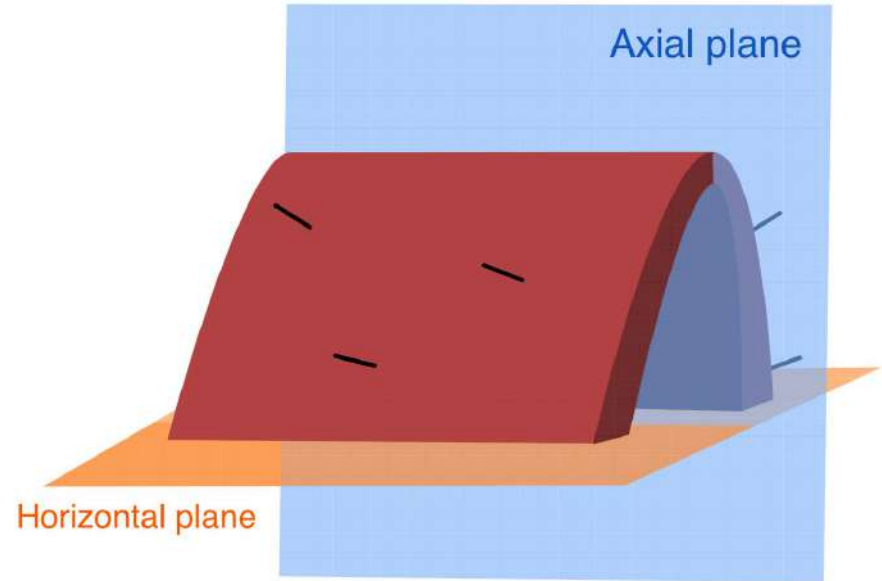
We need to rotate it back until the fold axis becomes horizontal.



Undipping

- Once the fold axis is Horizontal,
We rotate the limbs until the beds
become horizontal.

**The goal is to move the pole of
the limb to the centre of the
stereonet.**



Illustrations by Arya

Fold Test

To understand the temporal aspect of the fractures specifically, if there are different fracture sets which formed at different times wrt. the folding (**pre-folding**, **syn-folding**, or **post-folding**.)

Pre-Folding



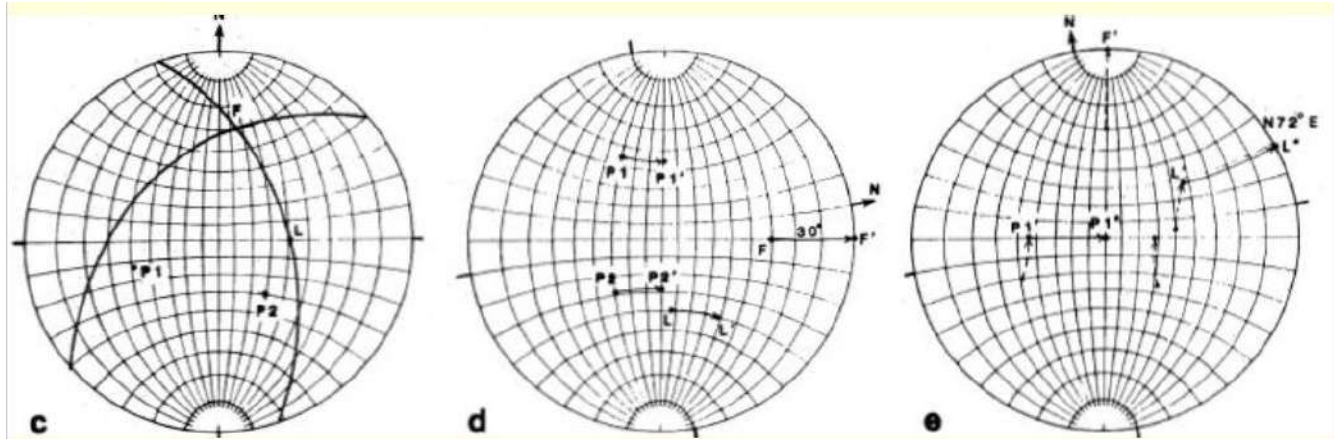
- Fracture orientation remains the **same** relative to the bedding before and after un plunging and unfolding.
- Fractures form better clustered pattern

Syn/Post-Folding



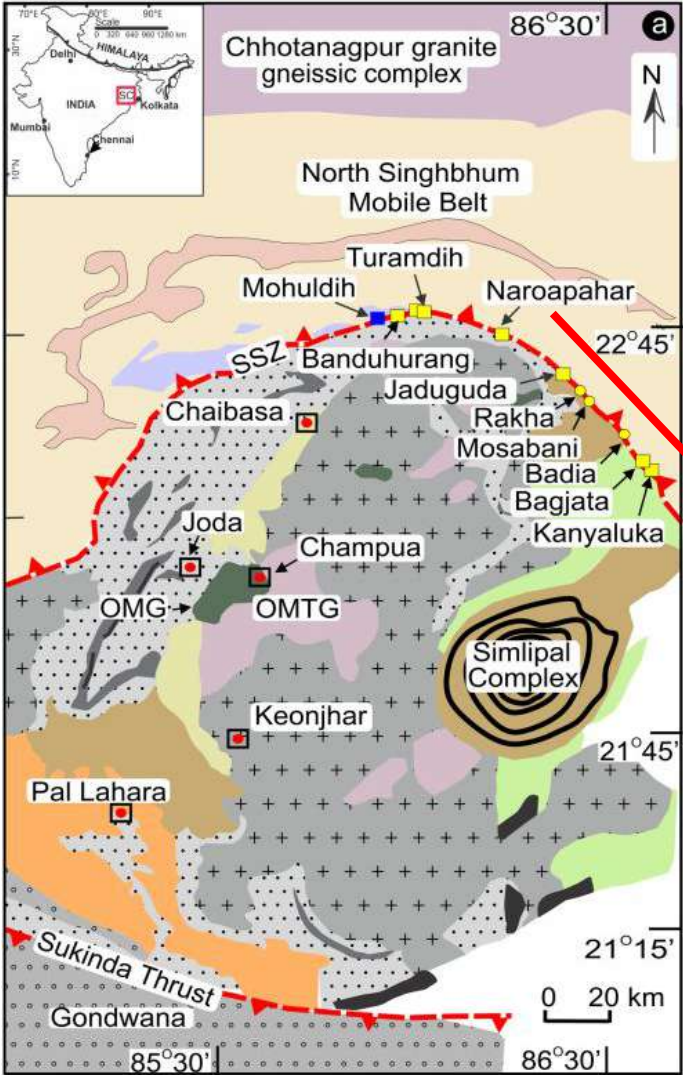
- Fracture orientation remains the **same** in geographic coordinates (absolute space) before and after un plunging and unfolding.
- Fractures don't form cluster.

Unfolding

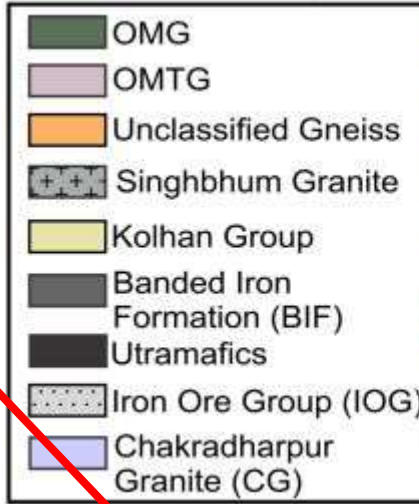


Steps:

- Bring F to E-W diameter; rotate F along a horizontal axis to bring it to the primitive. Plot F' as the rotated F. Similarly, all the other structures move concurrently along the small circles, by the same amount.
- Bring F' to the N-S diameter, it is now a horizontal rotation axis. Rotate east limb to the horizontal, by bringing P1' to P1''. L' should be horizontal when the bed is horizontal. So, L' moves to L''. L'' gives the original orientation of the sedimentary lineation; it trends N72E.



S. Patel, M. K. Ozha, Saha 1994



The NSMB records three deformation events (D1, D2, D3) and three corresponding metamorphic stages. Field structures like foliation, porphyroblasts, and fracture sets are the surface expression of the tectonic history (*Dasgupta et al., 2005*)

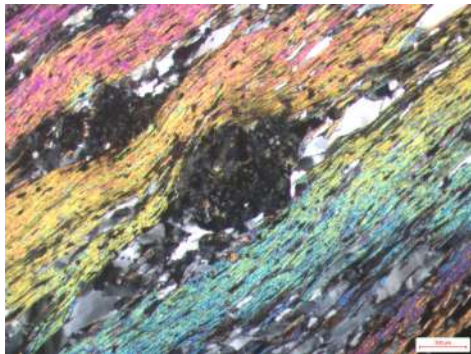
Metamorphic grade increases from greenschist facies at the limbs toward amphibolite facies at the hinge, **Galudih is on the limb**, placing it in the greenschist-lower amphibolite transition zone (*Chatterjee et al., 2007*)

Galudih

Petrological Analysis: What to look for in the field in these terms?

Identify mineral assemblages and rock types

- ▶ Frame and identify a deformation and metamorphic history
- ▶ Recognize reaction textures, which may point to >>> (growth, replacement, retrogression)
- ▶ Select equilibrium mineral pairs for **thermobarometry**
- ▶ Qualitative observations are converted into a quantitative P–T path



Xpl, Scale: 500 μm

- Optical petrography (PPL/XPL) at multiple magnifications enables identification of **mineral assemblages, modal proportions, and textural relationships**

Plane-Polarized Light (PPL)

Identify mineral color, cleavage, crystal habit, pleochroism, inclusions.
Assess grain size distribution and fabric elements.

Cross-Polarized Light (XPL)

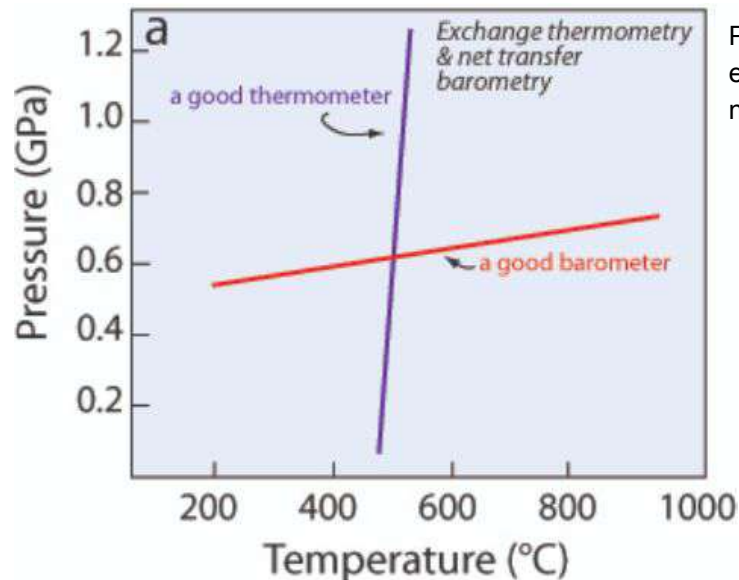
Birefringence colors, extinction angles, twinning patterns. Distinguish quartz, feldspar, micas, and amphiboles reliably.

Multiple Magnifications

Low mag (4×, 10×) for fabric, texture, porphyroblast-foliation relationships. High mag (40×) for grain-scale details.

Why Petrographic Analysis?: Using thin section as probes

- Thin section reveals the sequence of mineral reactions and metamorphic events, but does not provide quantitative temperature or pressure
- But textures alone cannot constrain depth or temperature of metamorphism
- Thermobarometry uses equilibrium thermodynamics on mineral compositions to calculate P–T conditions
- Selection is based on the equilibrium mineral assemblages!



Pressure-temperature sensitivity of exchange (thermometer) and net-transfer (barometer) reactions

Assumptions

- Minerals were in chemical equilibrium at the time of last equilibration
- Mineral compositions (especially if garnet zoning) are preserved and not fully reset during the retrograde phase

ELECTRON MICROPROBE ANALYSIS (EMPA) & NORMALIZATION

Table 1: Representative microprobe analyses of garnet, biotite, muscovite, and plagioclase in garnet–biotite–muscovite schist from Barru Complex.

- Quantifies compositional zoning in minerals (core–rim variation)
- Calculation of cations and unit formula

Mineral	Grt			Bt		Ms		Pl	
	core	mantle	rim [†]	matrix [‡]	matrix	matrix [‡]	matrix	matrix [‡]	matrix
SiO ₂	38.19	38.35	38.55	37.07	38.17	47.49	47.53	59.20	62.27
TiO ₂	0.01	0.23	0.00	1.44	1.08	0.48	0.38	0.12	0.19
Al ₂ O ₃	21.18	21.38	21.18	20.59	20.98	34.57	33.61	25.02	23.28
Cr ₂ O ₃	0.00	0.00	0.00	0.00	0.19	0.02	0.00	0.02	0.00
FeO	27.17	27.06	27.36	17.25	16.19	0.80	1.80	0.00	0.00
Fe ₂ O ₃	–	–	–	–	–	–	–	–	–
MnO	3.50	3.14	0.14	0.13	0.00	0.05	0.00	0.12	0.05
MgO	0.80	0.59	1.60	10.40	9.19	0.62	1.19	0.07	0.00
CaO	9.50	9.53	11.60	0.23	0.00	0.00	0.00	6.11	5.63
Na ₂ O	0.00	0.01	0.01	0.08	0.30	0.58	1.21	9.26	8.93
K ₂ O	0.00	0.00	0.13	7.21	9.03	10.19	9.33	0.25	0.01
Total	100.35	100.29	100.57	94.40	95.13	94.80	95.05	100.17	100.36
O	12	12	12	11	11	11	11	8	8
Si	3.04	3.04	3.03	2.76	2.83	3.16	3.16	2.65	2.76
Ti	0.00	0.01	0.00	0.08	0.06	0.02	0.02	0.00	0.01
Al	1.99	2.00	1.96	1.81	1.83	2.71	2.63	1.32	1.21
Cr	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00
Fe	1.81	1.80	1.80	1.07	1.00	0.04	0.10	0.00	0.00
Mn	0.24	0.21	0.01	0.01	0.00	0.00	0.00	0.00	0.00
Mg	0.09	0.07	0.19	1.15	1.01	0.06	0.12	0.00	0.00
Ca	0.81	0.81	0.98	0.02	0.00	0.00	0.00	0.29	0.27
Na	0.00	0.00	0.00	0.01	0.04	0.07	0.16	0.80	0.77
K	0.00	0.00	0.01	0.69	0.85	0.86	0.79	0.01	0.00
Total cation	7.97	7.94	7.99	7.60	7.64	6.94	6.98	5.09	5.01

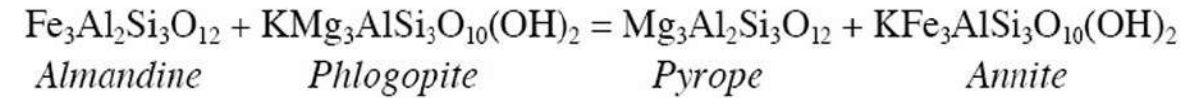
Oxide wt% → Moles → Oxygen normalization (available oxygen) → Cation proportions → Fe–Mg ratios / X-values → Thermobarometry

Geothermometry: Garnet-Biotite Geothermometer

Gibbs free energy minimization, linking composition to P-T conditions

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ + P\Delta V^\circ$$

$$\Delta G^\circ = -nRT \ln K \qquad \text{At eq. } \Delta G = 0; Q = K_d$$



$$K_{\text{Fe-Mg}}^{\text{Gt-Bt}} = \frac{X_{\text{Fe}}^{\text{Bt}} / X_{\text{Mg}}^{\text{Bt}}}{X_{\text{Fe}}^{\text{Gt}} / X_{\text{Mg}}^{\text{Gt}}} = \frac{(\text{Fe/Mg})^{\text{Bt}}}{(\text{Fe/Mg})^{\text{Gt}}}$$

$$\ln K_D = \frac{-\Delta H - P\Delta V}{3R} \left(\frac{1}{T} \right) + \frac{\Delta S}{3R}$$

Y

=

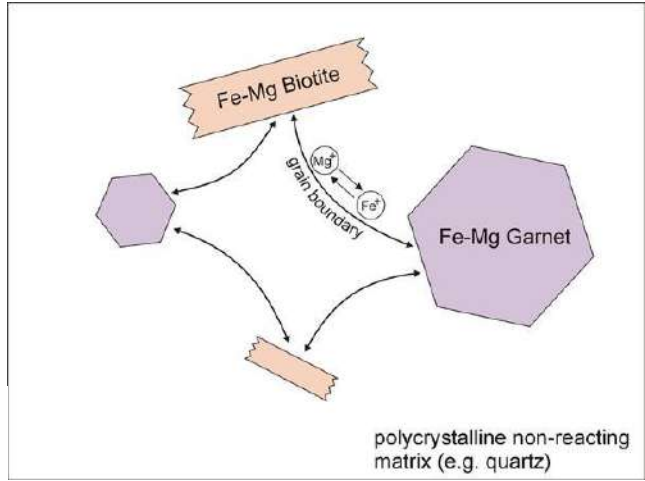
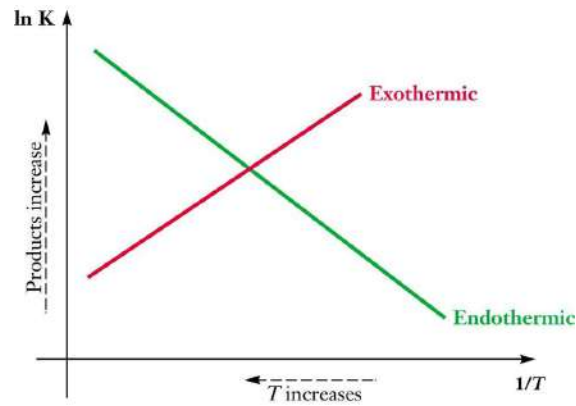
m

.

x

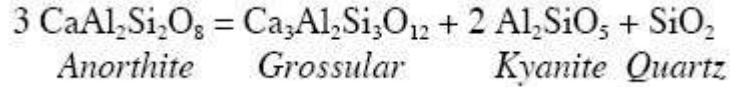
+

c



- Fe and Mg exchange between garnet and biotite.
- Net Transfer reaction: Large volume (ΔV) change;
- small entropy (ΔS) change

Garnet-Al₂SiO₅-Quartz-Plagioclase barometer

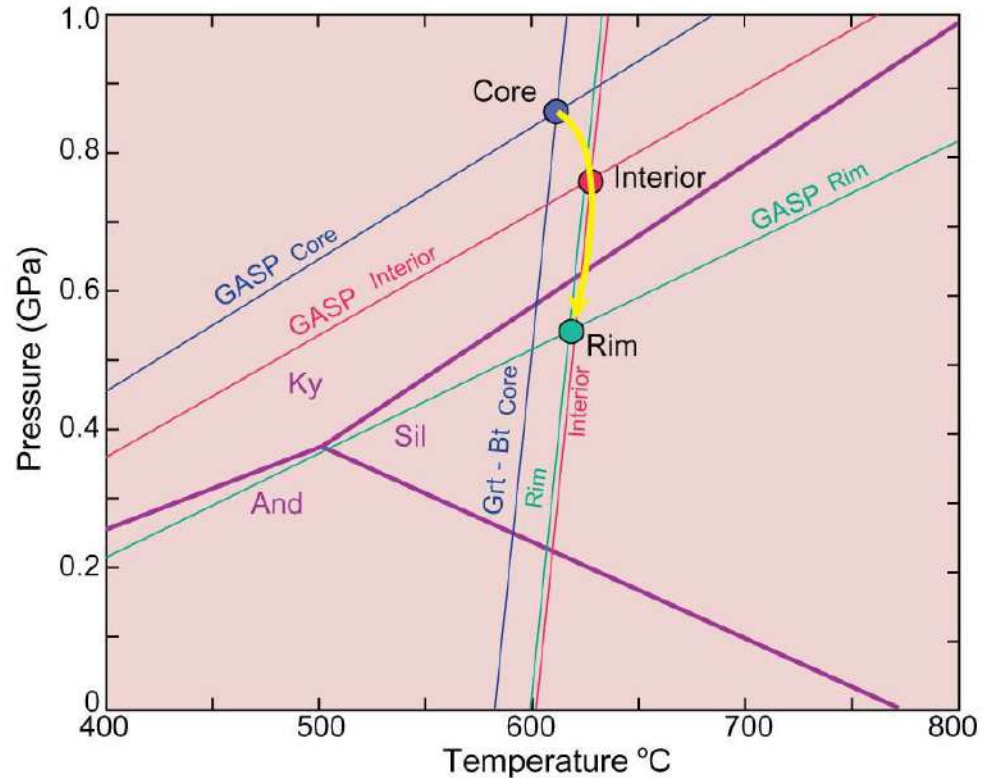


$$\ln K_{Ca}^{Gt-Pl} = \left(\frac{55,865}{R} \right) \frac{1}{T} - \frac{153.59}{R} + P \left(\frac{6.608}{R} \right) \frac{1}{T}$$

$$K_{Ca}^{Gt-Pl} = \frac{(a_{Grs}^{Gt})}{(a_{An}^{Pl})^3}$$

$$a_{Grs}^{Gt} = (\gamma_{Grs}^{Gt} X_{Grs}^{Gt})^3 = \left[1.03 \left(\frac{Ca}{Ca + Fe + Mg + Mn} \right)_{Gt} \right]^3$$

$$a_{An}^{Pl} = 1.35 (X_{Ca}^{Pl}) = 1.35 \left(\frac{Ca}{Ca + Na + K} \right)_{Pl}$$

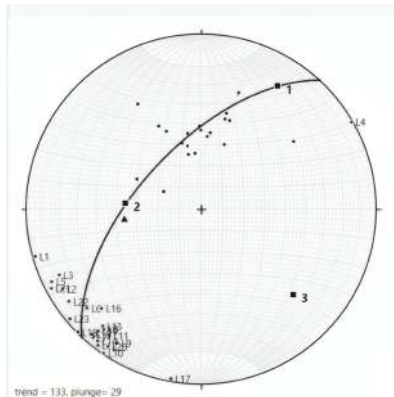


Results and discussion

Results

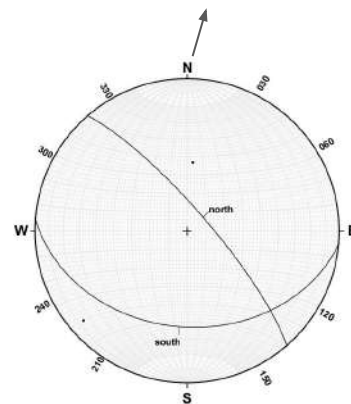
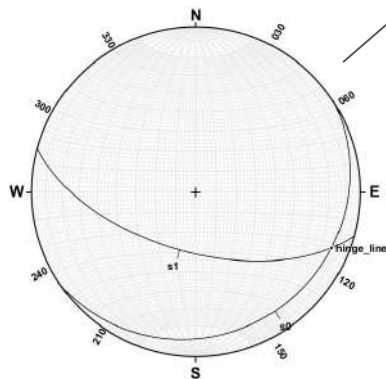
Chorinda

Determination of hinge line

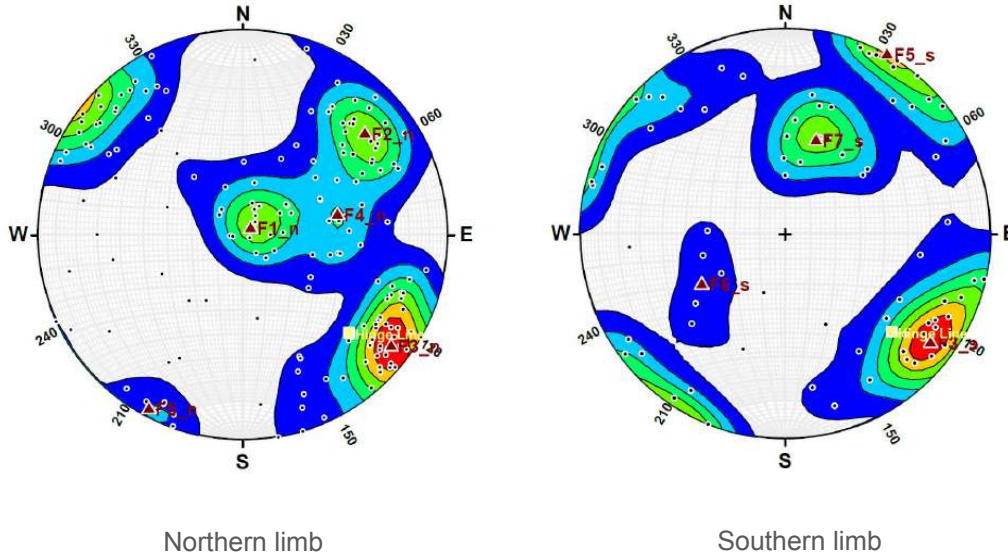


We got 3 different hinge line data:-

- From π -plot - 133/29
- From Bedding(So) - Clevage intersection(S1) - 112.5/11.9
- From Intersection of Limb (similar to β -plot) - 133.8/25.6 .



Fracture set based on the orientation

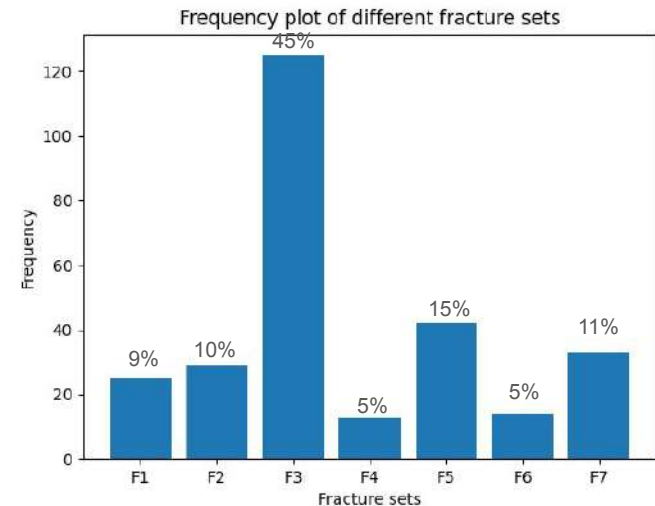


The plots above are of fractures set from the northern limb(left) and from the southern limb(right) with their mean.

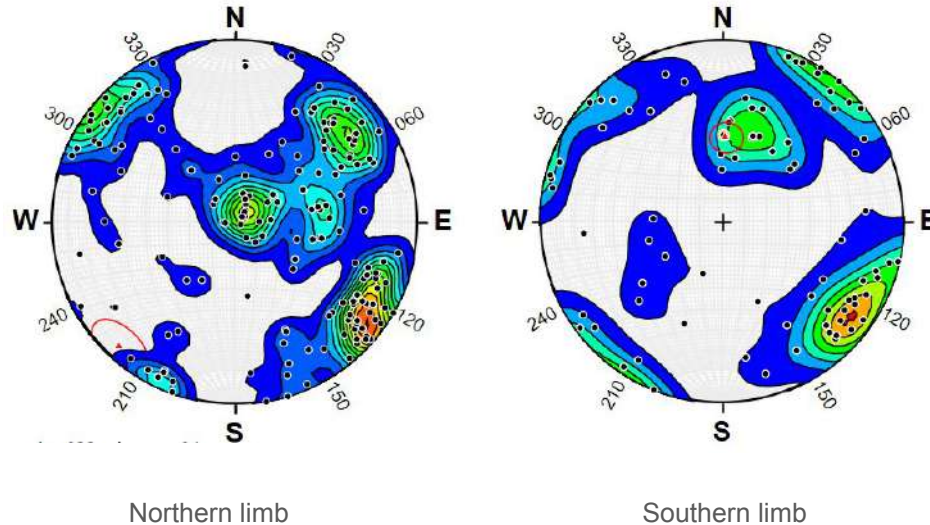
Stereoplot of the fractures from Chorinda were plotted and fracture with similar orientation were classified as different fracture set.

Key observations:-

- 7 Different fracture sets present.
- Fracture set F3 and F5 are present in both the limb.
- F3 is the most dominant fracture set.



Fracture set wrt bedding angle relationship

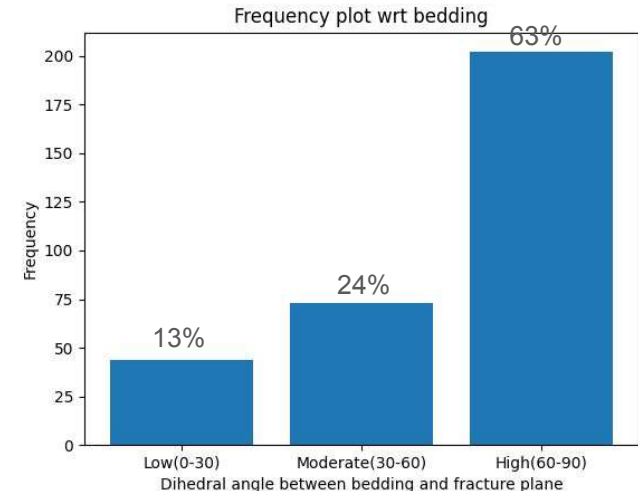


The frequency plot show high angle fractures are more than low and moderate angle fractures.

Key Observations:-

- There more fractures present in the northern limb than southern limb.
- High angle fractures are more dominated.

Clusters of fracture in the stereonet were categorized as low, moderate and high angle wrt bedding orientation. Where the bin size was taken from the maximum and minimum angle of the clusters



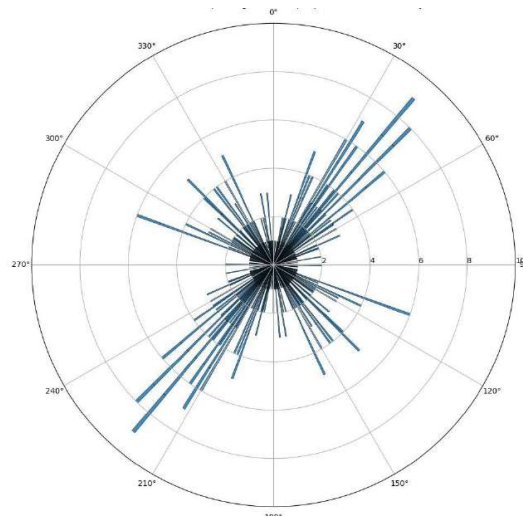
Fracture wrt the fold

Fracture that strike parallel to the trend of the regional structure(in this case fold) is called Longitudinal Fracture.

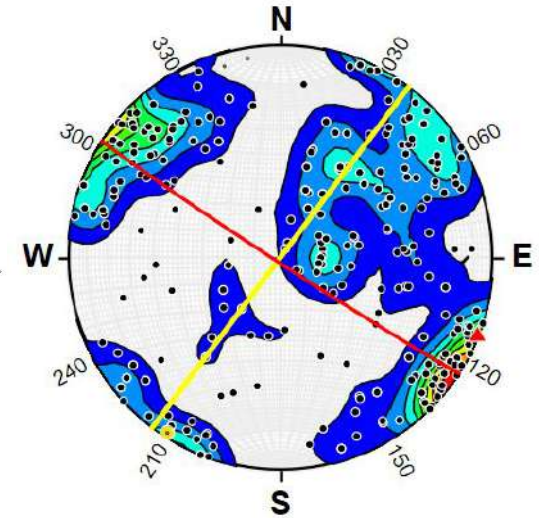
Fracture that strike perpendicular to the trend of the regional structure is called Transverse Fracture.

Key observation-

- The chorinda region is most dominated by the longitudinal and transverse fractures.
- The frequency of transverse fractures were more than longitudinal.
- Mamtani (2004) which was only joint shows similar plot.



Rose diagram of the strike of all the fractures



Contour plot of all the fracture with dominant orientation show with Red and Yellow lines

Shear fractures Characterisation

Location	Fracture set	Number of Shear Fractures	Position of the Location wrt the fold
n20	F2	4	Middle of the limb to Hinge on the northern limb
v10	F3	5	Middle of the limb to Hinge on the southern limb
v12	F3	8	Middle of the limb to Hinge on the southern limb
v12	F5	2	Middle of the limb to Hinge on the southern limb
v13	F7	1	Middle of the limb to Hinge on the southern limb

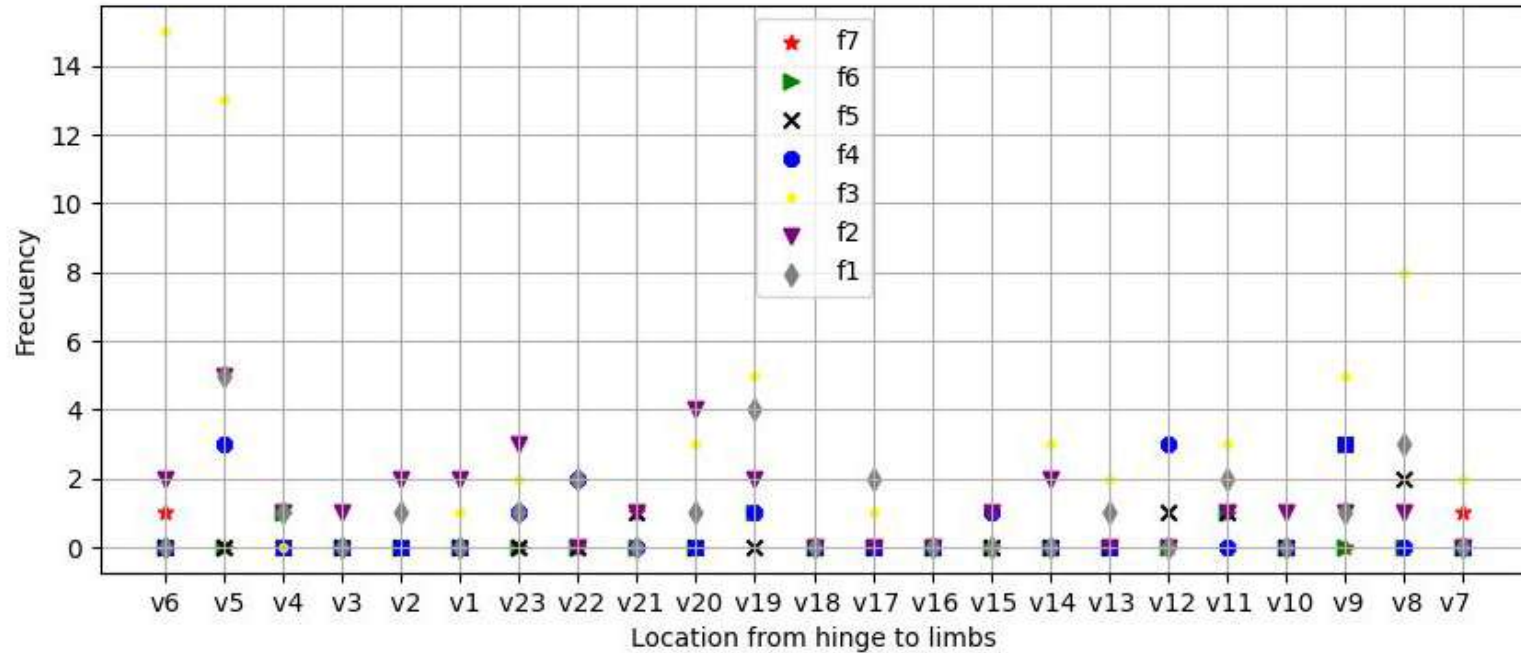
Key observations

- All the shear fractures are concentrated in locations in mid way from hinge to limb.
- F3 is a high angle and transverse fracture set.
- Location v10 and v12 are located where we saw a possible offset in the outcrops in southern limb.
- F2, F5 and F7 are low angle fracture sets which are present in the inner arc.

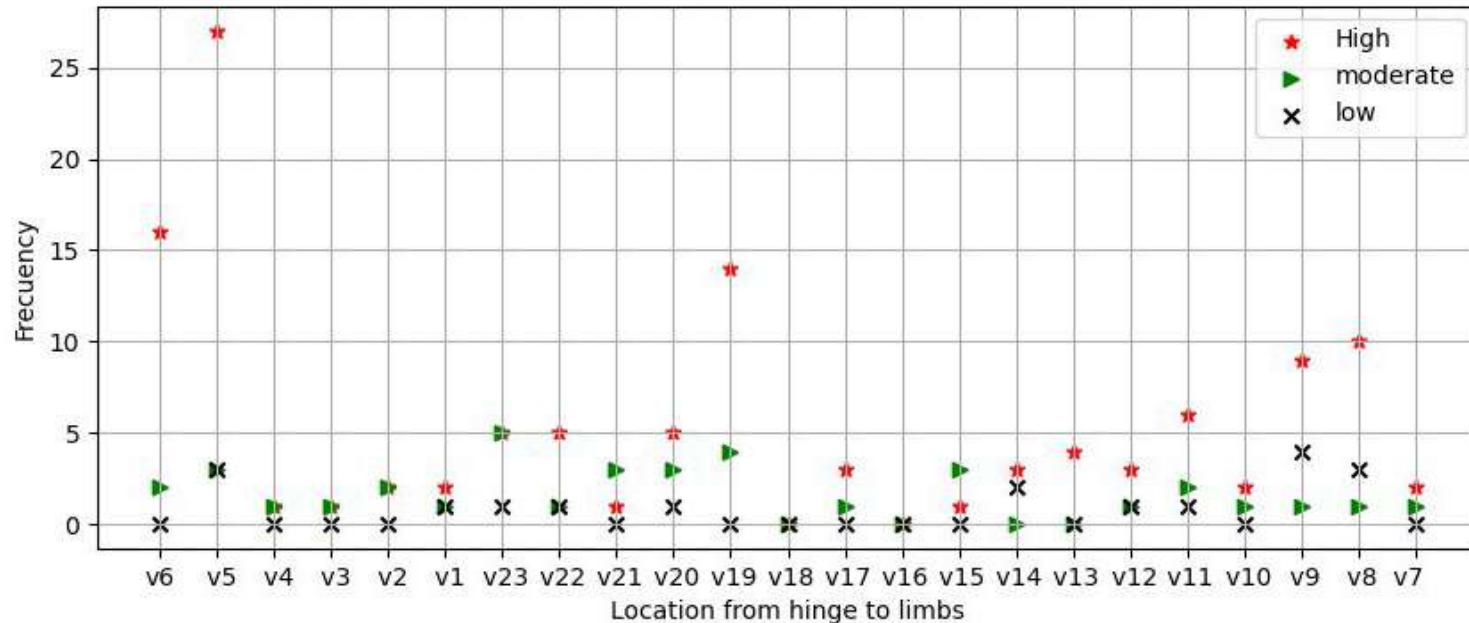
Fracture set characterization

Fracture Set	Present in	n	Mean Attitude	Vein count	Orientation wrt fold	Orientation wrt bedding
F1	North	25	149/07	0	Oblique	High(N)
F2	North	29	141/66	8	Longitudinal	Low(N)
F3	Both	125	217/80	6	Transverse	High(N), High(S)
F4	North	13	169/39	0	Oblique	Moderate(N)
F5	Both	42	296/86	1	Longitudinal	Low(N), High(S)
F6	South	14	328/39	0	Oblique	Mod(S)
F7	South	33	109/39	0	Longitudinal	Low(S)

Spatial variation of fracture sets from hinge



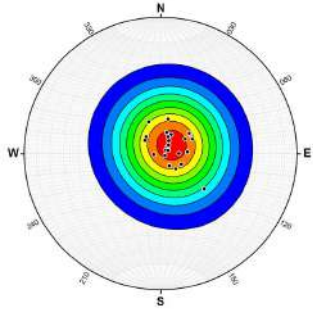
Spatial variation of low, moderate and high angle from hinge



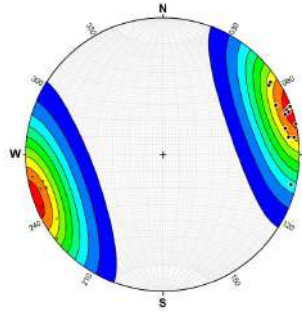
What did we infer from our fracture data?

- F3 are high angle, transverse and have higher frequency, they could be possibly post folding.
- All transverse fractures are high angle. A point to remember.
- We did observe veins in field, which were present in F2, F3 and F5, where F2 has the highest veins to fracture ratio. Indicating that it was more conducive to fluid, maybe have formed temporally earlier.
- F2 is also low angle and longitudinal fracture set.
- All the low angle shear fractures were present in the inner arc.
- High angle shear fractures (F3) are present on location v10 and v12, where we saw possible offsets of outcrops, this may imply F3 was post folding.
- All the low angle fractures are also longitudinal. But F5(S) is a longitudinal high angle fracture set. An interesting point to remember.
- F1, F2, F4, F6 and F7 are only present in one limb, could imply they are pre to syn folding.

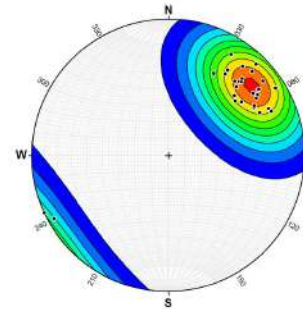
Fold test of different fracture sets



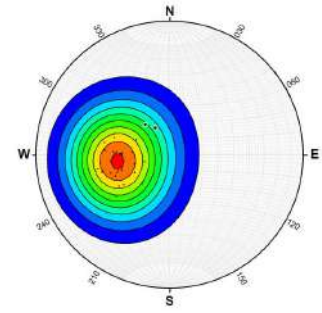
F1 finite



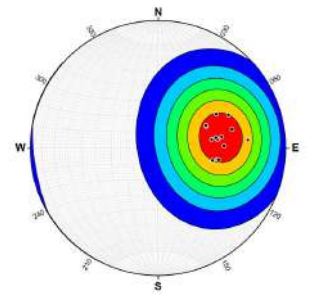
F1 undeformed



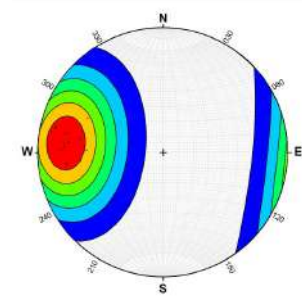
F2 finite



F2 undeformed

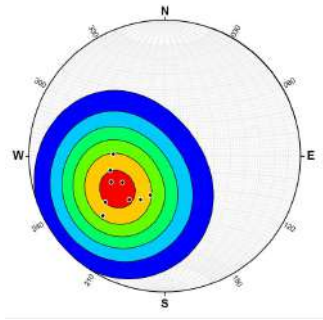


F4 finite

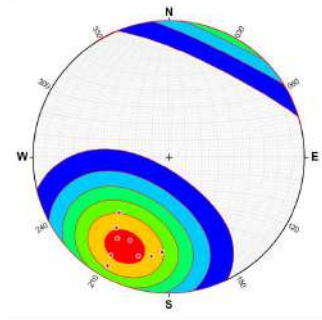


F4 undeformed

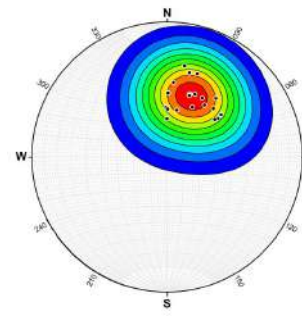
Sets of northern Limb



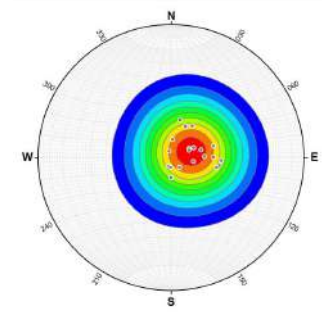
F6 finite



F6 undeformed

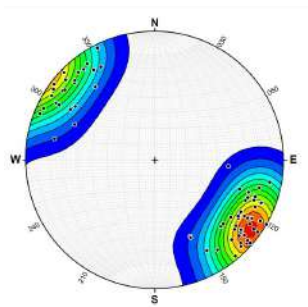


F7 finite

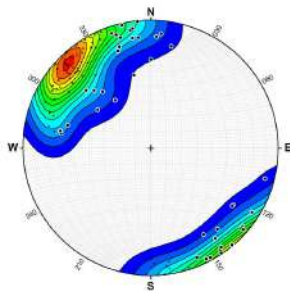


F7 undeformed

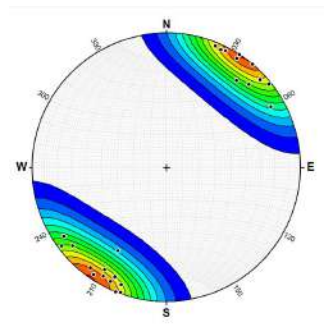
Set of southern Limb



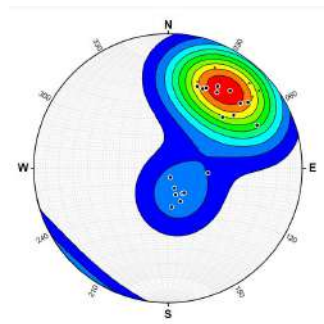
F3 finite



F3 undeformed



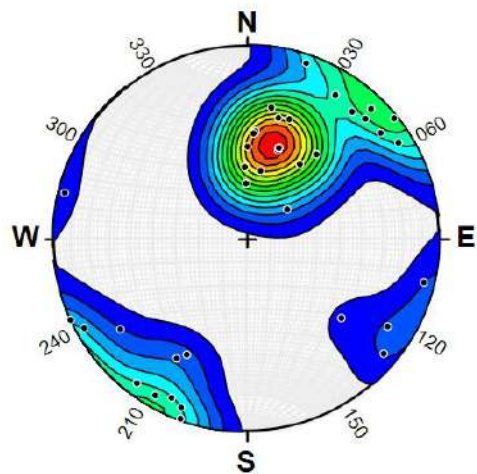
F5 finite



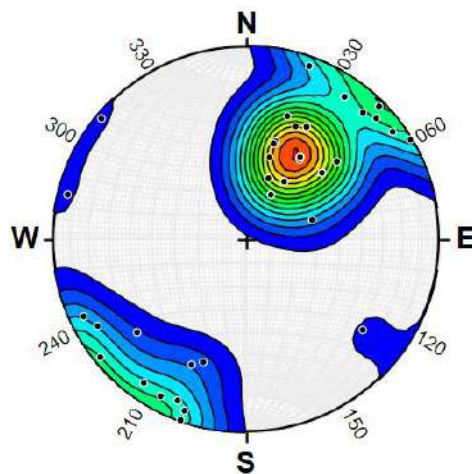
F5 undeformed

Sets of both limbs

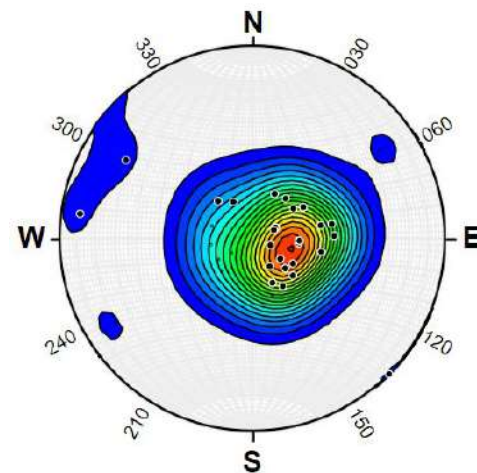
Low angle



Finite

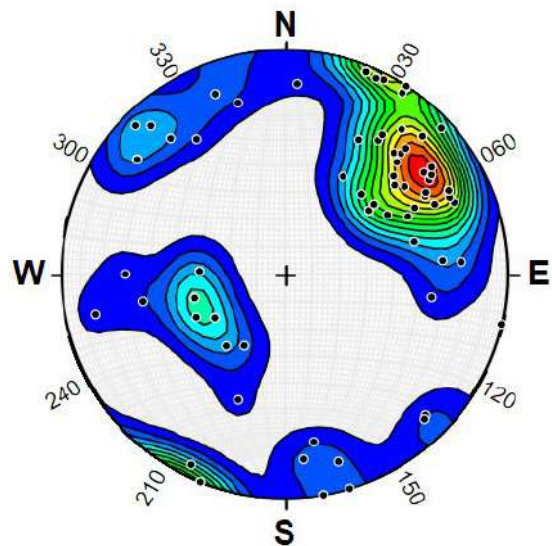


Unplunged

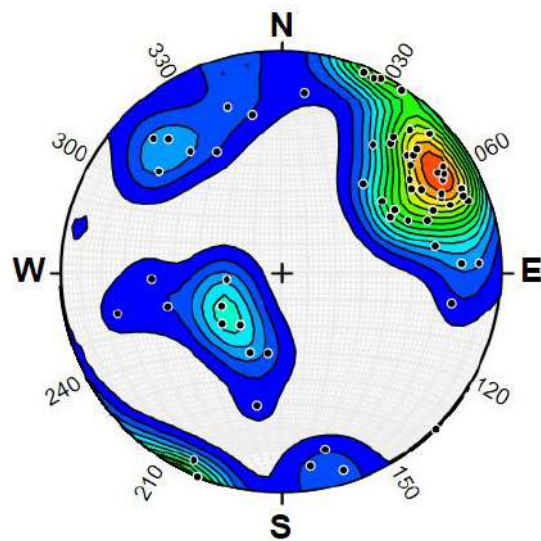


Unfolded

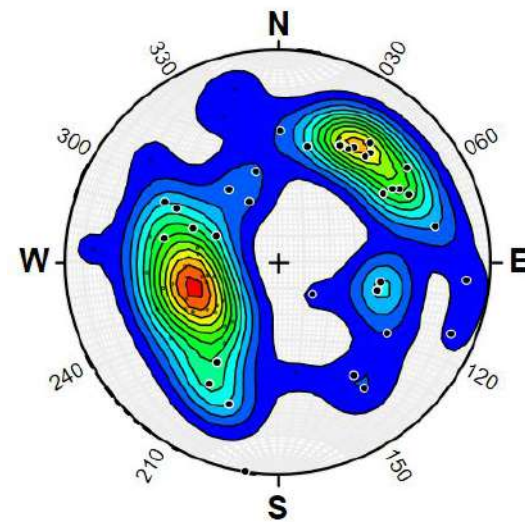
Moderate angle



Finite

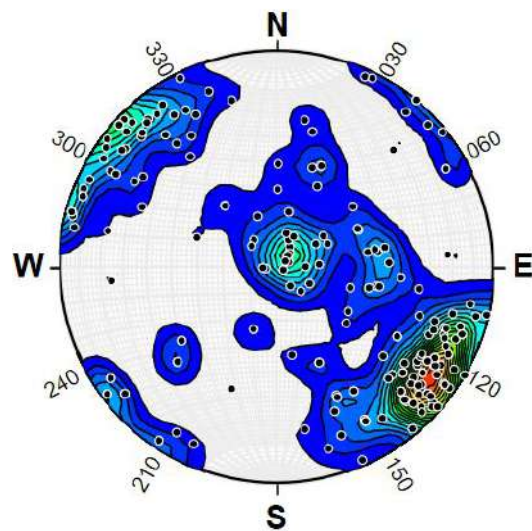


Unplunged

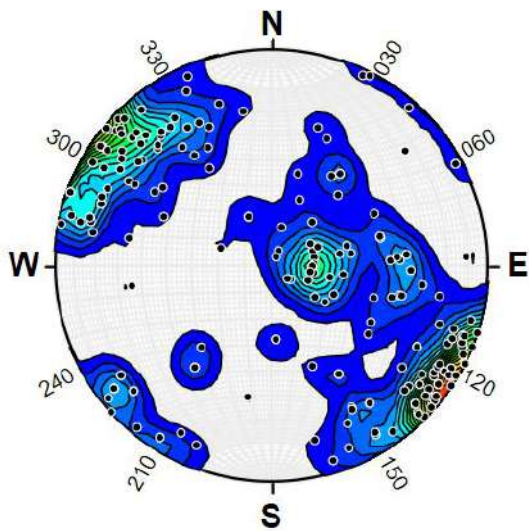


Unfolded

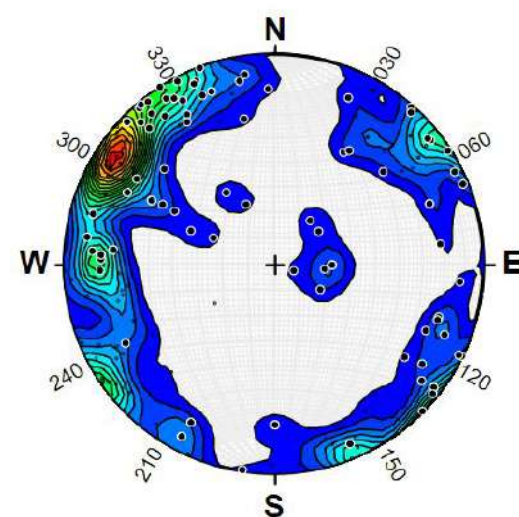
High angle



Finite



Unplunged



Unfolded

Folding Mechanism

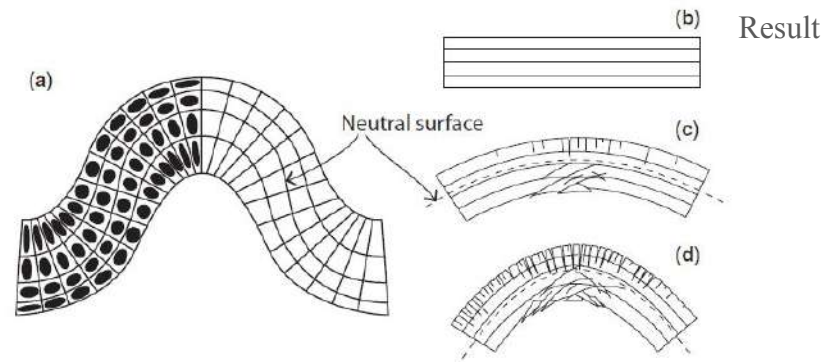


Fig. 1. (a) The strain distribution within a buckle formed in a homogeneous, isotropic layer such as a massive, unbedded limestone or sandstone bed. (b) The undeformed layers prior to folding. (c, d) High strains are concentrated in the hinge region which is therefore the site of most intense fracture formation. From Ramsay (1967).

Cosgrove (2015)

Mechanisms of folding- 1) Orthogonal Flexure 2) Flexural slip

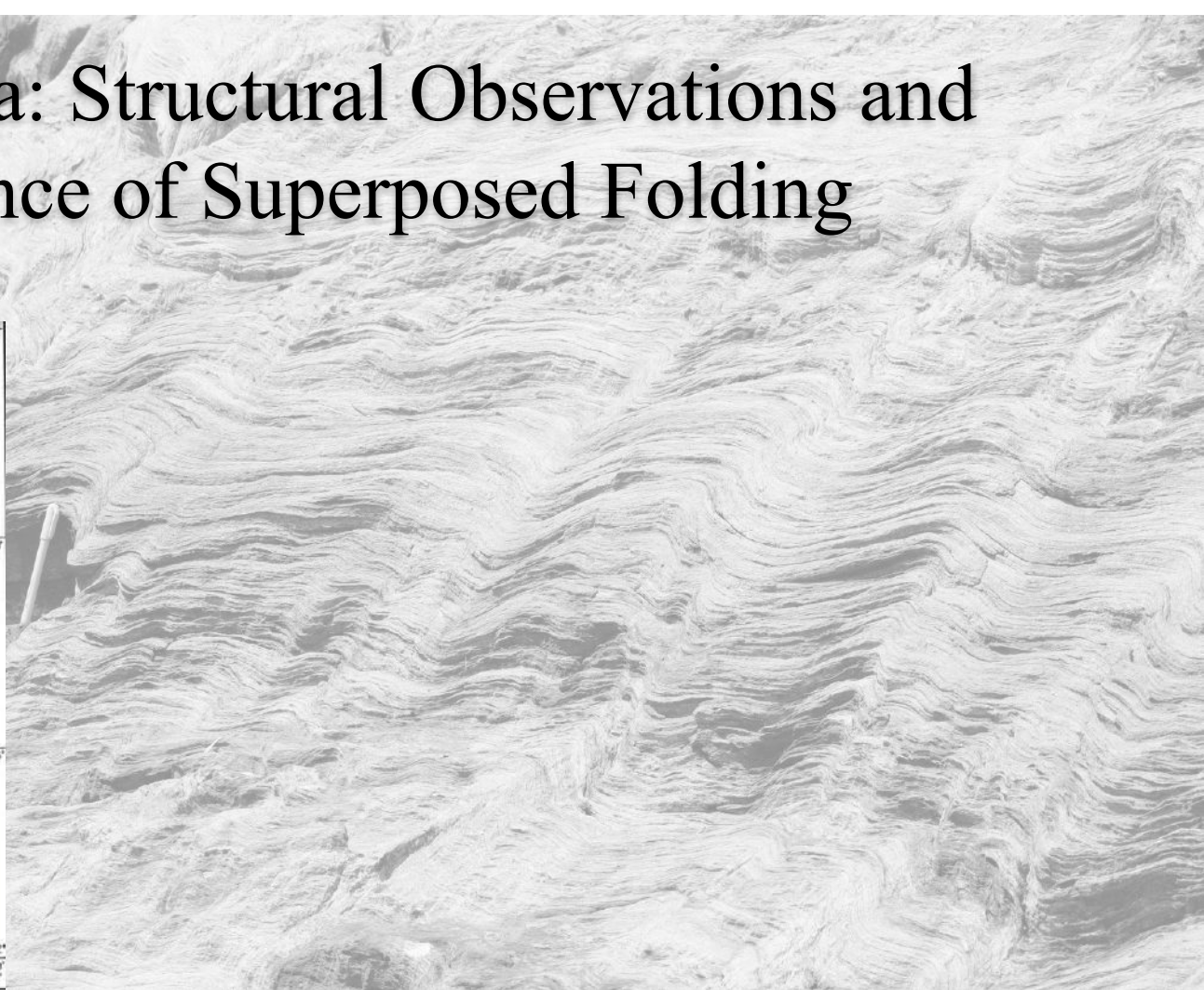
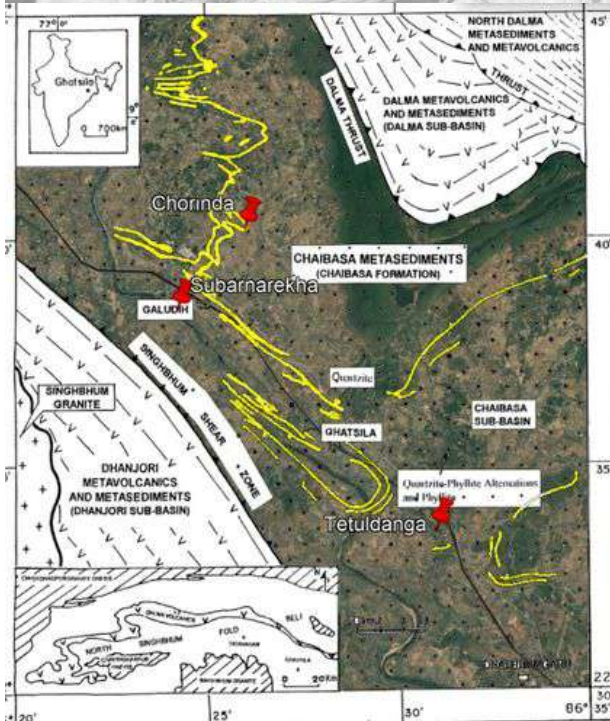
- Why not flexural slip?

No evidence of bedding parallel shear

- Why orthogonal flexure?

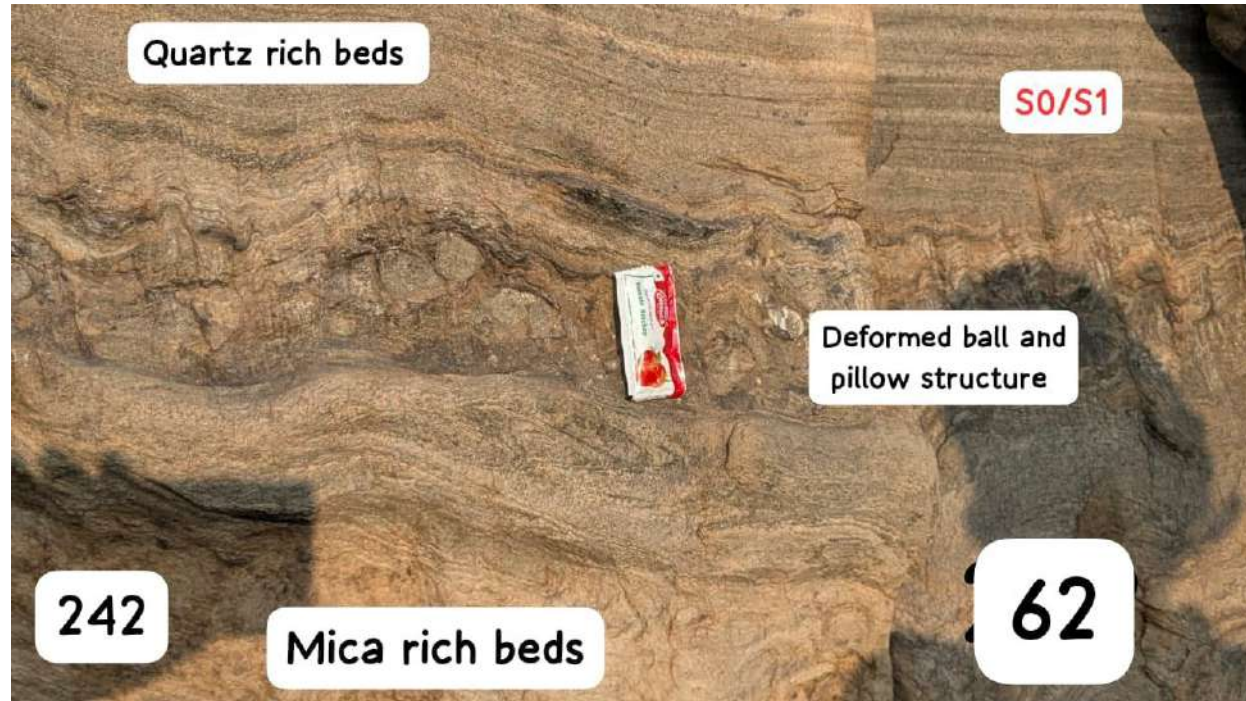
Quartzite band being the marker bed- High angle fractures in outer arc, syn folding fold test!

Tetuldanga: Structural Observations and Evidence of Superposed Folding



What was observed at Tetuldanga?

- Well exposed metasedimentary outcrops
- Distinct compositional layering identifying primary bedding (S0)
- Alternate Quartz-rich and mica-rich bands
- Deformed sedimentary structures (ball and pillow structures) observed.



How were primary and secondary fabrics identified?

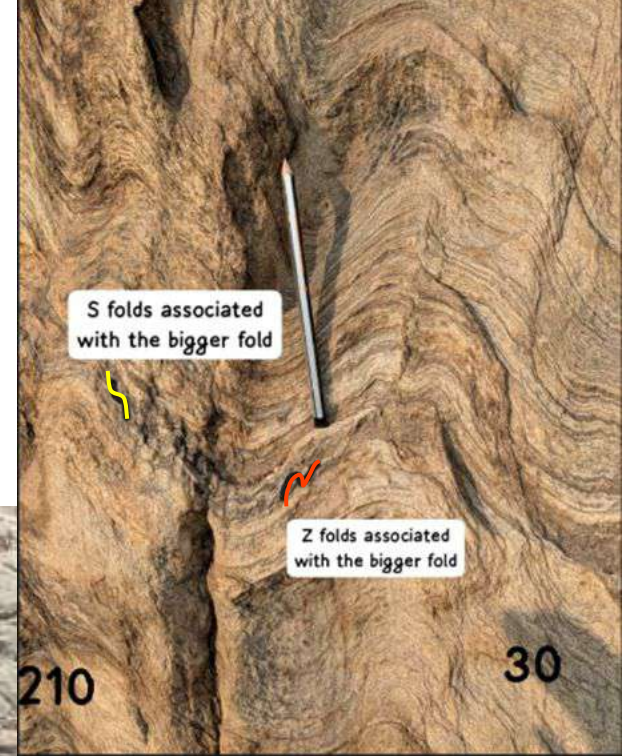
- Compositional layering interpreted as **S0**
- A penetrative foliation parallel to layering interpreted as **S1**
- **Later cleavages locally overprint earlier fabrics**

Interpretation: Multiple planar fabrics indicate progressive deformation.

What minor structures were observed?

- **Two** sets of crenulation cleavages observed
- **Cleavage sets** occur at **high angle** / near perpendicular orientation
- Minor **S-type** and **Z-type** folds present in different domains

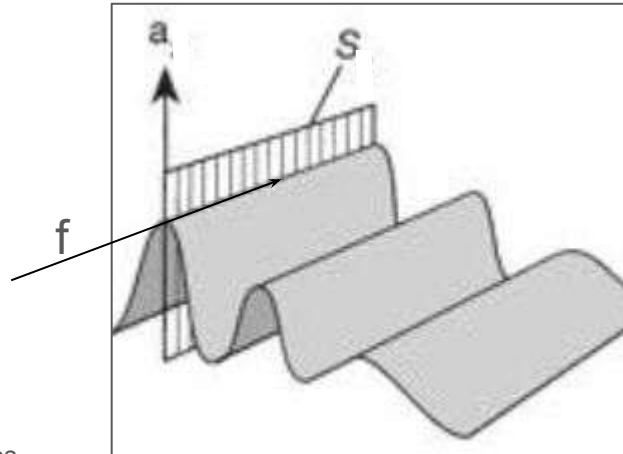
Interpretation: Variation in minor fold vergence and multiple cleavages suggest more than one overprinting event



Superposed folding, Theory

Superposed folding - Older folds in the rock being refolded by consequent / subsequent deformation events. Resulting in structures called “Interference patterns”

“Interference patterns are characterized by angle between S_1 and a_2 , and the angle between f_1 and S_2 ”

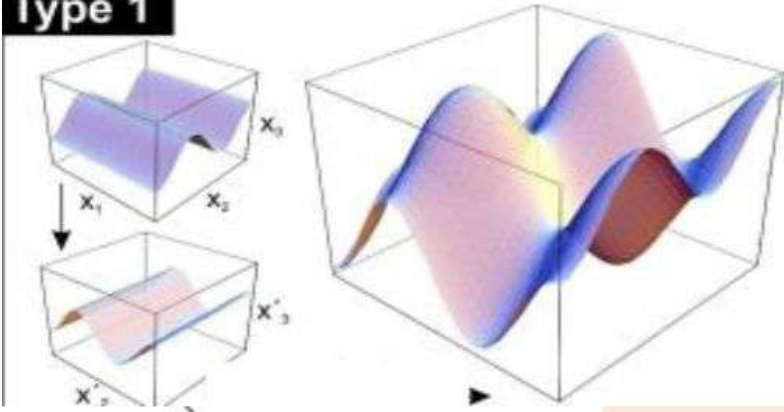


a- net slip directing
S- fold axis
f- hinge line orientation

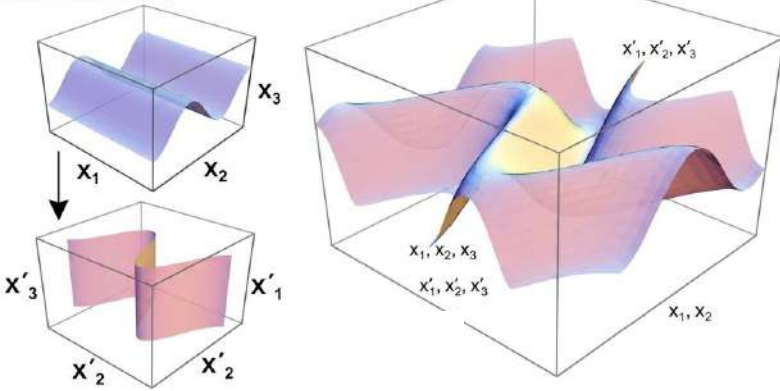
Interference Patterns

If the wavelengths of superposing folds are equal, we can have three end members.

Type 1



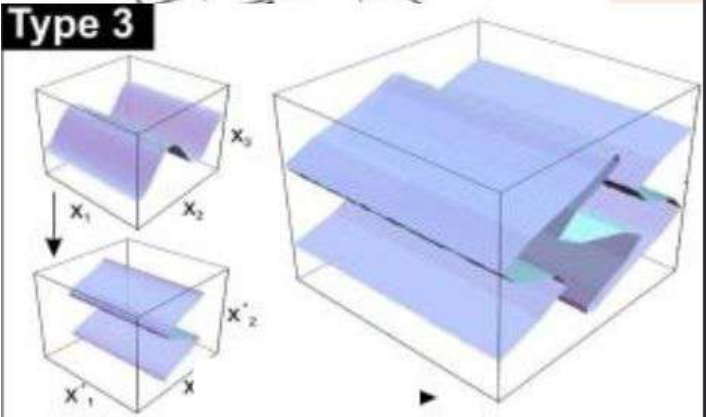
Type 2



A2 S1 angle - low
F1 s2 angle - high



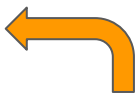
Type 3



A2 S1 angle - high
F1 s2 angle - high



A2 S1 angle - high
F1 s2 angle - low



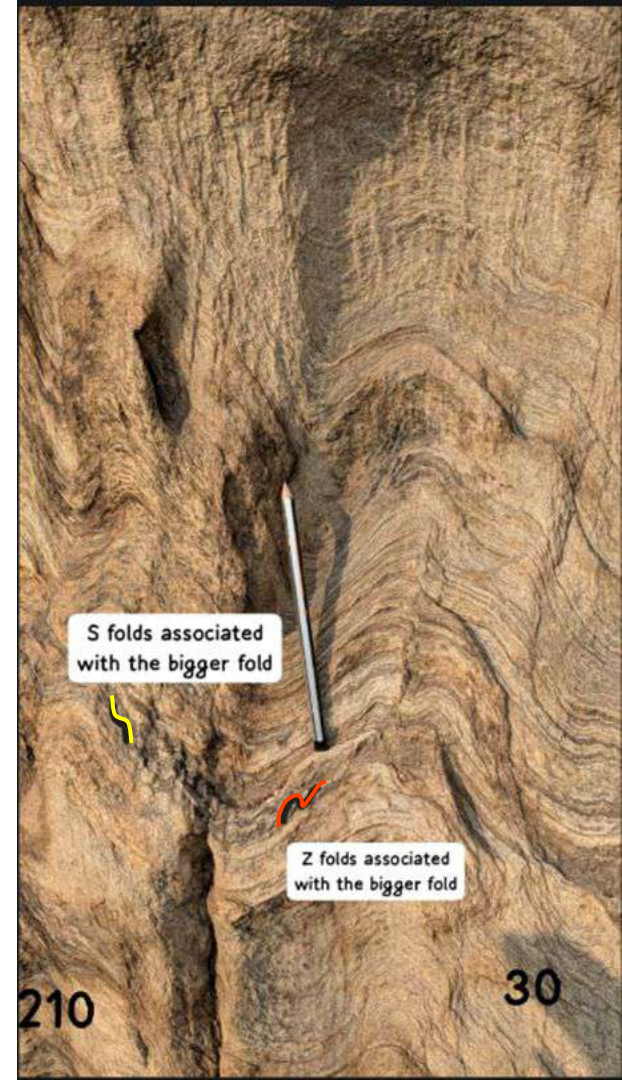
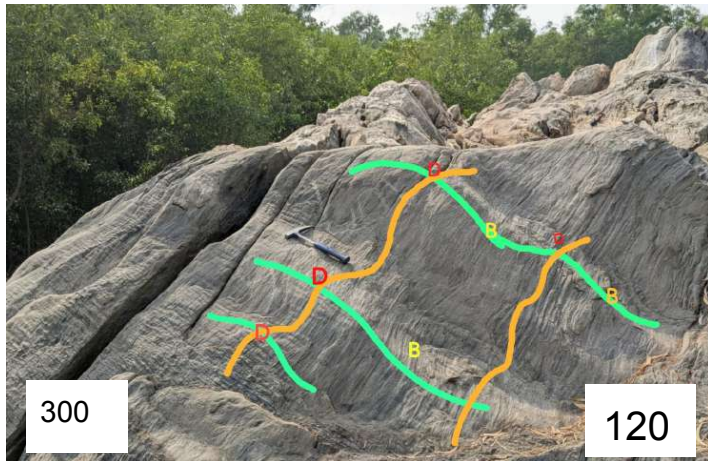
Credits - Superposed folding,
ES4101, slides.

How do we know multiple folding events occurred?

Parasitic Minor folds

+

Interference Patterns



Evidence of Superposed Folding Observed at Tetuldanga

Mean Attitudes of the Pucker axis lineation(F) and Foliations(S)

There are at least 4 folds and their indirect hinge line orientations.

Tetuldanga Data Set			
Hinge line (Trend/Plunge/Rake)(°)		Foliation plane (Dip/Dip Direction)(°)	
f1	322/23/63	S1	79/030
f2	315/35/70	S2	75/240
f3	140/65/45	S3	75/217
f4	056/25/45	S4	75/165

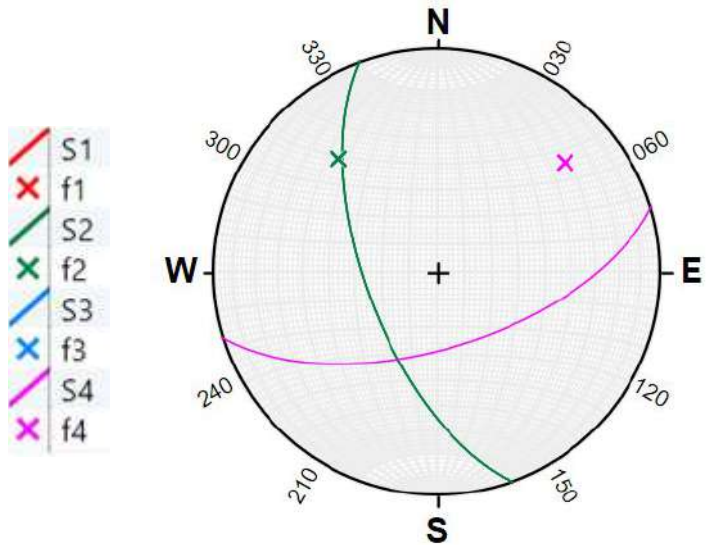
What fold interference patterns were recognised?

Type 1 – Dome and Basin feature

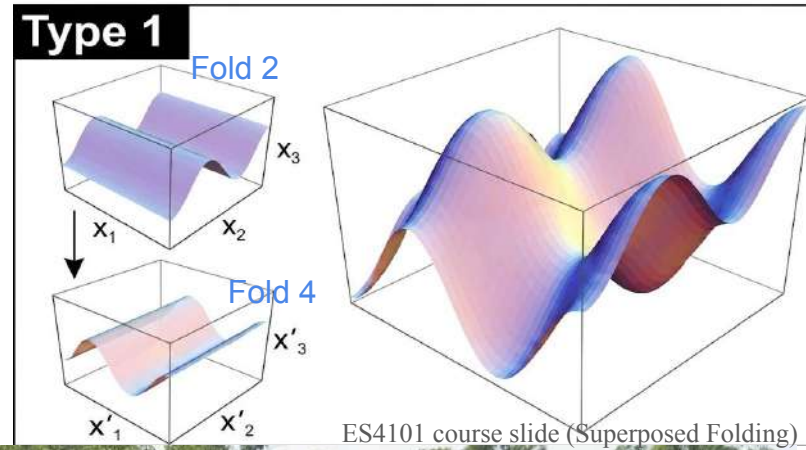
Hinge lines and axial planes are at high angle to each other for the 2 folds

f2 → 315°/35°/70° S2 → 75°/240°

f4 → 056°/25°/45° S4 → 75°/165°



Fold 2 + Fold 4



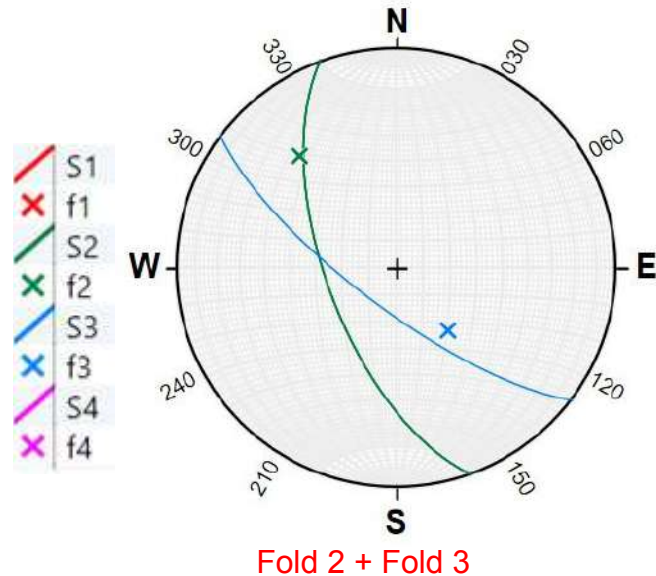
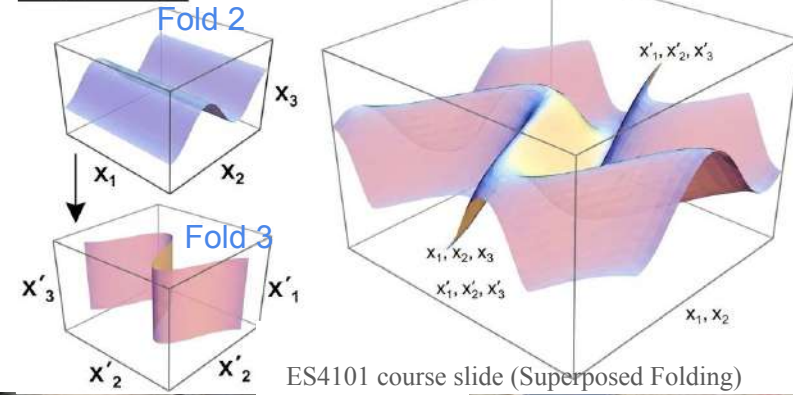
Type 2 – Interference Pattern

Hinge lines are at high angle and axial planes are at high angle for the 2 folds.

f2 → 315°/35°/70° S2 → 75°/240°

f3 → 140°/65°/45° S3 → 75°/217°

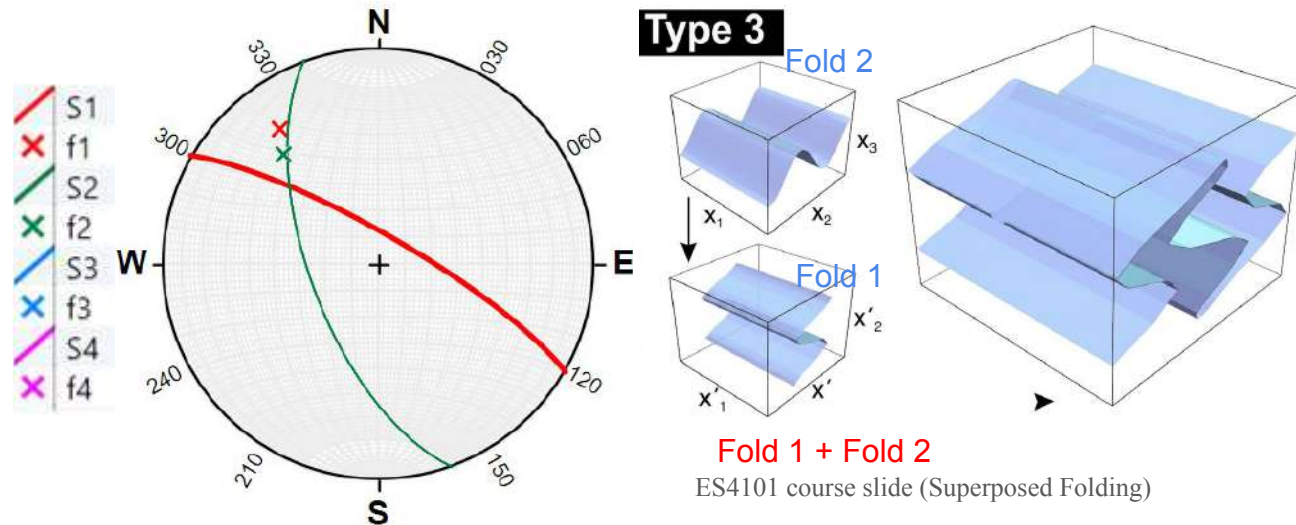
Type 2



Type III – Hook Structures

Hinge lines are at low angle to each other
and axial planes are at a higher angle to
each other

f2 → 315°/35°/70° S2 → 75°/240°
f1 → 322°/23°/63° S1 → 79°/030°



River Section



Types of Foliations Observed

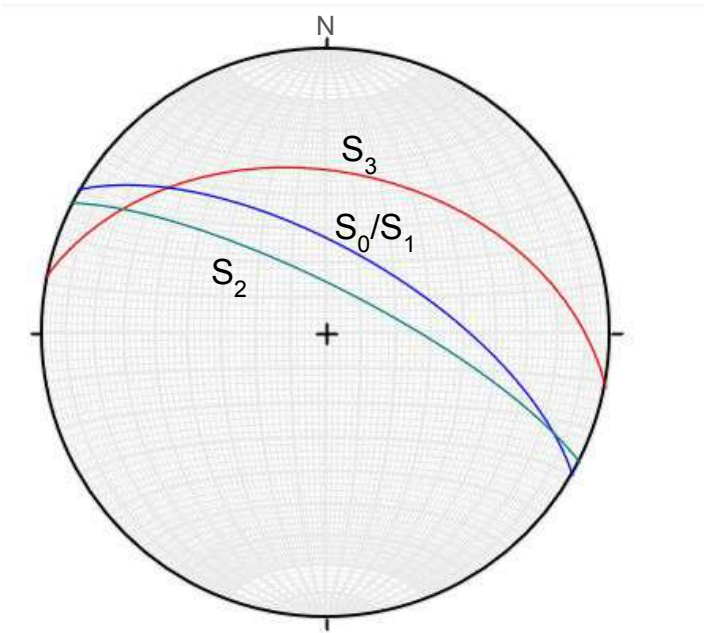


Fig 1: S_0/S_1 (scale(pencil):14cm)



Fig 2: S_2 (scale(pencil):14cm)

Orientation of Foliations



Foliations	Strike	Dip	Dip Directions
S_0/S_1 (mean)	300°	66°	030°
S_2 (mean)	297°	77°	027°
S_3 (mean)	281°	43°	011°

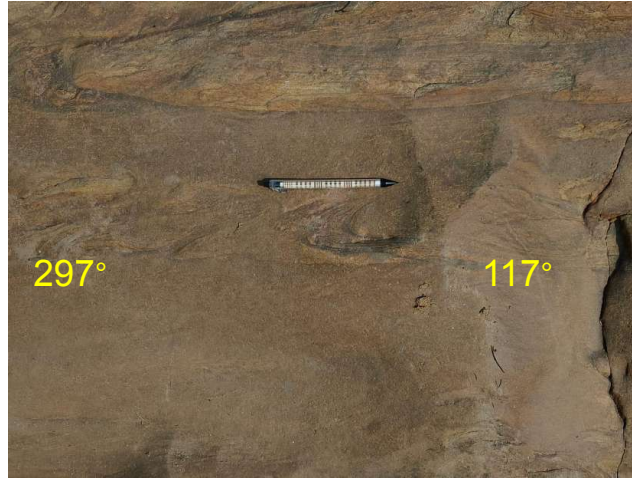
Fig 3: Stereo plot showing orientations of foliations

Types of Lineations Observed

- Isoclinal fold hinges
- Striping lineations
- Pucker Axis
- Mineral Lineations

Types of Lineations Observed

Isoclinal fold hinges



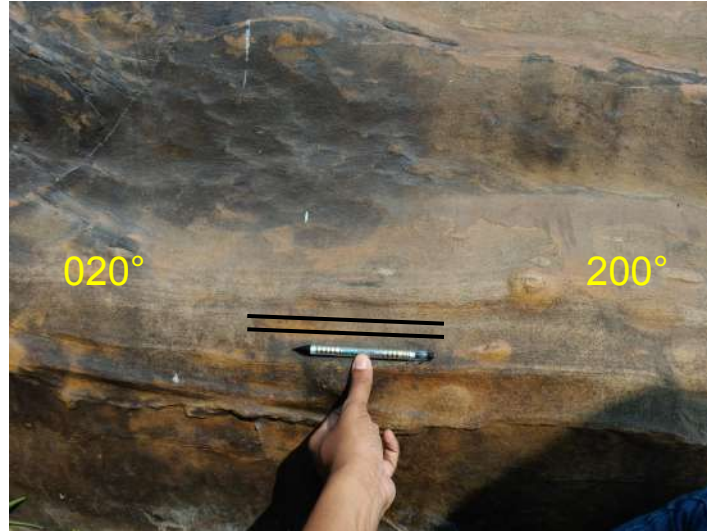
Isoclinal Hinges (scale(pencil size):14cm)



Isoclinal Hinges (scale(Brunton compass):12cm)

Types of Lineations Observed

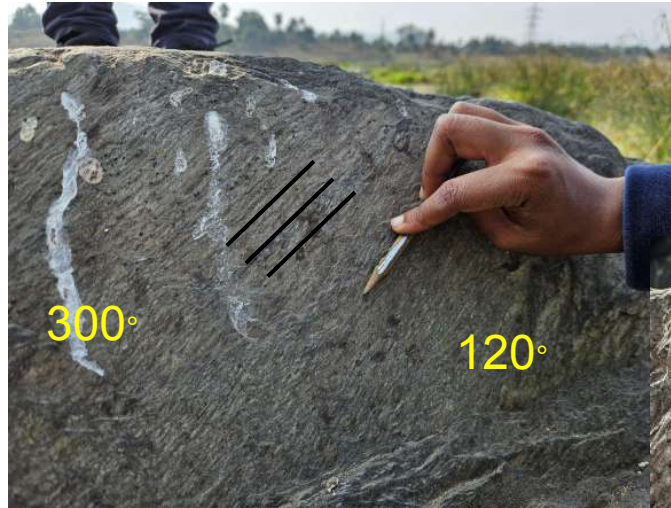
Striping lineations



Striping lineation (pencil size):14cm)

Types of Lineations Observed

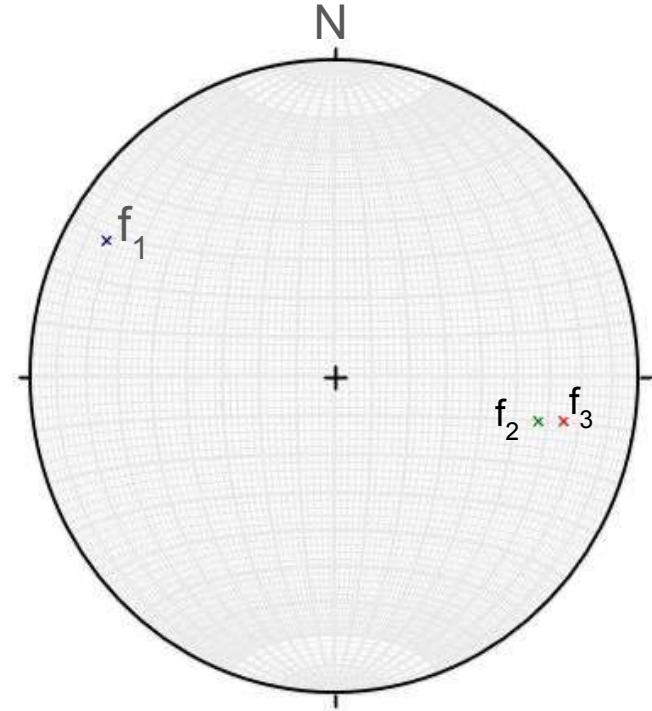
Pucker Axis



Types of Lineations Observed

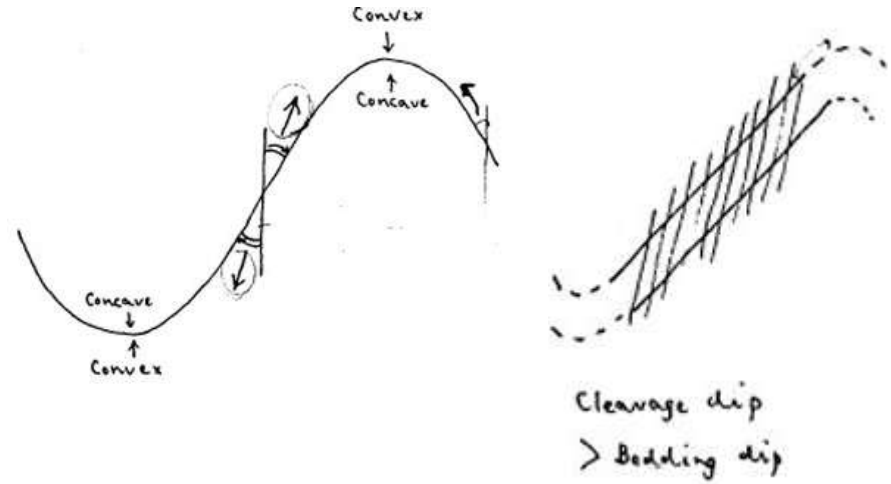
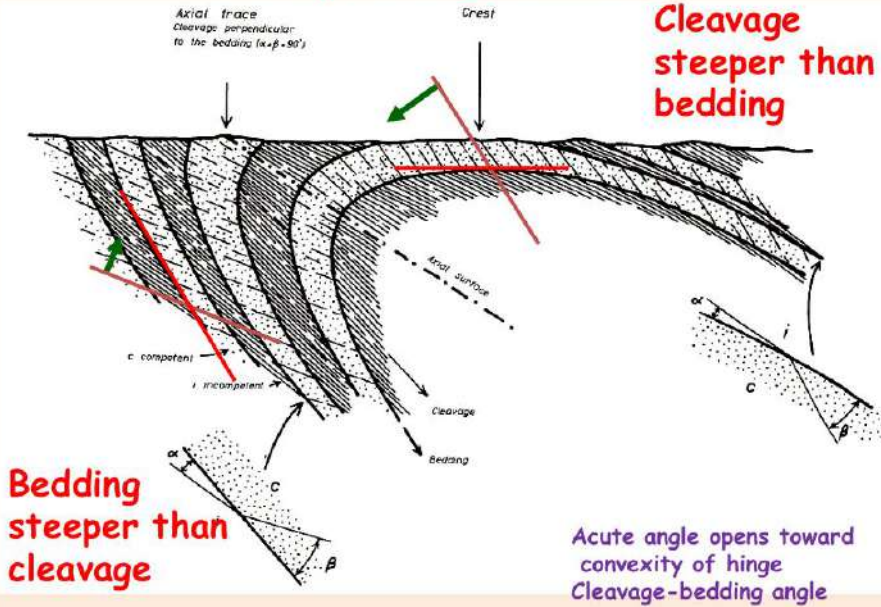
- Isoclinal fold hinges
- Striping lineations
- Pucker axis
- Mineral Lineations

Fold Axes	Trend	Plunge
f_1 (isoclinal hinges)	300°	16°
f_2	102°	32°
f_3	101°	24°
Striping lineations	149°	20°



Structural Facing

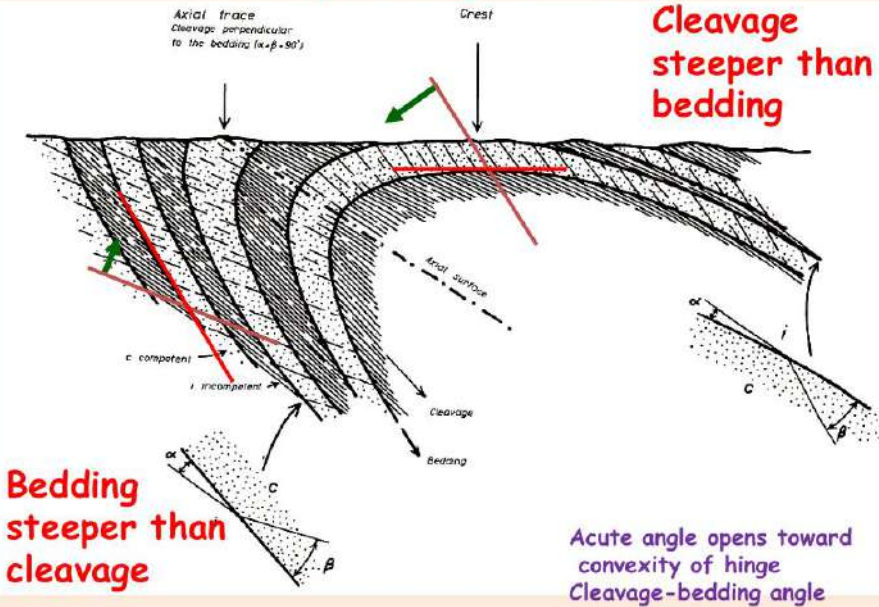
Cleavage Bedding Relationship



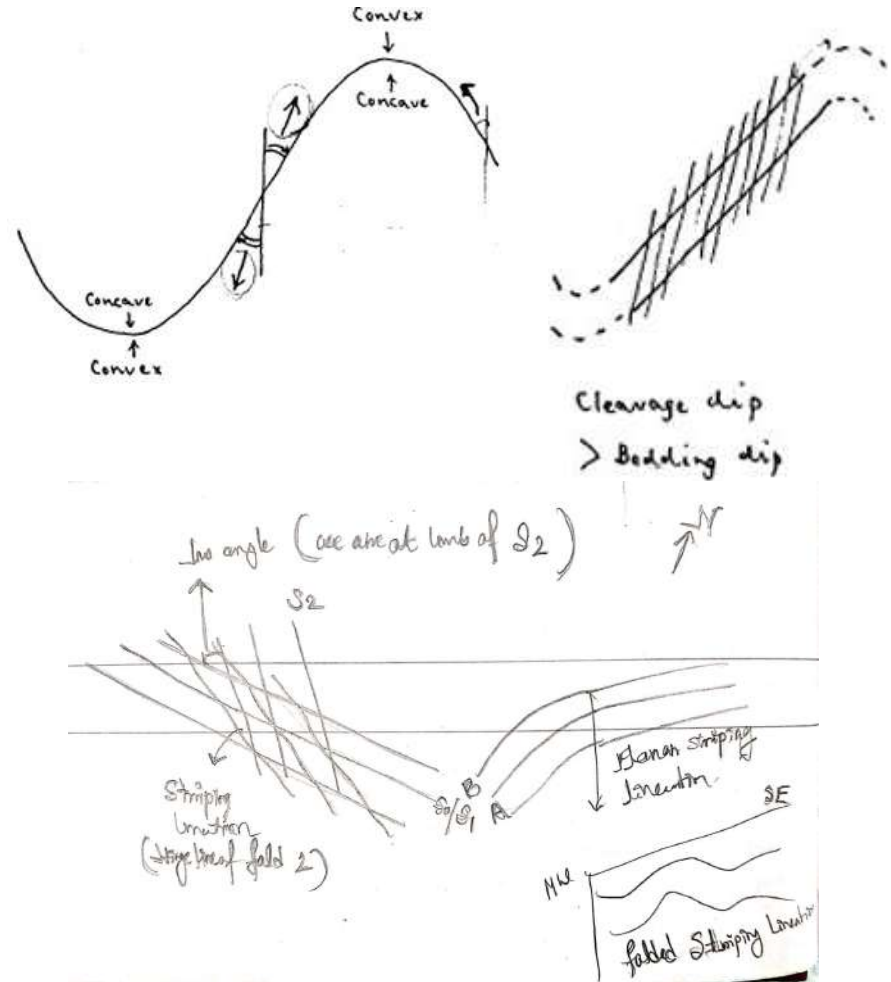
Reference: ES4101 course slide (lecture_41-42)
Chapter 12, Fossen; Chapter 11, Twiss & Moores, Paschier & Trouw

Structural Facing

Cleavage Bedding Relationship



Reference: ES4101 course slide (lecture_41-42)
Chapter 12, Fossen; Chapter 11, Twiss & Moores, Paschier & Trouw



Structural Facing

Cleavage Bedding Relationship

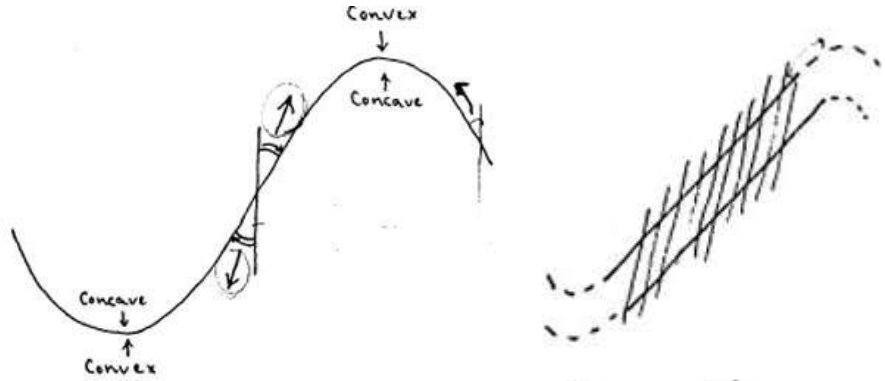
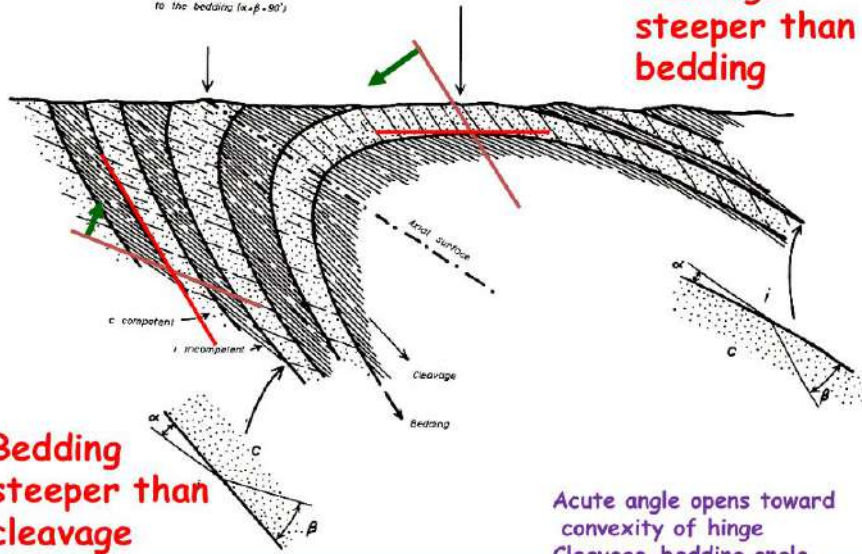
Axial trace
Cleavage perpendicular
to the bedding ($\alpha + \beta = 90^\circ$)

Crest

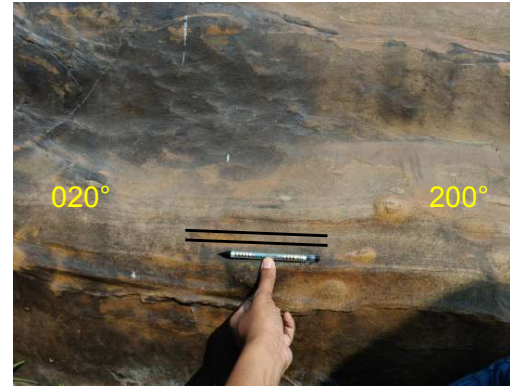
**Cleavage
steeper than
bedding**

**Bedding
steeper than
cleavage**

Acute angle opens toward
convexity of hinge
Cleavage-bedding angle



Cleavage dip
> Bedding dip



Reference: ES4101 course slide (lecture_41-42)

Chapter 12, Fossen; Chapter 11, Twiss & Moores, Paschier & Trouw

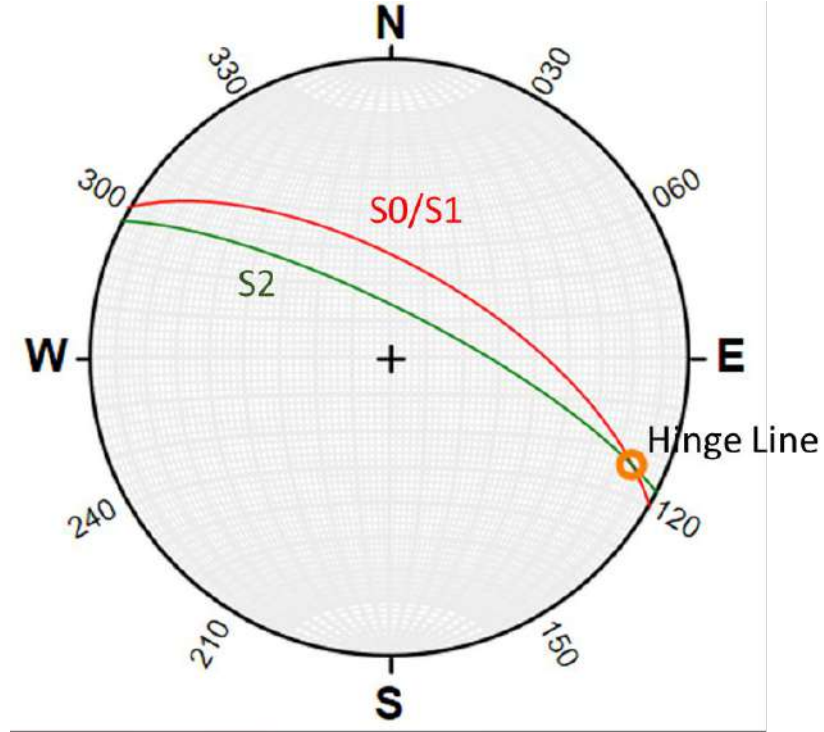
Hinge line orientation

Hinge Line-14°/114°

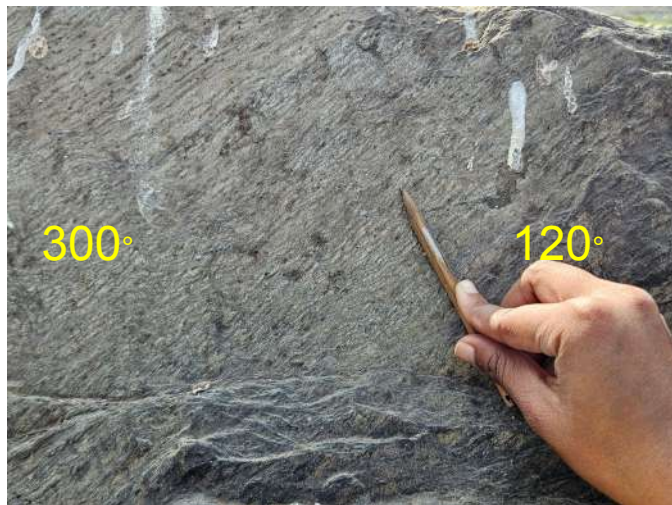
Consistent with what we have found from the
Chorinda i.e. 12°/112°

Criteria:

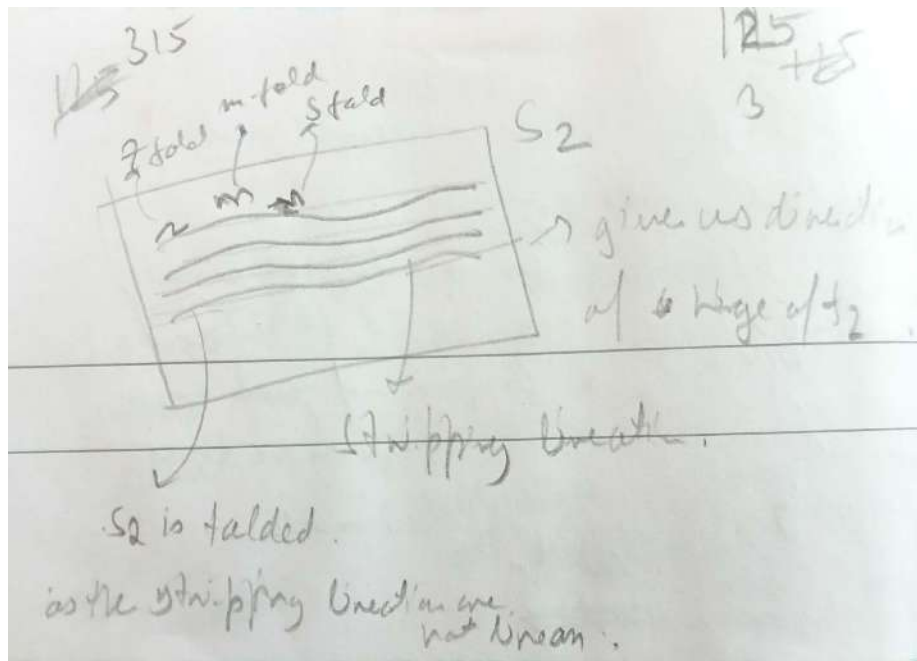
- Cleavage -Bedding intersection
- Pucker axis orientations
- Isoclinal hinge orientations



Minor Folds

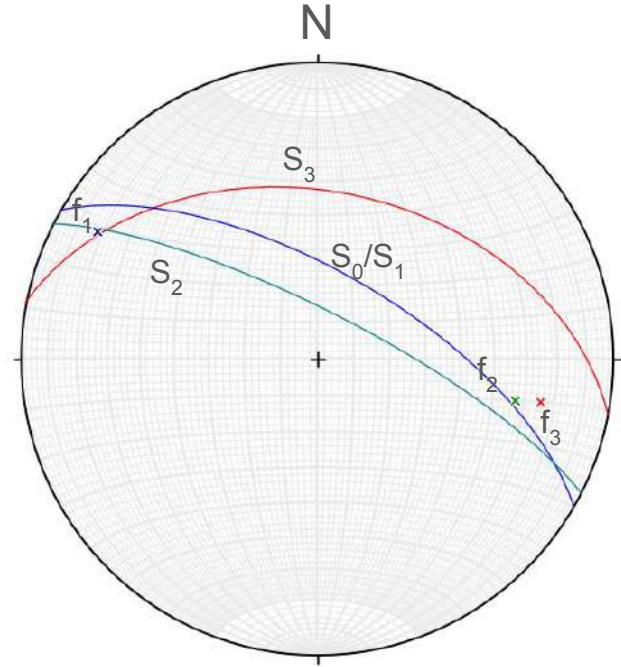


Pucker axis



Evidence of Superposed folding

- f_2 & S_3 at high angle
- f_3 and S_2 at low angle
- No end members of superposed foliations were observed.

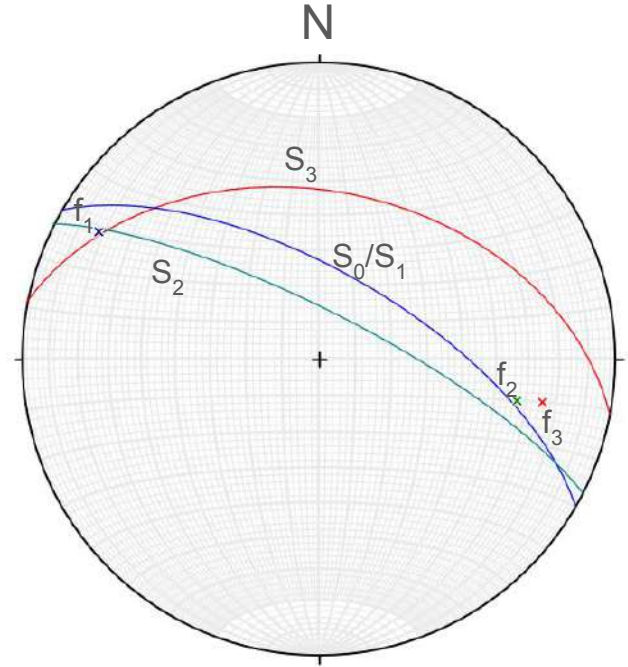


Evidence of Superposed folding

- f_2 & S_3 at high angle
- f_3 and S_2 at low angle
- No end members of superposed foliations were observed.

Reasons??

- Different Wavelength or intensity of folding
- Heterogeneous rock properties
- Structural overprinting



Which lithology develop the cleavage better?

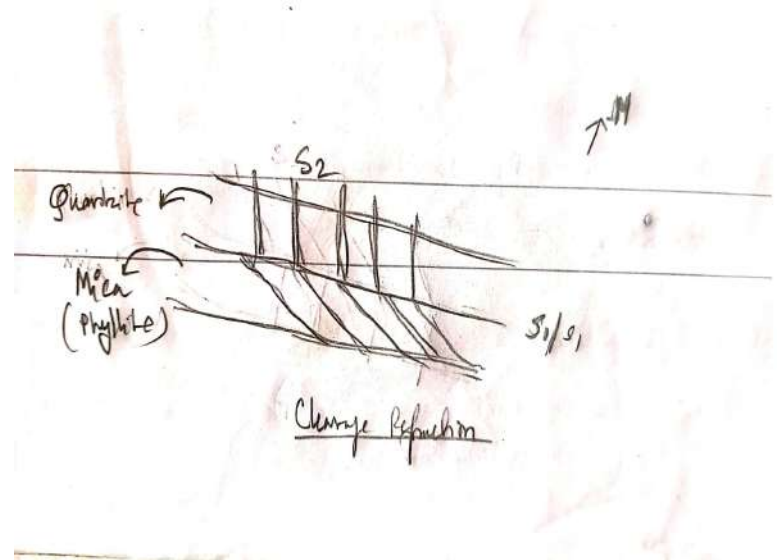
Two dominant types of lithology
observed-

Quartzite(quartz rich)-competent layer

Phyllite(mica rich)-incompetent layer

Phyllite developed cleavage better

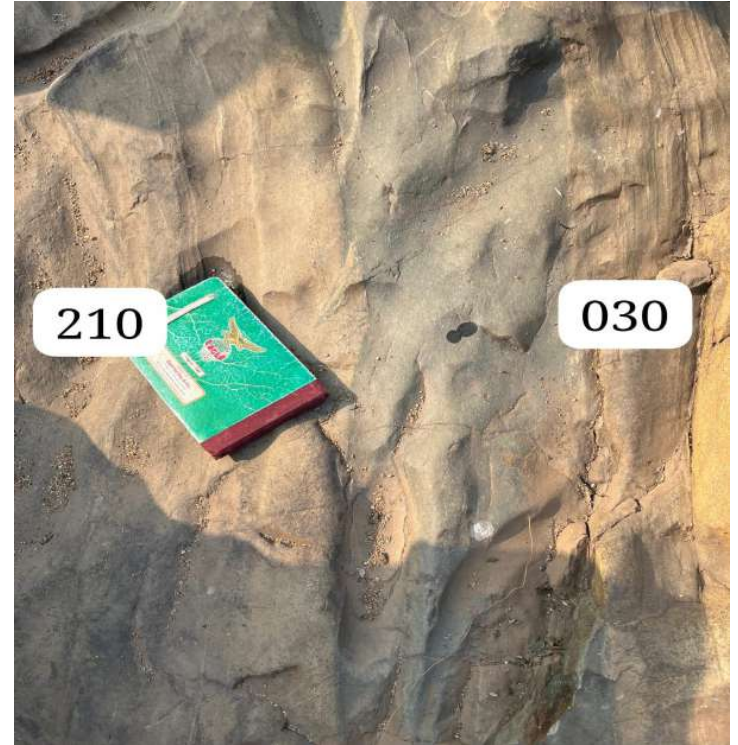
Cleavage refraction???



Evidence of Igneous Rock: Dyke

- Dyke cuts across country rock at high angle → younger intrusion
- Orientation:
 - Trend: East–West
 - Dip: Towards North
- Colour: Greenish to black
- Medium-grained, massive
- No foliation → no significant metamorphic deformation

Discordant Structure



Dyke (Notebook: length-15.7cm, width-10.5cm)

Petrographic Analysis

Of rocks from:

- Chorinda
- Subarnarekha river section
- Tetuldanga
- Dyke at the Subarnarekha River section



Petrography of Chorinda Rocks:



Field Appearance

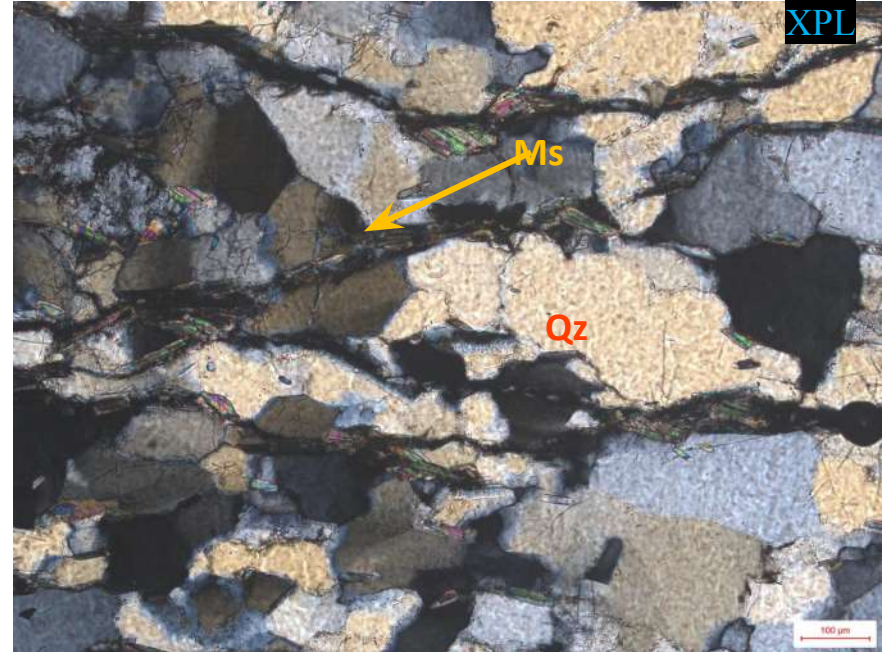


Thin section view

Observations

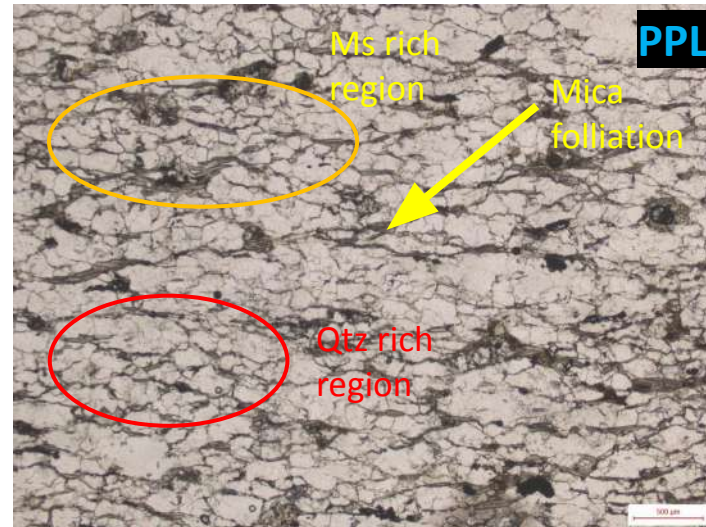
Minerals Present:

1. Quartz
2. Muscovite



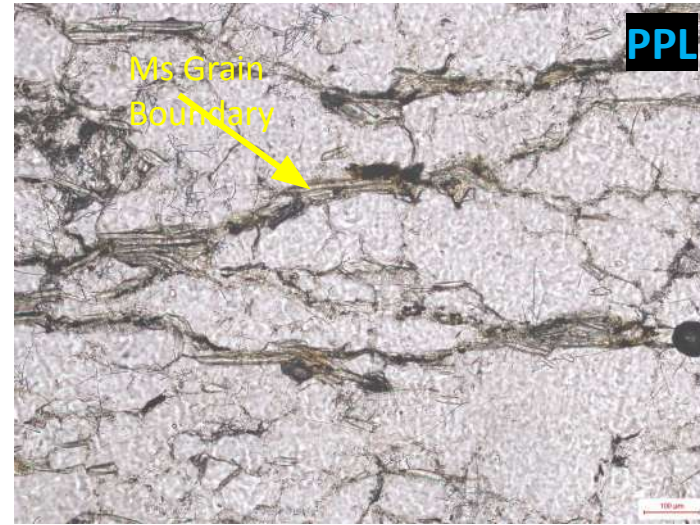
Observations: Significant Textures

- ❑ Grains are predominantly **anhedral to subhedral**
- ❑ **Little grain size variation**, with overall **small to medium grain size**
- ❑ **Mica content variation** corresponds with field-scale observations
- ❑ Presence of **crude foliation**
- ❑ Quartz exhibits **strong undulose extinction**



Observations: Significant Textures

- ❑ Stretched quartz and muscovite grains are prominent
- ❑ Quartz appears less **stretched** relative to muscovite
- ❑ **Broken/fractured quartz grains** are observed
- ❑ Grain boundaries are commonly **defined by aligned muscovite**



Observations vs Interpretation

Observation	Interpretation
Muscovite Grains are more stretched compared to quartz	Strong ductile deformation and mineral alignment as quartz grains are more resistive
Crude foliation	Uni-directional strain during deformation
Strong undulose extinction in quartz	Sign of intense strain
Broken quartz grains	Intense deformation along the direction of foliation
Muscovite-defined grain boundaries	Mica-controlled foliation fabric

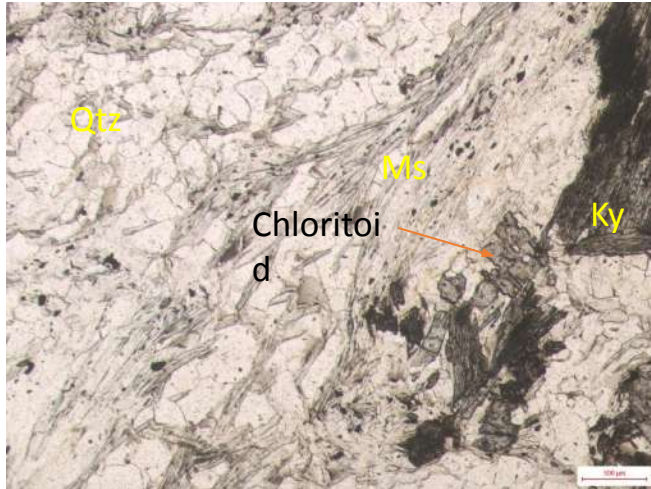
Based on the minerals present, the name of the rock should be: **Quartzite**

Protolith: **Pelitic bulk** (SiO_2 , K_2O , Al_2O_3)

Metamorphism: Can not be identified due to a lack of mineral variability

Petrography of Tetuldanga rocks:

Mineral assemblage: **Qtz + Ms + Ky + Chloritoid**



PPL Photomicrograph of a thin section of the rock.

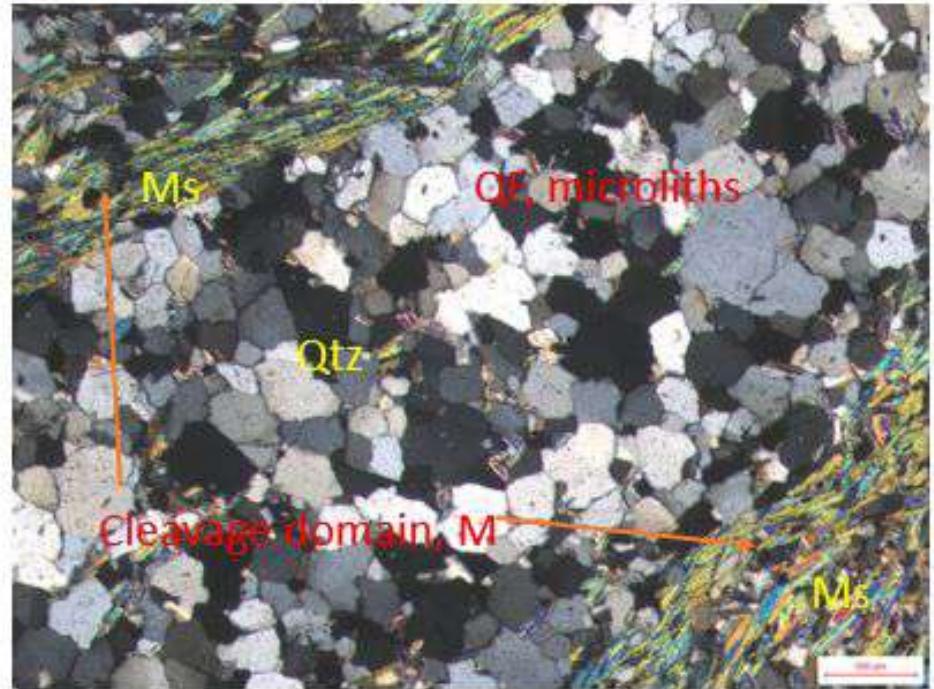


Field Photograph

Observations:

Significant textures: By studying the thin section of tetuldanga rocks, we have observed that

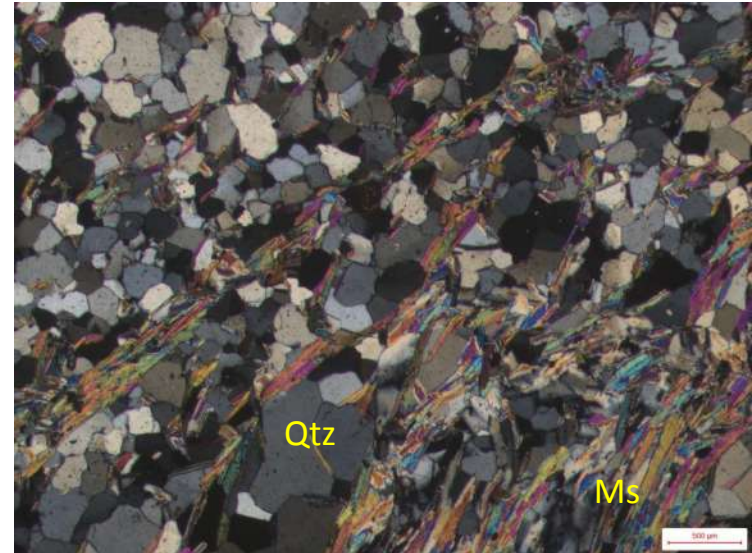
- Coarse-grained muscovite grains define the **foliation** here.
- There are **separate zones dominated by muscovite and quartz**, defining a **disjunctive cleavage**.
- We can also observe that the **quartz grains are not elongated here**, but are **irregular in shape at places**, showing **no undulose extinction**.
- **Muscovites in quartz microliths are fine-grained and form at a high and low angle to the foliation.**



Observations:

Significant textures: By studying the thin section of tetuldanga rocks, we have observed that

- Quartz grains show a **grain size variation** here.
- It also shows **granoblastic texture**, making $\sim 120^\circ$ with the boundary of other grains.

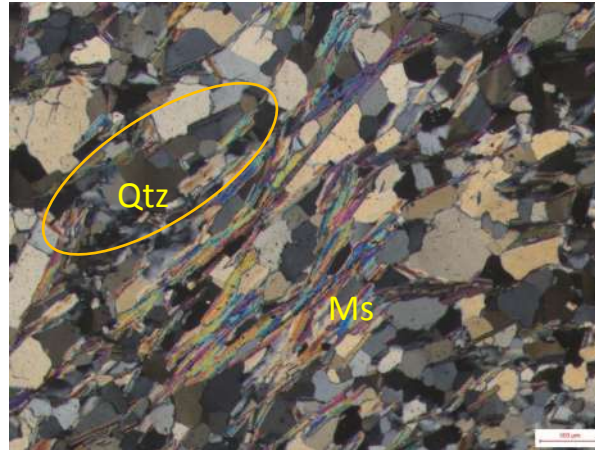


XPL

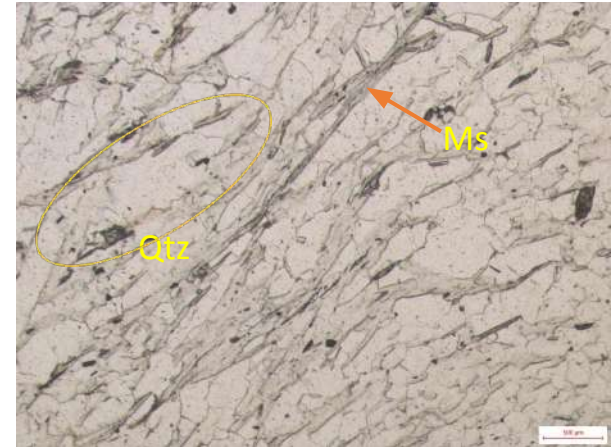
Observations:

Significant textures: By studying the thin section of tetuldanga rocks, we have observed that

- **Quartz grains elongated in the direction of foliation** defined by coarse-grained muscovite grains. Quartz Subgrain **formation** is observed here with a suture boundary.
- Some quartz grains **show undulose extinction** here.
- Muscovite grains bend around quartz aggregates.



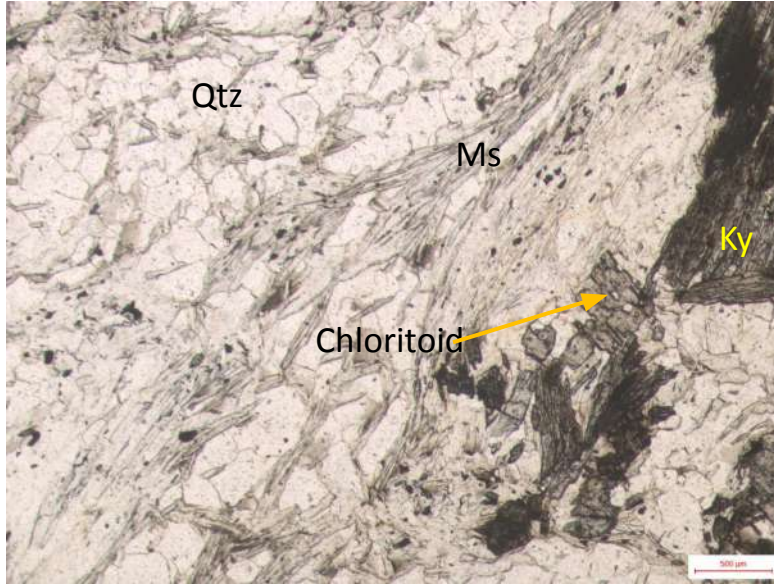
XPL



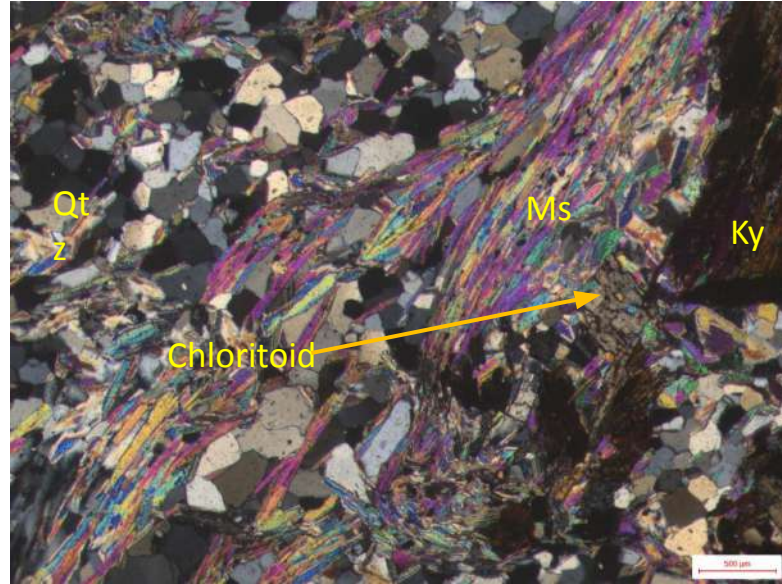
PPL

Observations:

Significant textures: By studying the thin section of tetuldanga rocks, we have observed that



PPL

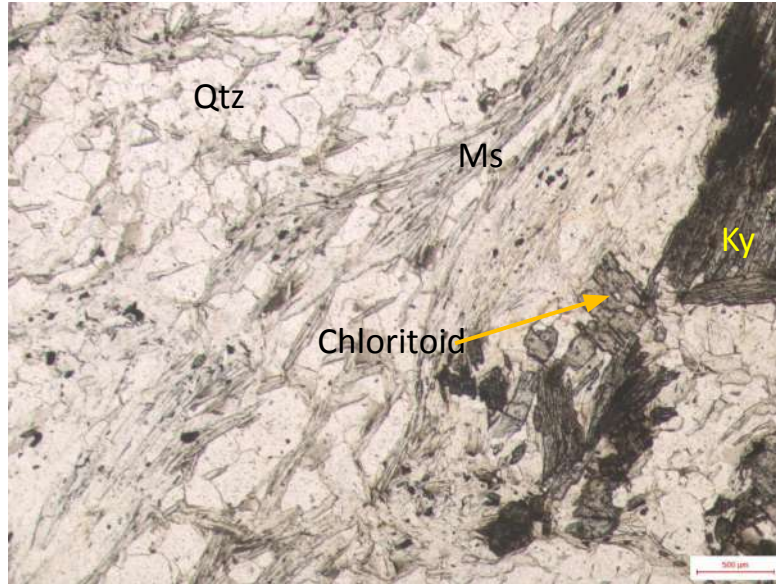


XPL

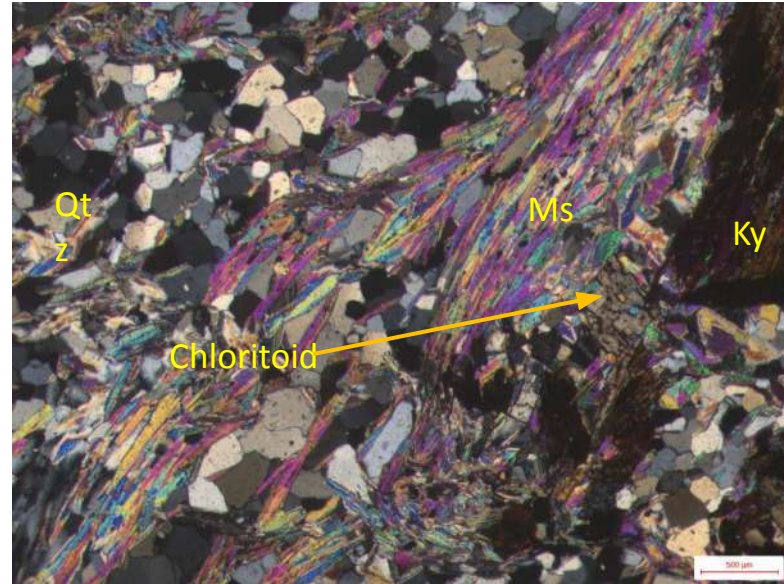
- Kyanite is formed with an orientation parallel to foliation, while chloritoid has an orientation at a high angle to foliation.

Observations:

Significant textures: By studying the thin section of tetuldanga rocks, we have observed that



PPL



XPL

- The **foliation is bent** at some places.

Observation vs Inference

Rock name: **Quartz muscovite Schist**

Observation:

- **Separate zones dominated by muscovite and quartz**, defining a **disjunctive cleavage**.

Inference:

- This indicates the deformational strain on the rock led to **pressure dissolution and recrystallisation**, leading to the formation of separate zones of quartz and mica. This indicates the presence of fluids in the rock during deformation,

Protolith: Pelitic Bulk (SiO_2 , K_2O , Al_2O_3 , FeO , MgO , MnO)

Observation vs Inference

Rock name: **Quartz muscovite Schist**

Observation:

- **Muscovite grains define the foliation and also show a bend in foliation. Fine-grained Muscovites in quartz microliths are forming at a high and low angle to the foliation.**

Inference:

- Due to a **directional deformation event**, muscovite grains have elongated in a particular direction and **formed the foliation.**
- Muscovite forming at a low angle with the foliation indicates that it formed when the foliation was developing.
- The muscovite grains forming at a high angle with the foliation indicate the fabric is associated with another deformation event, which led to the bending of foliation.

Observation vs Inference

Rock name: **Quartz muscovite Schist**

Observation:

- **Kyanite** is formed with an orientation **parallel to foliation**, while **chloritoid** has an orientation at **a high angle to foliation**.

Inference:

- This is also witnessed for kyanite and chloritoid.
- Presence of Ky also indicates **metamorphism up to upper greenschist facies to lower amphibolite facies with pelitic bulk composition**.

Observation vs Inference

Rock name: **Quartz muscovite Schist**

Observation:

- Quartz grains showing **subgrain formation**, suture boundary, **elongation in the direction of foliation and undulose extinction** at some places, and again showing **granoblastic texture, no undulose extinction, irregularly shaped and grain size variation** at some places.

Inference:

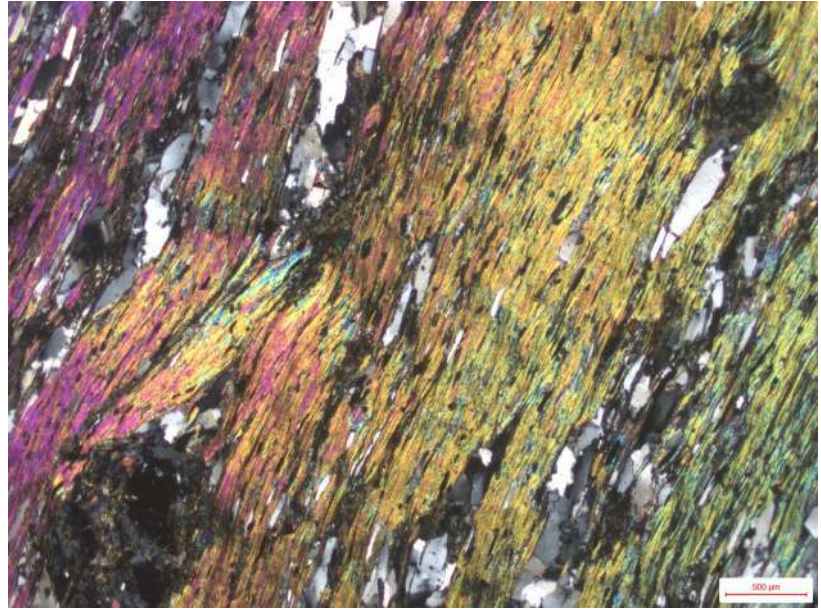
- This is because the deformation is manifested by quartz grains differently at different places.
- The former one indicates stretching of quartz grains in the direction of foliation, leading to a **change in shape and atomic arrangement of the grains**.
- The latter one indicates **dynamic recrystallisation**, formation of newer strain-free grains of quartz. This indicates the presence of fluids in the rock during deformation

The rocks had **muscovite and quartz** as major minerals and had **evidence of fluids in them** (at least while deformation), thus the deformation was manifested to a **higher degree in these rocks** than in the Chorinda rocks.

Petrography of Riversection Rocks:



Field appearance

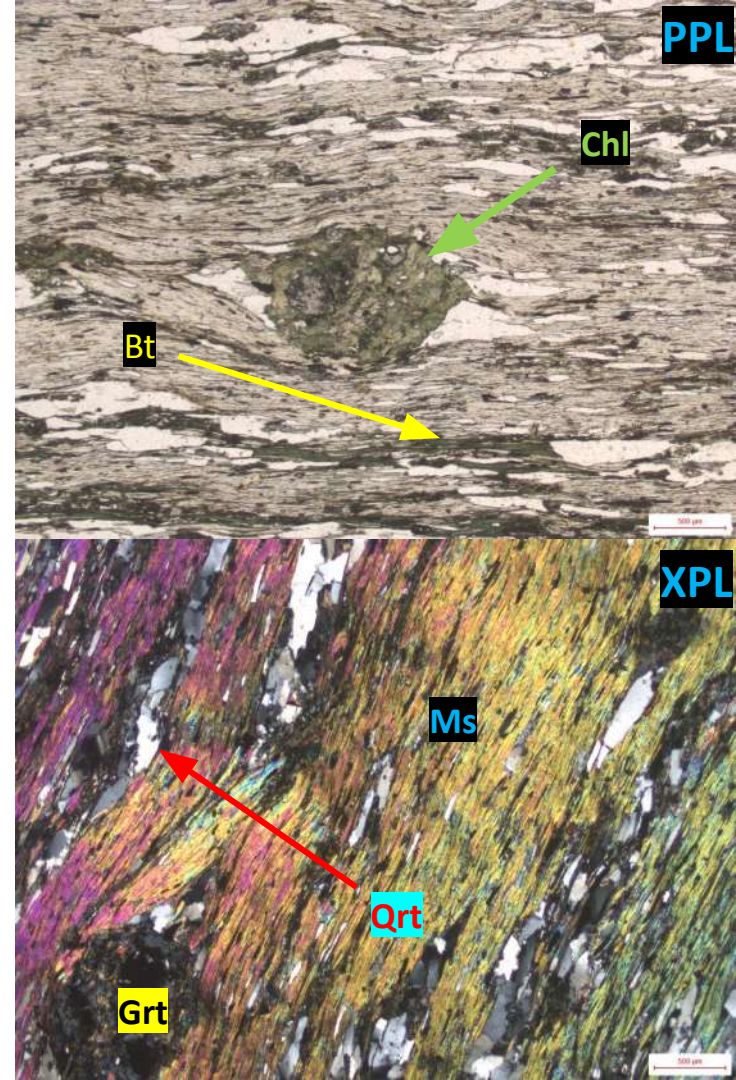


Thin section View

Observations

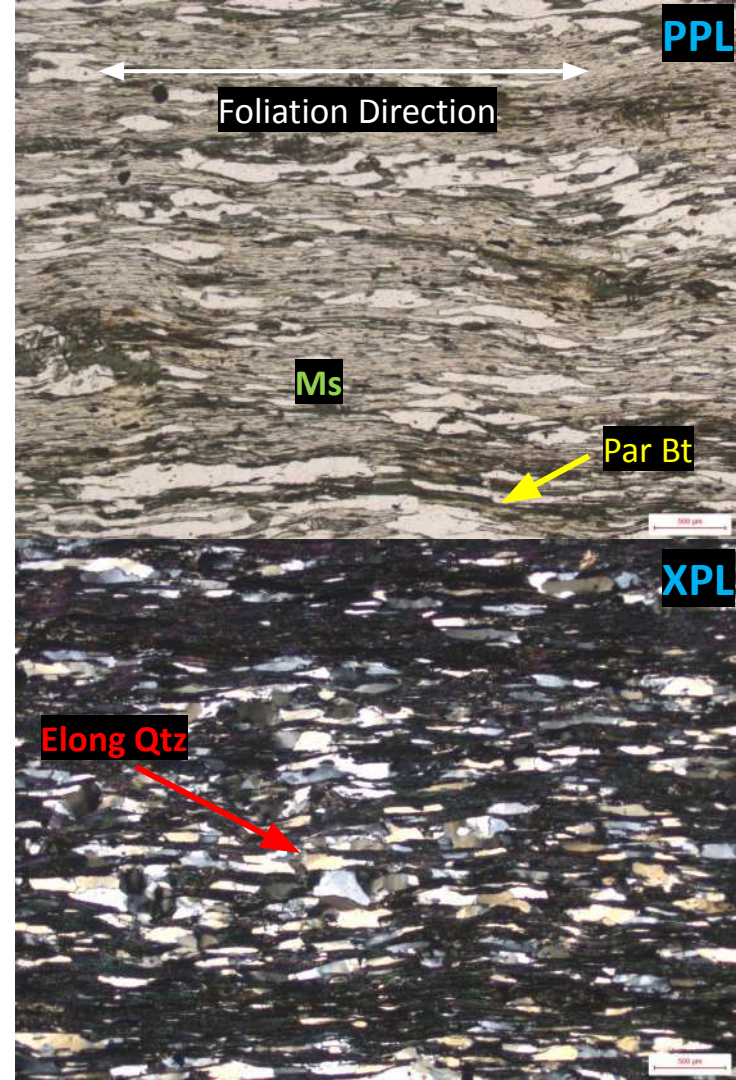
Minerals Present:

1. Muscovite
2. Quartz
3. Garnet
4. Chlorite
5. Biotite



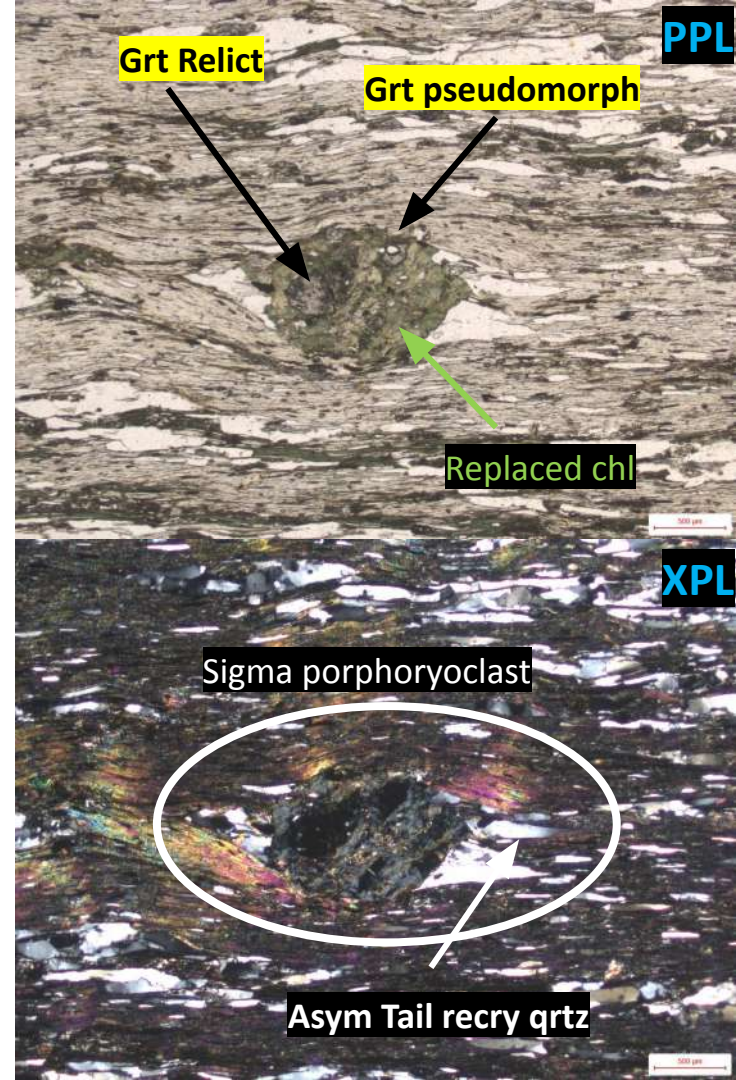
Observations: Significant Textures

- ❑ **Muscovite is the dominant mineral phase**, forming the major matrix
- ❑ Strongly developed unidirectional **predominant foliation**
- ❑ Biotite grains locally align parallel to muscovite
- ❑ **Elongated quartz grains** are parallel to foliation plane



Observations: Significant Textures

- ❑ Presence of **garnet pseudomorphs**
- ❑ Original garnet is partially to completely **replaced by chlorite**
- ❑ **Sigma-shaped porphyroclasts** are present
- ❑ Sigma clasts are surrounded by **recrystallized quartz in pressure shadows/tail regions**
- ❑ Tail geometry is asymmetrical



Observations vs Interpretation

Observation	Interpretation
Muscovite-dominated matrix	Strong foliation development
Elongated quartz	Unidirectional strain
Garnet pseudomorphs	Pre-kinematic Garnet
Chlorite replacement of garnet	Retrogression
Sigma porphyroblasts	Asymmetrical strain
Quartz tails	Dynamic recrystallization

Based on the minerals present, the name of the rock should be: **Garnet-bearing Quartz–Muscovite–Biotite Schist**
Protolith: **Pelitic bulk** (SiO₂, K₂O, Al₂O₃, CaO, FeO, MgO)
Metamorphic facies: **Garnet zone, lower amphibolite facies**

Lithological and Fluid-interaction Manifestation on River-Section Rock

Lithological manifestation:

- The **Subarnarekha river section** contains the highest proportion of **mica-rich minerals**
- **Mica is less resistant, platy and mechanically weak**

Maximum visible foliation

Fluid manifestation:

- The **river section** shows **chloritization of garnet, recrystallized quartz tails, and retrograde mineral replacement**
- These features indicate **significant fluid infiltration, hydration, and fluid-assisted retrogression during deformation**

Maximum fluid–rock interaction / retrograde overprint

Petrography of Dyke rocks:

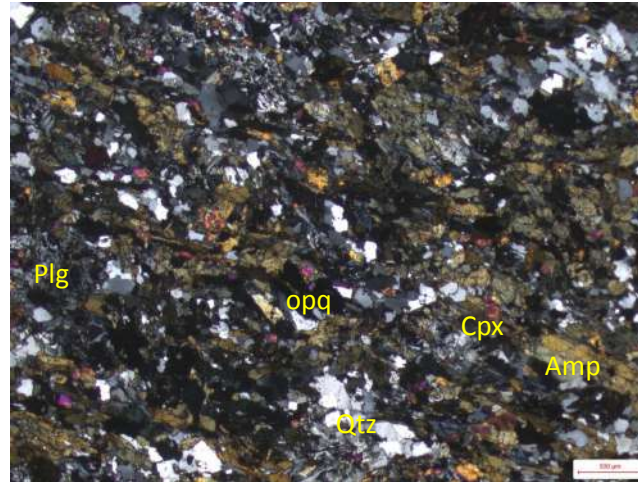
Location: Dyke intrusion at the Subarnarekha river bank

Mineral assemblage: Cpx + Amp + Qtz + Plg + Sericite + Chlorite + Opaque minerals

- Mafic bulk composition
- Intrusive contact - contact aureole



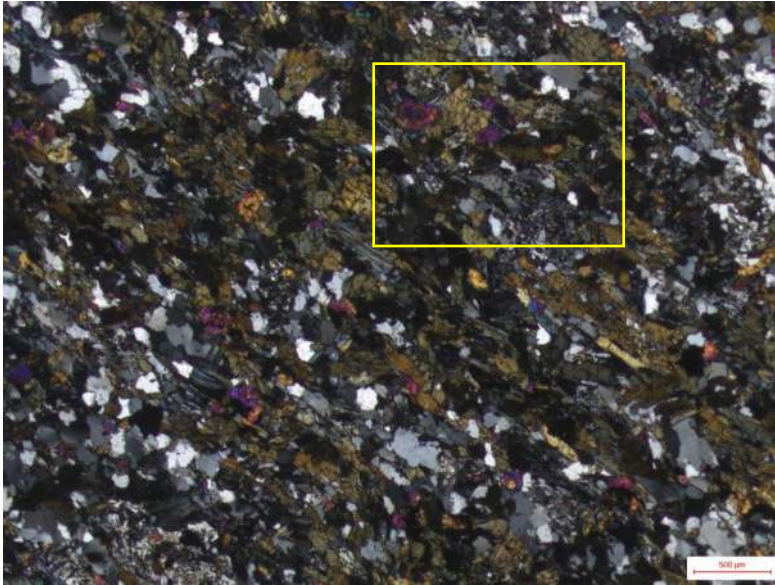
Field Photograph



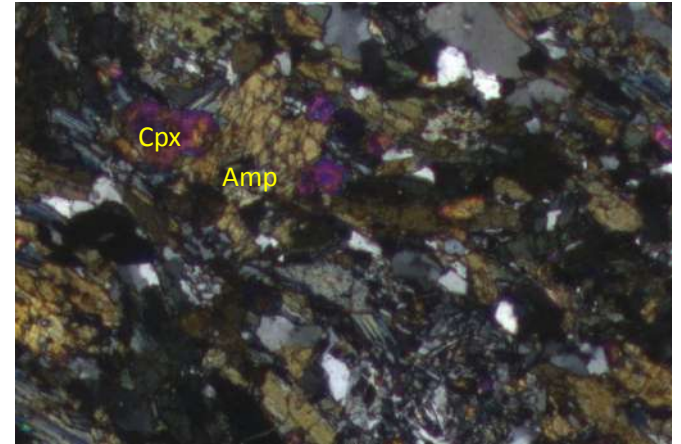
XPL Photomicrograph of a thin section of the rock.

Observations:

Significant textures: By studying the thin section of dyke rocks, we have observed that



XPL

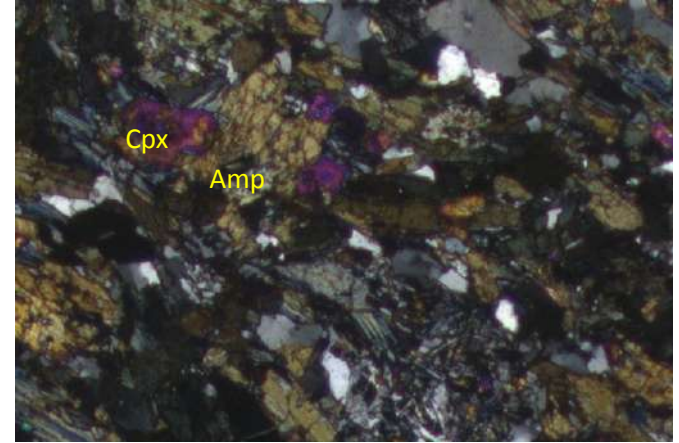


- The Cpx being replaced by Amp, forming a **pseudomorphic texture**.

Observations:

Significant textures: By studying the thin section of dyke rocks, we have observed that

- The Cpx being replaced by Amp, forming a **pseudomorphic texture**.



Observations:

Significant textures: By studying the thin section of dyke rocks, we have observed that



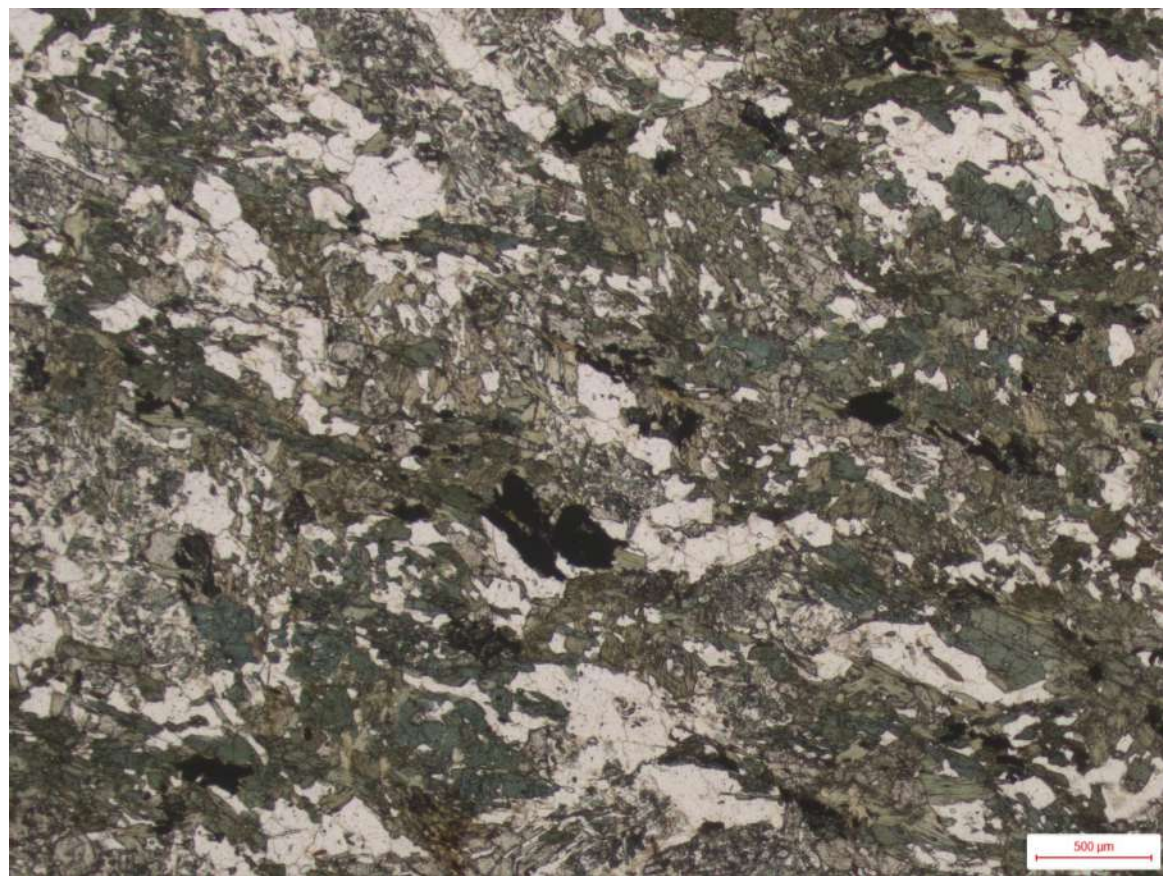
- The plagioclases present are mostly **sericitized**; however, in some place it tend to preserve the lamellar twinning and euhedral shape (now pseudomorphed by sericite and chlorite) faintly.

Observations:

Significant textures: By studying the thin section of dyke rocks, we have observed that

- The plagioclases present are mostly **sericitized**; however, in some place it tend to preserve the lamellar twining and euhedral shape (now pseudomorphed by sericite and chlorite) faintly.





Observation vs Inference

Observation:

- The amphiboles replacing the Cpx, forming a pseudomorphic texture.
- The plagioclases are now pseudomorphed by sericite and chlorite.

Inference:

- Both of the observations in dyke rocks indicate that the rocks have undergone **metamorphism**, qualitatively, up to **medium to upper green schist facies** because of the formation of amphibole (not Hbl) and chlorite and had **presence of fluid** during metamorphism.

Protolith: Mafic Bulk (FeO, MgO, SiO₂, CaO, K₂O, Al₂O₃)

Manifestation of deformation for different lithologies of rocks:

Manifestation of deformation depends on two factors when we have similar deformation (assuming similar pressure and temperature) going on:

- Lithology
- Fluid presence

Manifestation of deformation for different lithologies of rocks:

- ❖ Lithology very much influences the manifestation of deformation in the rocks.
- ❖ A rock with micas will show many evident signs of deformation, as they are less resistant to bending and stretching.
- ❖ But a rock with harder minerals like quartz will show resistance to deformation, thus recording fewer effects of it.

Location □ Rock Lithology

- Chorinda □ Quartzite
- Tetuldanga □ Quartz Muscovite Schist
- River section □ Garnet Quartz Muscovite Schist

Manifestation of deformation for different lithologies of rocks:

Location □ Lithology

- Chorinda □ Quartzite
 - Tetuldanga □ Quartz Muscovite Schist
 - River section □ Garnet Quartz Muscovite Schist
-
- ❖ The river section rocks had the maximum micas, followed by tetuldanga rocks, and lastly chorinda rocks. Whereas the chorinda rocks had maximum quartz, followed by tetuldanga rocks and lastly riversection rocks.
 - ❖ Thus, the manifestation of deformation recorded in river section rocks is maximum, followed by tetuldanga rocks and finally chorinda rocks.

Manifestation of deformation for different lithologies of rocks:

- ❖ The presence of fluid weakens the rocks and affects how they deform.
- ❖ Usually this leads to dynamic recrystallisation, pressure solution and metamorphism (at appropriate P-T conditions).
- ❖ Metamorphism is seen in all the rocks to a similar degree, but fluid-related alteration is seen in the tetuldanga rocks, followed by the riversection rocks.
- ❖ Teltuldanga rocks witness dynamic recrystallisation and disjunctive cleavage formation.
- ❖ Riversection and dyke rocks show fluid-based mineral alteration (garnet to chlorite, plagioclase to sericite, and chlorite and clinopyroxene to amphibole).

Manifestation of deformation for different lithologies of rocks:

- ❖ Metamorphism is seen in all the rocks, but fluid-related alteration is seen in the tetuldanga rocks, followed by the riversection rocks and dyke.
- ❖ Teltuldanga rocks witness dynamic recrystallisation and disjunctive cleavage formation.
- ❖ Riversection and dyke rocks show fluid-based mineral alteration (garnet to chlorite, plagioclase to sericite, and chlorite and clinopyroxene to amphibole).

Therefore, the tetuldanga, river section, and dyke rocks have exhibited different degrees of deformation due to the presence of fluid.

Geothermobarometry

It estimates the temperature and pressure of metamorphism using mineral equilibrium and chemistry.

Thermometry using Garnet–Biotite System

Logic of Method

Temperature is estimated via the Fe–Mg exchange distribution coefficient between garnet and biotite:

$$K_D = \frac{(Fe/Mg)_{garnet}}{(Fe/Mg)_{biotite}}$$

If Fe/Mg ratio changes → Temperature changes
Calibrated experimentally (Holdaway, 2000)

Interpretation:

- $\ln K$ describes Fe–Mg distribution between the two minerals
- $1/T$ term shows the reaction is temperature dependent
- Constants are from experimental thermodynamic calibration

What We Did

Mineral compositions from Table 1 (Setiawan et al., 2014) were used to calculate Fe/Mg ratios.

These were substituted into the garnet–biotite geothermometer equation to obtain temperature values, which were then plotted to derive P–T conditions.

$$\ln K_{Fe-Mg}^{Gt-Bt} = -\left(\frac{52,108}{3R}\right)\frac{1}{T} + \frac{19.51}{3R} - P\left(\frac{0.238}{3R}\right)\frac{1}{T}$$

Credit - Slides Geothermobarometry

Geothermobarometry

Barometry using Garnet–Plagioclase–Kyanite–Quartz System

Concept

Pressure conditions of metamorphism are estimated using geobarometry.

In the garnet–plagioclase–kyanite–quartz system, the distribution of Ca between garnet and plagioclase is pressure-dependent.

Reaction: Anorthite ↔ Grossular + Kyanite + Quartz

As this reaction shifts with pressure, mineral compositions record the pressure conditions during metamorphism.

Geobarometer Equation (Newton & Haselton, 1981):

$$\ln K_{Ca}^{Gt-Pl} = \left(\frac{55,865}{R} \right) \frac{1}{T} - \frac{153.59}{R} + P \left(\frac{6.608}{R} \right) \frac{1}{T}$$

Credit - Slides Geothermobarometry

Logic of Method

Pressure is calculated using the Ca distribution coefficient:

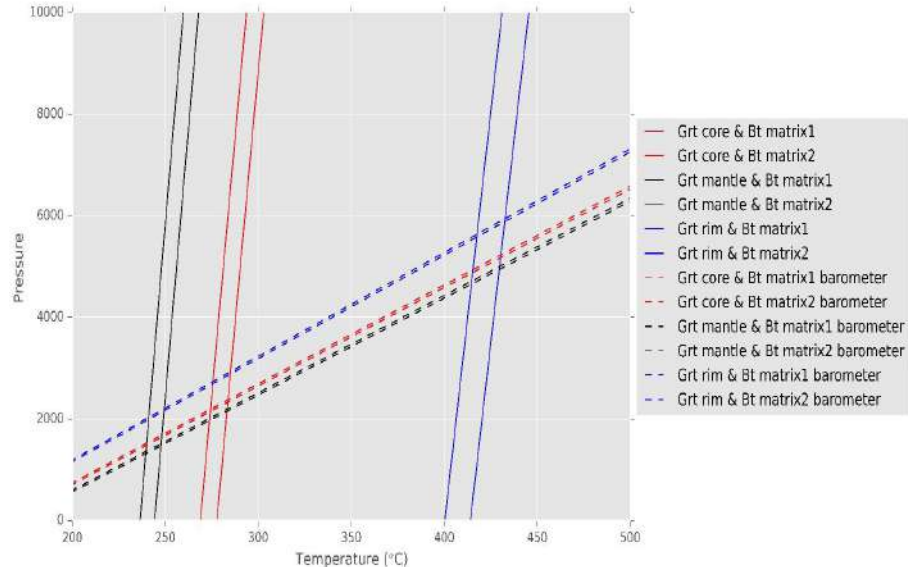
$$K_{Ca}^{Grt-Pl} = \frac{a_{grs}^{Grt}}{(a_{An}^{Pl})^3}$$

If Ca distribution changes → Pressure changes
Calibrated experimentally (Newton & Haselton, 1981)

$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ (gas constant)
 P is in bars, not kilobars!
 T is in Kelvins, not °C!!

P–T Diagram: Geothermobarometry Results

Garnet–Biotite System | Pressure–Temperature Conditions



Data - (Setiawan et al., 2014)

Interpretation

- This graph represents the Pressure–Temperature (P–T) conditions derived using garnet–biotite thermometry and garnet–plagioclase geobarometry.
- The solid steep lines represent temperature (thermometer) estimates, while the dashed gently sloping lines represent pressure (barometer) estimates.
- The intersection of thermometer and barometer lines gives the equilibrium P–T condition of the rock, which qualitatively falls within the upper greenschist to amphibolite facies.
- Different colors (core, mantle, rim) represent zoning in garnet, indicating how mineral composition changes during metamorphism.
- Minerals grow from core to rim, so we interpret the P–T path in that direction, showing the progressive evolution of metamorphic conditions.
- The barometer lines were obtained using permutations and combinations of grossular content from garnet at core, mantle, and rim with plagioclase compositions.
- From the graph, peak metamorphic conditions are observed at around 400–430°C and 4–5 kbar, which likely represent the maximum temperature and pressure experienced by the rock.

Discussion

- What do the results tell us?
- Limitations of the work we have done so far
- Comparison with the previous literature cited

Central discussion idea

- The three locations are not separate observations
- They represent different expressions of the same progressive deformation history.

One deformation history, three expressions

- Chorinda: mainly records brittle deformation through systematic fracture sets.
- River Section: records ductile deformation through foliations, lineations, minor folds and crenulations.
- Tetuldanga: gives the evidence of superposed folding and repeated refolding.

Petrography: confirms that deformation is recorded differently depending on mineralogy and lithology, because the same regional deformation is expressed differently - quartz-rich and mica-rich rocks respond differently to stress.

Limitations of the Interpretation

- **Discontinuous exposure:** Vegetation, weathering and human activity limited continuous tracing of structures between localities.
- **Measurement uncertainty:** Strike, dip, fracture and lineation readings may include small errors due to uneven or weathered surfaces.
- **2D outcrop view:** Many folds and interference patterns were observed on exposed surfaces, so their full 3D geometry is not completely constrained.
- **Relative timing only:** Fold generations, veins and dyke intrusion are interpreted from cross-cutting and overprinting relations, not absolute dating.
- **Correlation between localities:** Chorinda, River Section and Tetuldanga show comparable deformation styles, but direct structural continuity is not fully proven.
- **Geometrical Assumptions:** The connection between two consecutive exposed outcrops during fold reconstruction were done solely considering geometrical effects.

Comparison with previous literature

- **Ghosh & Sengupta, 1987:** Reported progressive ductile deformation and repeated folding in the Singhbhum Shear Zone.
Our work: Tetuldanga and River Section also show refolding, crenulations and overprinting, but in the framework of the NSMB.
- **Mamtani et al., 2004:** Studied fold-related joints in Galudih quartzites.
Our work: Chorinda quartzite fractures are systematic and likely fold-controlled, not random.
- **Mazumder, 2005:** Explained the regional background of Proterozoic sedimentation and later deformation in Singhbhum.
Our work: Our S_0 bedding and later foliation fit this framework of deformation, but not very precisely.
- **Roy & Matin, 2020:** Discussed small-scale structures and lithology-controlled deformation in the SSZ.
Our work: Quartz-rich and mica-rich rocks in our area preserve deformation differently.

Conclusion

Conclusion: Chorinda Fold Mapping

Structural Geometry of the Chorinda Fold

- Structural mapping at a scale of **1:2500** over **~0.186 km²** successfully constrained the fold geometry using a combination of strike and dip data. **Quartzite** beds acted as excellent **marker** horizons because of their competence, continuity and clear exposure.
- The mapped structure is an **open, asymmetric, plunging antiform** with an interlimb angle of **~72.5°** and fold axis plunging **25–29° toward ~133°**.
- **Opposite** dip directions of the two limbs confirm that the fold is **not overturned** and preserves upright to steeply inclined geometry.
- Chorinda therefore represents the best exposure for recognising the **first-order** regional fold architecture.

Conclusion: Chorinda Hinge Line & Reliability

Reliability of Fold Axis Measurements

- Hinge orientation was estimated using **multiple methods** such as π plot, limb intersection, striping lineations and Bedding-axial planar cleavage.
- π plot (**29°/133°**) and limb-intersection (**25°/134°**) gave the more consistent but not that reliable for our superposed fold.
- Striping lineation-based methods showed minor deviation due to influence of local hinge-zone deformation but is more reliable as directly measured.
- Final interpreted hinge line is therefore **moderately plunging toward the southeast (~133°)**.

Conclusion: Chorinda Fracture Mapping

Joint Sets and Brittle Deformation

- **Seven** fracture sets (F1–F7) were recognised and **grouped** into longitudinal, transverse and oblique sets relative to the fold axis.
- Longitudinal sets are **parallel** to fold trend, transverse sets **cut across** the structure, while oblique sets represent intermediate orientations.
- Frequency analysis shows that **transverse and high-angle fractures are most dominant** in the area.
- These fractures represent a later brittle **overprint** developed after the major ductile folding event.

Conclusion: Fluid Flow Through Fractures

Structural Control on Veining

- Veins were mainly observed along fracture sets **F2, F3 and F5**, showing that **not all fractures** were **equally** favourable for fluid flow.
- Set **F2 (141/66)** had the highest **vein-to-fracture** ratio, indicating the best permeability and connectivity.
- These fractures likely acted as **pathways** for hydrothermal or metamorphic ***fluids*** during late-stage tectonic activity.
- The study shows that structural **discontinuities** strongly control alteration, ***mineralisation*** and fluid migration.

Conclusion: Chorinda ↔ River Section Correlation

- The River Section exposes the **continuation** of the Chorinda structural system, with **similar** fold trends, hinge orientation and preservation of S₀ and S₁ surfaces.
- However, the River Section is **richer in mica**-bearing schists, so additional structures such as S₂, S₃ cleavages, lineations and shear indicators are **better** developed.
- A **discordant dyke** cutting the foliation records a younger intrusive event after deformation.
- Together, both localities represent the **same** tectonic history expressed **differently** due to lithological variation.

Conclusion: Chorinda ↔ Tetuldanga Correlation

- Chorinda records the **first-order** regional antiform with measurable limbs, hinge line and fracture overprint.
- Tetuldanga preserves the same tectonic history at **smaller scale** through repeated refolding, **multiple** foliations and interference patterns.
- Structures that are **subtle or poorly** preserved at Chorinda become **clearly** visible at Tetuldanga due to stronger mica-rich layering.
- Thus, Tetuldanga can be viewed as the **mesoscale** expression of the **broad**er fold system mapped at Chorinda.

Conclusion: River Section ↔ Tetuldanga Correlation

- Both River Section and Tetuldanga are **rich** in mica-bearing rocks and therefore preserve foliations and crenulation cleavages better than Chorinda.
- River Section shows **lineations**, shearing and metamorphic overprint, while Tetuldanga shows multiple generations of fold interference more clearly.
- In both areas, S_0, S_1 are preserved and **later fabrics overprint** earlier structures.
- Together they provide strong evidence for **progressive ductile** deformation within the **same tectonic** regime.

Conclusion: Tetuldanga Outcrop

Best Exposure of Polyphase Deformation

- Tetuldanga contains **alternating** quartz-rich and mica-rich **layers**, creating strong **competence contrast** and **excellent preservation** of structures.
- Bedding-parallel S_0/S_1 surfaces, later foliations, crenulations and minor folds are all **clearly visible** in a single outcrop.
- Multiple hinge orientations and **repeated overprinting** indicate several deformation phases.
- It is therefore the most suitable locality for understanding **superposed folding** in the study area.

Conclusion: Fold Interference Patterns

Evidence of Superposed Folding

- Tetuldanga preserves *three classic interference* geometries: **Type III hook structures**, **Type II arrowhead patterns**, and **Type I dome-and-basin forms**.
- These patterns formed by **progressive refolding** of earlier folds by younger fold generations (f1 to f4).
- The **coexistence** of *multiple interference* styles proves that deformation was polyphase rather than a single event.
- **At least four** recognisable folding stages affected the rocks.

Conclusion: Lithology Controls Deformation Style

Why Structures Differ Between Localities ?

- Although all outcrops belong to the **same** tectonic belt and **similar** metamorphic grade, structural **expression** varies strongly with lithology.
- Quartzite-rich Chorinda preserves **large folds** well but shows **weaker cleavage** development.
- Mica-rich River Section preserves foliations, shearing and lineations more **effectively**.
- Mixed quartz–mica layering at Tetuldanga gives the **best** preservation of **interference** patterns and multiple foliations.

Conclusion: Regional Structural Evolution

Tectonic History & Folding Mechanism Interpreted

- Sedimentation of the **Chaibasa** Formation produced original bedding (S_0), later followed by burial, metamorphism & development of early foliation (S_1) via LPS.
- Continued **contraction** generated major **orthogonal flexure** folding and later **refolding** events (f_1 to f_4), producing cleavages and interference structures.
- Subsequent **brittle** deformation formed fractures and veins, followed by **dyke** intrusion cutting earlier fabrics.
- Final **uplift** and **retrogression** preserved the present structural **architecture** of the North Singhbhum **Orogen**.

Final Overall Conclusion

Integrated Outcome of the Fieldwork

- Chorinda revealed the regional **fold** geometry, **fracture** mapping showed late brittle tectonics, River Section preserved metamorphic **shearing**, and Tetuldanga displayed **polyphase** folding most clearly.
- Thin section studies confirmed that deformation is recorded from **outcrop** scale down to **microscopic** scale through recrystallisation, pressure shadows, foliation and retrogression.
- The **Galudih** region (including all locations) therefore preserves a **complete tectonic** transition from sedimentation to metamorphism, ductile folding, brittle fracturing and magmatic intrusion.
- This field area is an excellent **natural laboratory** for understanding **crustal** deformation in the Singhbhum region.

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